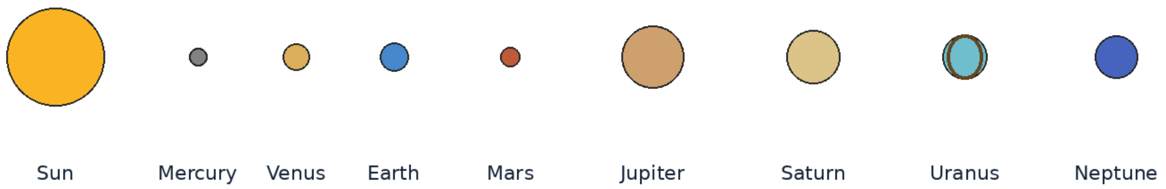
OFFICIAL READING MATERIAL

Indian Space Olympiad

ISO 2026

Foundation Level · Classes I-IV

The planets are shown in order, not to scale.



Published by Indian Space School

Revised textbook edition · Scientific status reviewed through 29 June 2026

Publication and Editorial Note

This handbook is original educational material prepared for the Foundation Level syllabus of the Indian Space Olympiad 2026. Its organisation is inspired by the clarity of well-designed school textbooks: concepts are introduced in sequence, illustrations are tied directly to the explanation, and questions appear after the lesson rather than interrupting the main text.

The book is written for Classes I-IV. Younger learners may read with a parent or teacher, discuss the illustrations and answer orally. Older learners can complete the written exercises independently. Scientific terms are introduced only when they help explain an idea accurately.

This edition distinguishes completed achievements from missions that are still under development. Current-affairs material should be checked again immediately before any later reprint.

Safety Note

Never look directly at the Sun. Ordinary sunglasses, smoked glass, exposed film and homemade filters do not protect the eyes. Solar observation requires certified equipment and responsible adult or expert supervision.

Preface

A child does not need to travel to a laboratory to begin studying space. A changing shadow, the Moon seen on several evenings, the direction of sunrise and a clear night sky can all become starting points for scientific thought.

The purpose of this book is not to fill memory with isolated facts. It is to help learners notice patterns, ask sensible questions, compare possibilities and explain an answer using evidence. Olympiad preparation becomes stronger when a learner understands why an answer is correct and why the other options are not.

The chapters therefore move from familiar observations to larger ideas. Earth and sky patterns lead to the Solar System, stars and galaxies. Everyday communication and weather lead to satellites. Pushes, pulls, light and motion lead to rockets and spaceflight. Reasoning chapters then show how the same habits of observation can be used in unfamiliar questions.

How to Use This Book

- **Read the lesson first.** Do not begin with the questions. Follow the explanation and examine the figure or example.
- **Say the idea in your own words.** After each section, explain the main point without looking at the page.
- **Perform only safe activities.** Record the date, time, direction and observation. Do not look directly at the Sun.
- **Attempt the exercises without help.** Then check the answer key and correct the reason, not only the option letter.
- **Review by unit.** The unit review combines ideas from several lessons and is closer to an Olympiad test.

Suggested Weekly Routine

Day 1	Read one lesson and discuss the figure.
Day 2	Complete the observation or activity.
Day 3	Answer the short questions.
Day 4	Attempt the MCQs and explain every choice.
Day 5	Review key words and one earlier chapter.

Foundation Level Syllabus

01. Fundamentals of Astronomy: What is Astronomy?, The Sky Around Us, Day and Night, Sunrise and Sunset, Directions, Seasons, Observing the Sky, Stars Visible at Night, Introduction to Constellations, Astronomy in Everyday Life, Simple Astronomical Observations.

02. The Solar System: Our Sun, The Eight Planets, Dwarf Planets, Order of the Planets, Natural Satellites, Asteroids, Comets, Meteors and Meteorites, Characteristics of the Planets.

03. Stars, Galaxies & the Universe: What are Stars?, Brightness of Stars, The Nearest Stars, The Milky Way Galaxy, Other Galaxies, Understanding the Universe.

04. Earth & Moon: Earth: Our Home Planet, Continents and Oceans, Earth’s Rotation, Earth’s Revolution, Day and Night Revisited, Seasons Revisited, Phases of the Moon, Solar and Lunar Eclipses, Tides.

05. Satellites & Space Technology: Natural and Artificial Satellites, Communication Satellites, Weather Satellites, GPS and Navigation, Satellites in Everyday Life.

06. Rockets & Launch Vehicles: What is a Rocket?, How Rockets Reach Space, Launch Pads, Astronauts, Reusable Rockets.

07. ISRO & India’s Space Missions: Introduction to ISRO, Chandrayaan Missions, Mangalyaan, Aditya-L1, Gaganyaan, Great Indian Space Scientists, India's Achievements in Space.

08. Space Exploration: Who are Astronauts?, Space Stations, Moon Exploration Missions, Mars Exploration Missions, Famous Space Missions.

09. Basic Physics for Space Science: Gravity, Force, Motion, Light, Shadows, Heat, Simple Machines, Air and Atmosphere.

10. Scientific Reasoning & Logical Aptitude: Observation Skills, Picture Reasoning, Pattern Recognition, Classification, Sequencing, Simple Mathematical Reasoning, Logical Thinking.

11. General Space Awareness: Current Space Events, Famous Astronauts, Planets in the News, Important Space Discoveries, Amazing Space Facts.

The sequence follows the official ISO 2026 Foundation Level syllabus supplied for Classes I-IV.

Contents

The units and chapters below follow the official Foundation Level syllabus.

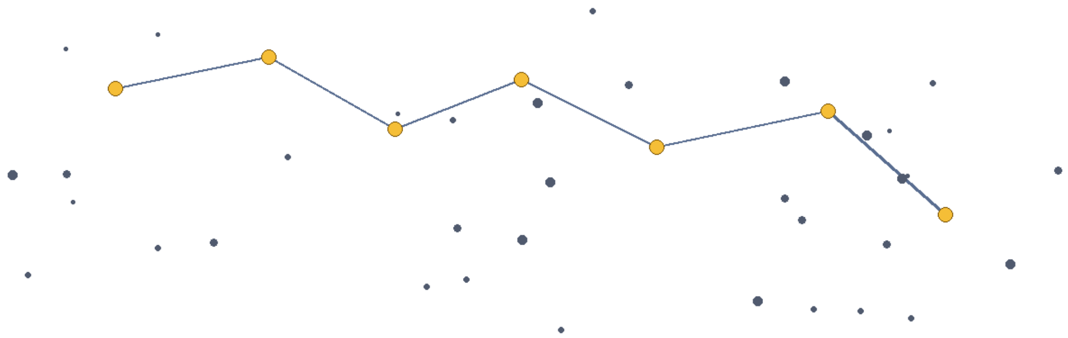
Contents · Continued

Contents · Continued

UNIT 1

Fundamentals of Astronomy

The sky is the first science laboratory available to every child. In this unit, learners observe daily and seasonal patterns, use directions correctly and begin to explain what they see with evidence.



Constellation lines are imagined patterns; the stars can be at very different distances.

Chapters in this unit

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 1 · CHAPTER 1

What is Astronomy?

Learning outcomes

- Explain what Astronomy is using simple scientific language.
- Use the words astronomy and astronomer correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Astronomy is the scientific study of objects and events beyond Earth, including the Sun, Moon, planets, stars, galaxies and the universe. Astronomers learn by observing light, measuring patterns and testing explanations.

Understanding the idea

Astronomy is a science. It uses evidence, careful measurements and repeated observations instead of guesses.

Many astronomers use telescopes, cameras, computers and spacecraft, but useful astronomy can also begin with unaided-eye observations.

Astronomy asks questions such as: What is an object made of? How does it move? How far away is it? How did it form?

Space science connects astronomy with physics, mathematics, engineering, chemistry and Earth science.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

Do not learn astronomy, astronomer, observation separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

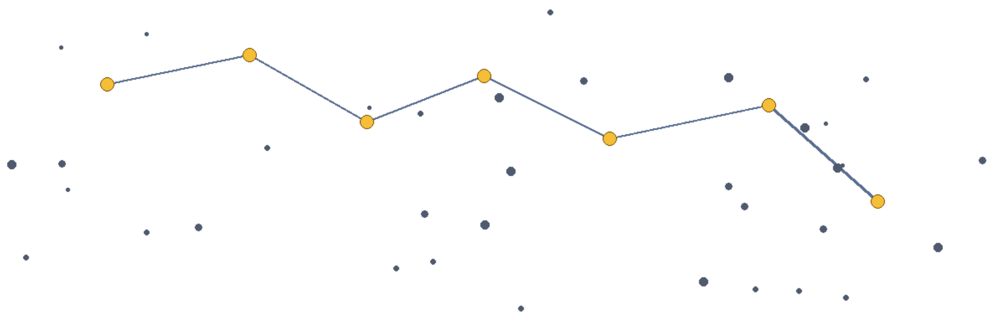


Figure 1.1 A simple star pattern. The lines are imagined guides; the stars may be at very different distances.

Example

A child who records the Moon at the same time for several evenings is doing a simple astronomical investigation.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Make a “sky scientist” notebook. Write the date, time, direction and one thing you notice in the sky.

Think and discuss

Why is writing the time and direction important when two people compare sky observations?

What you have learnt

- Astronomy is a science. It uses evidence, careful measurements and repeated observations instead of guesses.
- Many astronomers use telescopes, cameras, computers and spacecraft, but useful astronomy can also begin with unaided-eye observations.
- Astronomy explains space by using observations and evidence.

EXERCISES · WHAT IS ASTRONOMY?

Exercises

A. Choose the correct answer

1. Which statement about what is astronomy? is correct?

A. Astronomy is only the naming of stars.

B. Astronomy is the scientific study of space and objects beyond Earth.

C. Astronomy is a type of storytelling without evidence.

D. Astronomy studies only weather on Earth.

2. What does the word “astronomy” mean in this lesson?

A. information gathered by carefully watching or measuring

B. facts or measurements used to support an explanation

C. the scientific study of space and objects beyond Earth

D. a scientist who studies the universe

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.

B. A child who records the Moon at the same time for several evenings is doing a simple astronomical investigation.

C. A glass is filled with water.

D. A pencil is sharpened.

4. Which word means “the scientific study of space and objects beyond Earth”?

A. astronomer

B. observation

C. astronomy

D. evidence

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.

B. Change several conditions at the same time and compare nothing.

C. Ignore the date, time, direction and evidence.

D. Make a “sky scientist” notebook. Write the date, time, direction and one thing you notice in the sky.

6. Which sentence gives the best summary of this lesson?

A. Astronomy can be understood without observation, evidence or comparison.

B. Every object or event connected with astronomy behaves in exactly the same way.

C. The main idea of astronomy is unrelated to science or everyday experience.

D. Astronomy explains space by using observations and evidence.

B. Answer in one or two sentences

1. In one sentence, explain what is astronomy?
2. Write two important facts about astronomy
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is writing the time and direction important when two people compare sky observations?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Astronomy is one of the oldest sciences because people have watched the sky for thousands of years. Some telescopes study invisible kinds of light, such as radio waves, ultraviolet light and X-rays.

UNIT 1 · CHAPTER 2

The Sky Around Us

Learning outcomes

- Explain the sky around us using simple scientific language.
- Use the words atmosphere and horizon correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

The sky is the view we get when we look outward from Earth. It appears like a dome, but there is no solid roof above us. We see the atmosphere, clouds and objects in space along different lines of sight.

Understanding the idea

During daytime, sunlight scattered by the atmosphere makes most of the sky look blue.

At night, the side of Earth where we live faces away from the Sun, so the sky becomes dark enough for many stars to be seen.

Objects appear to rise in the east and set in the west mainly because Earth rotates from west to east.

The visible sky changes with time, season, location, weather and light pollution.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

The vocabulary of this chapter - atmosphere, horizon, scatter - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

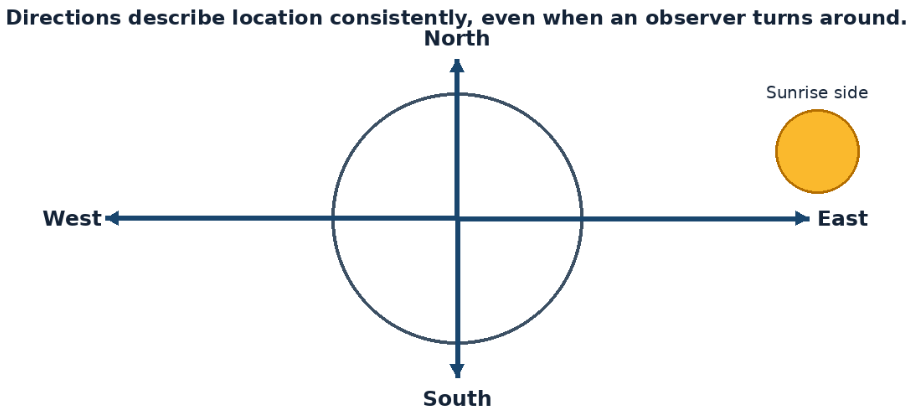


Figure 1.2 The four cardinal directions provide a fixed way to describe where an object appears.

Example

A bright streetlight can make faint stars difficult to see even when the sky is cloudless.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Stand in a safe open place and sketch the horizon. Mark buildings, trees and the directions where the Sun rises and sets.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Why might a village sky show more stars than the sky above a brightly lit city?

What you have learnt

- During daytime, sunlight scattered by the atmosphere makes most of the sky look blue.
- At night, the side of Earth where we live faces away from the Sun, so the sky becomes dark enough for many stars to be seen.
- The sky is our view through the atmosphere and into space; it is not a solid dome.

EXERCISES · THE SKY AROUND US

Exercises

A. Choose the correct answer

1. Which statement about the sky around us is correct?

A. The sky is a solid blue roof.
B. Stars disappear forever during daytime.
C. The visible sky changes with time, place, weather and artificial light.
D. Every person on Earth sees exactly the same sky at the same time.

2. What does the word "atmosphere" mean in this lesson?

A. extra artificial light that brightens the night sky
B. the line where the land or sea seems to meet the sky
C. the layer of gases surrounding Earth
D. to send light in many directions

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A pencil is sharpened.
C. A notebook is closed on a table.
D. A bright streetlight can make faint stars difficult to see even when the sky is cloudless.

4. Which word means "the layer of gases surrounding Earth"?

A. horizon
B. scatter
C. light pollution
D. atmosphere

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Stand in a safe open place and sketch the horizon. Mark buildings, trees and the directions where the Sun rises and sets.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. The sky around us can be understood without observation, evidence or comparison.
B. The sky is our view through the atmosphere and into space; it is not a solid dome.
C. Every object or event connected with the sky around us behaves in exactly the same way.
D. The main idea of the sky around us is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain the sky around us.
2. Write two important facts about the sky around us.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why might a village sky show more stars than the sky above a brightly lit city?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The blue sky is not a blue wall. It is sunlight scattered in Earth's air. Astronauts in orbit see a black sky because there is very little air around them to scatter sunlight.

UNIT 1 · CHAPTER 3

Day and Night

Learning outcomes

- Explain day and night using simple scientific language.
- Use the words rotation and axis correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Day and night happen because Earth rotates on its axis. The half facing the Sun has daylight, while the half turned away from the Sun has night.

Understanding the idea

Earth's axis is an imaginary line through the North and South Poles.

Earth completes one rotation in about 24 hours, giving us a day-night cycle.

The Sun does not travel around Earth each day. Earth's turning makes the Sun appear to move across the sky.

Different places experience sunrise and sunset at different times because Earth is round and rotating.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

The important words in this chapter include rotation, axis, daylight. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

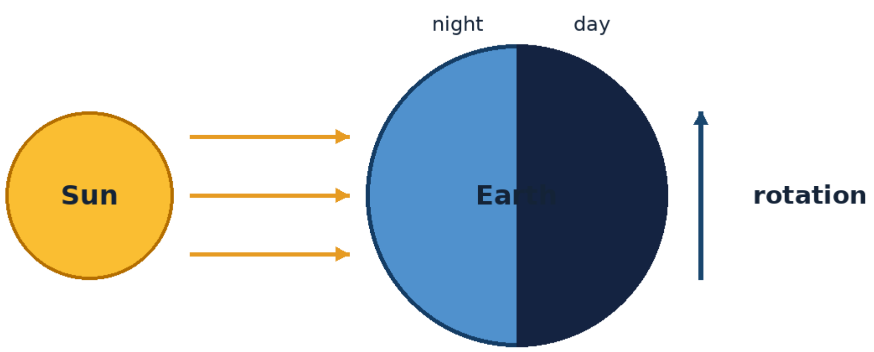


Figure 1.3 Earth's rotation carries places into sunlight and then into darkness.

Example

When it is afternoon in India, it may be morning, night or another time in countries farther east or west.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Darken a room, shine a torch on a ball and slowly turn the ball. Place a sticker "home" on it and watch day change to night.

Think and discuss

If Earth stopped rotating but still went around the Sun, would our usual 24-hour day-night cycle continue?

What you have learnt

- Earth's axis is an imaginary line through the North and South Poles.
- Earth completes one rotation in about 24 hours, giving us a day-night cycle.
- Earth's rotation causes the daily pattern of day and night.

EXERCISES · DAY AND NIGHT

Exercises

A. Choose the correct answer

1. Which statement about day and night is correct?

A. Earth becomes dark because the Sun switches off.
B. Day and night are caused by the Moon blocking the Sun every evening.
C. Day and night happen because clouds cover half of Earth.
D. Day and night are caused by Earth rotating on its axis.

2. What does the word "rotation" mean in this lesson?

A. the part of the day when a place receives sunlight
B. an imaginary line about which an object rotates
C. the spinning of an object around its axis
D. a pattern that repeats

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A pencil is sharpened.
C. A glass is filled with water.
D. When it is afternoon in India, it may be morning, night or another time in countries farther east or west.

4. Which word means "the spinning of an object around its axis"?

A. rotation
B. axis
C. daylight
D. cycle

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Darken a room, shine a torch on a ball and slowly turn the ball. Place a sticker "home" on it and watch day change to night.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Day and night can be understood without observation, evidence or comparison.

B. Every object or event connected with day and night behaves in exactly the same way.
C. The main idea of day and night is unrelated to science or everyday experience.
D. Earth's rotation causes the daily pattern of day and night.

B. Answer in one or two sentences

1. In one sentence, explain day and night.
2. Write two important facts about day and night.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. If Earth stopped rotating but still went around the Sun, would our usual 24-hour day-night cycle continue?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: At every moment, roughly half of Earth is lit by the Sun and roughly half is in darkness. One full rotation of Earth is also called a solar day when measured from one noon to the next.

UNIT 1 · CHAPTER 4

Sunrise and Sunset

Learning outcomes

- Explain sunrise and sunset using simple scientific language.
- Use the words sunrise and sunset correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Sunrise is when the Sun first appears above the eastern horizon. Sunset is when it disappears below the western horizon. These are apparent motions caused mainly by Earth's rotation.

Understanding the idea

Because Earth rotates from west to east, the Sun appears to move from east to west across the sky.

The exact sunrise and sunset positions shift through the year because Earth is tilted and revolves around the Sun.

Sunrise and sunset times differ from place to place and from season to season.

It is unsafe to stare at the Sun, even at sunrise or sunset, because intense sunlight can damage the eyes.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

Notice how the terms sunrise, sunset, apparent motion appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

Directions describe location consistently, even when an observer turns around.

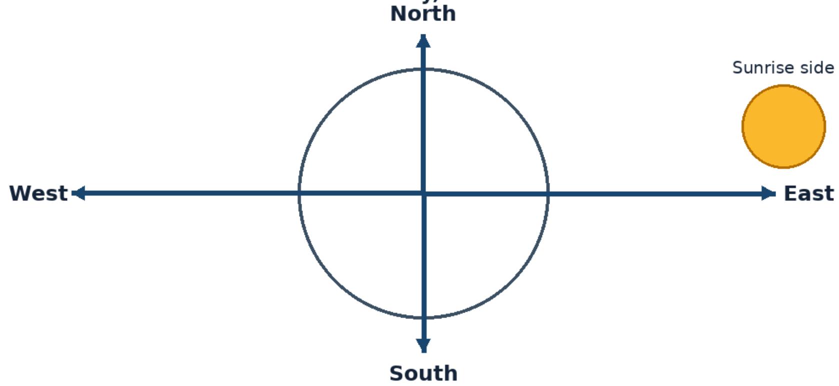


Figure 1.4 The four cardinal directions provide a fixed way to describe where an object appears.

Example

A school shadow may point in different directions in the morning and afternoon because the Sun's apparent position changes.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

For one week, note the time of sunset from a safe place. Compare whether it becomes earlier or later.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Why should sunrise observations use the horizon and a clock rather than looking directly at the Sun?

What you have learnt

- Because Earth rotates from west to east, the Sun appears to move from east to west across the sky.
- The exact sunrise and sunset positions shift through the year because Earth is tilted and revolves around the Sun.
- The Sun's daily east-to-west motion is apparent motion caused by Earth's rotation.

EXERCISES · SUNRISE AND SUNSET

Exercises

A. Choose the correct answer

1. Which statement about sunrise and sunset is correct?

A. The Sun appears to rise in the east and set in the west because Earth rotates.
B. Sunrise happens in the west everywhere.
C. Sunset occurs because the Sun becomes smaller.
D. The Sun circles Earth once every day.

2. What does the word "sunrise" mean in this lesson?

A. the appearance of the Sun above the eastern horizon
B. the disappearance of the Sun below the western horizon
C. motion that seems to happen because the observer or Earth is moving
D. the partly lit time before sunrise or after sunset

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A notebook is closed on a table.
C. A school shadow may point in different directions in the morning and afternoon because the Sun's apparent position changes.
D. A pencil is sharpened.

4. Which word means "the appearance of the Sun above the eastern horizon"?

A. sunset
B. sunrise
C. apparent motion
D. twilight

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. For one week, note the time of sunset from a safe place. Compare whether it becomes earlier or later.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Sunrise and sunset can be understood without observation, evidence or comparison.
B. The Sun's daily east-to-west motion is apparent motion caused by Earth's rotation.
C. Every object or event connected with sunrise and sunset behaves in exactly the same way.
D. The main idea of sunrise and sunset is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain sunrise and sunset.
2. Write two important facts about sunrise and sunset.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why should sunrise observations use the horizon and a clock rather than looking directly at the Sun?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The Sun can appear flattened near the horizon because Earth's atmosphere bends its light. After the Sun sets, the sky may remain bright for a while. This period is called twilight.

UNIT 1 · CHAPTER 5

Directions

Learning outcomes

- Explain directions using simple scientific language.
- Use the words cardinal directions and compass correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Directions help us describe where an object is located. The four main, or cardinal, directions are north, south, east and west.

Understanding the idea

A compass needle points approximately north-south because it responds to Earth's magnetic field.

East is the general direction of sunrise, but the exact point of sunrise changes through the year.

Maps usually place north at the top, south at the bottom, east to the right and west to the left.

Astronomers use direction and height above the horizon to tell others where to look.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

Do not learn cardinal directions, compass, magnetic field separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

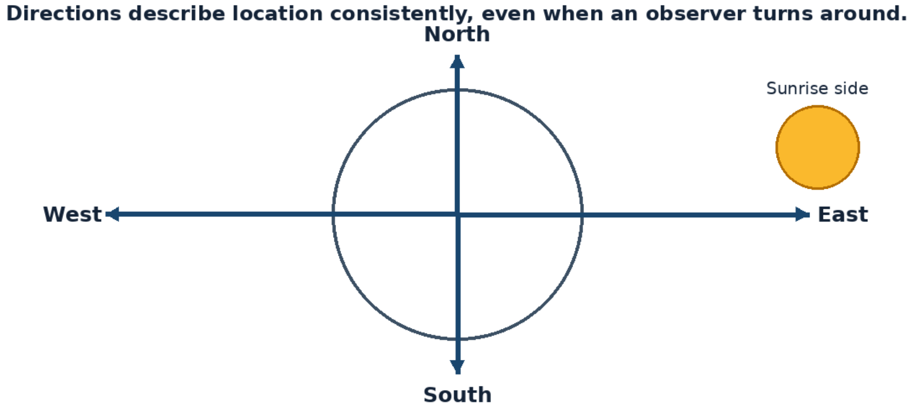


Figure 1.5 The four cardinal directions provide a fixed way to describe where an object appears.

Example

"Look low in the western sky after sunset" is more useful than saying "look over there."

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Draw a compass rose on paper. Use a real compass with an adult and rotate the paper until north matches north.

Think and discuss

Why is "the Moon is on my left" not a reliable direction for another observer?

What you have learnt

- A compass needle points approximately north-south because it responds to Earth's magnetic field.
- East is the general direction of sunrise, but the exact point of sunrise changes through the year.
- Use fixed directions such as north, south, east and west when recording sky observations.

EXERCISES · DIRECTIONS

Exercises

A. Choose the correct answer

1. Which statement about directions is correct?

A. North, south, east and west are the four cardinal directions.
B. A compass always points east-west.
C. Directions change whenever a person turns around.
D. Up, down, near and far are the four cardinal directions.

2. What does the word "cardinal directions" mean in this lesson?

A. an instrument used to find direction
B. north, south, east and west
C. an invisible region where magnetic forces act
D. the direction halfway between north and east

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. "Look low in the western sky after sunset" is more useful than saying "look over there."
C. A pencil is sharpened.
D. A notebook is closed on a table.

4. Which word means "north, south, east and west"?

A. compass

B. magnetic field
C. cardinal directions
D. north-east

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Draw a compass rose on paper. Use a real compass with an adult and rotate the paper until north matches north.

6. Which sentence gives the best summary of this lesson?
A. Directions can be understood without observation, evidence or comparison.
B. Every object or event connected with directions behaves in exactly the same way.
C. The main idea of directions is unrelated to science or everyday experience.
D. Use fixed directions such as north, south, east and west when recording sky observations.

B. Answer in one or two sentences

1. In one sentence, explain directions.
2. Write two important facts about directions.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is “the Moon is on my left” not a reliable direction for another observer?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: North-east, south-east, south-west and north-west are called intermediate directions. The Pole Star appears close to north for observers in much of the Northern Hemisphere.

UNIT 1 · CHAPTER 6

Seasons

Learning outcomes

- Explain seasons using simple scientific language.
- Use the words season and tilt correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Seasons are yearly changes in weather and daylight patterns. They are caused mainly by Earth’s tilted axis as Earth revolves around the Sun, not by Earth moving much closer to or farther from the Sun.

Understanding the idea

Earth’s axis is tilted by about 23.5 degrees compared with its path around the Sun.

When a hemisphere tilts toward the Sun, sunlight is more direct and days are generally longer, producing warmer conditions.

When a hemisphere tilts away, sunlight is less direct and days are generally shorter.

The Northern and Southern Hemispheres have opposite seasons at the same time.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

The vocabulary of this chapter - season, tilt, hemisphere - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

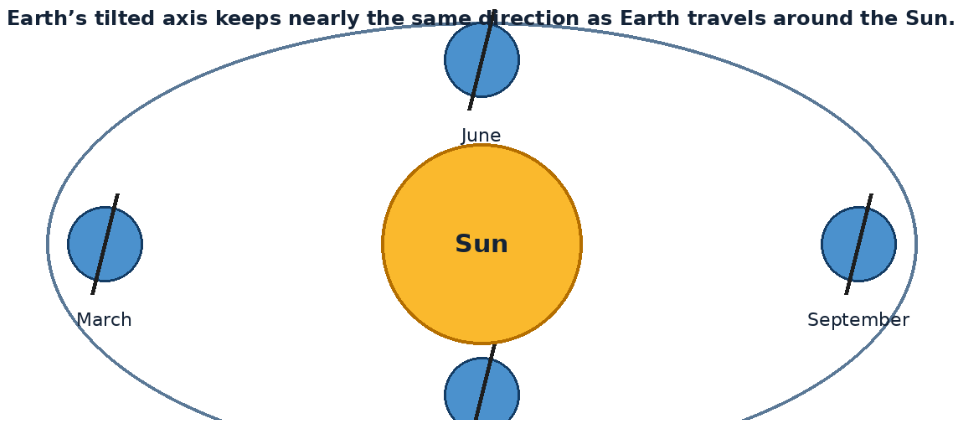


Figure 1.6 Earth travels around the Sun while its tilted axis keeps nearly the same direction in space.

Example

When it is summer in India, countries far south such as Australia may be in winter.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Shine a torch straight down on paper, then at a slant. Compare how concentrated the light looks.

Think and discuss

Why does the slanted light spread over a larger area and warm it less strongly?

What you have learnt

- Earth's axis is tilted by about 23.5 degrees compared with its path around the Sun.
- When a hemisphere tilts toward the Sun, sunlight is more direct and days are generally longer, producing warmer conditions.
- Seasons come from Earth's tilt and revolution around the Sun.

EXERCISES · SEASONS

Exercises

A. Choose the correct answer

1. Which statement about seasons is correct?

A. Seasons are caused by the Moon changing shape.
B. Every place on Earth has the same season at the same time.
C. Seasons happen because Earth is much nearer the Sun in summer.
D. Earth's tilt and revolution cause the seasons.

2. What does the word "season" mean in this lesson?

A. a slant from an upright position
B. sunlight arriving more nearly straight onto a surface
C. one half of Earth
D. a part of the year with a typical pattern of weather and daylight

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A pencil is sharpened.
C. When it is summer in India, countries far south such as Australia may be in winter.
D. A glass is filled with water.

4. Which word means "a part of the year with a typical pattern of weather and daylight"?

A. tilt

B. hemisphere
C. direct sunlight
D. season

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Shine a torch straight down on paper, then at a slant. Compare how concentrated the light looks.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?
A. Seasons can be understood without observation, evidence or comparison.
B. Seasons come from Earth's tilt and revolution around the Sun.
C. Every object or event connected with seasons behaves in exactly the same way.
D. The main idea of seasons is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain seasons.
2. Write two important facts about seasons.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why does the slanted light spread over a larger area and warm it less strongly?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Earth is actually slightly closer to the Sun during Northern Hemisphere winter than during its summer. Places near the equator usually have smaller changes in daylight length than places near the poles.

UNIT 1 · CHAPTER 7

Observing the Sky

Learning outcomes

- Explain observing the sky using simple scientific language.
- Use the words dark adaptation and cloud cover correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Good sky observation means looking safely, patiently and systematically. Scientists record when, where and under what conditions an observation was made.

Understanding the idea

Choose a safe place with a clear view and adult supervision, especially after dark.

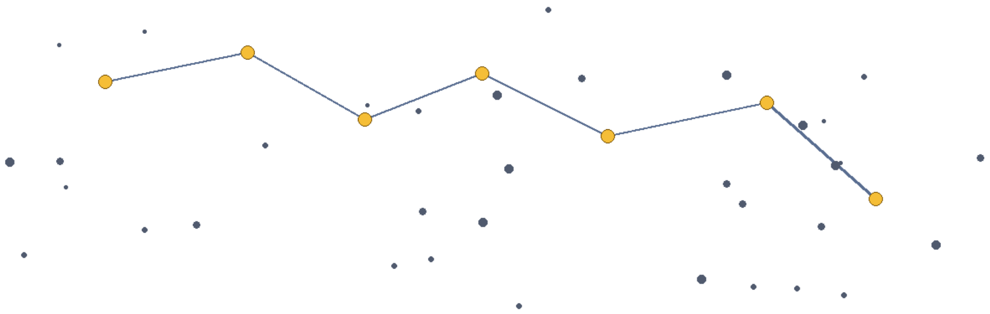
Record the date, time, direction, weather, cloud cover and brightness of nearby lights.

Use red light at night when possible because it disturbs dark-adapted vision less than bright white light.

Never look directly at the Sun or through binoculars or a telescope without a certified solar filter fitted correctly by an expert.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

The important words in this chapter include dark adaptation, cloud cover, record. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.



Constellation lines are imagined patterns; the stars can be at very different distances.

Figure 1.7 A simple star pattern. The lines are imagined guides; the stars may be at very different distances.

Example

Two sketches of the Moon are more useful when each includes the date, time and direction.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Observe the same part of the sky for ten minutes. List changes such as moving clouds, aircraft, satellites or stars becoming visible.

Think and discuss

How can another student repeat your observation if you forgot to write the direction?

What you have learnt

- Choose a safe place with a clear view and adult supervision, especially after dark.
- Record the date, time, direction, weather, cloud cover and brightness of nearby lights.
- Safe, useful observations include time, place, direction and conditions.

EXERCISES · OBSERVING THE SKY

Exercises

A. Choose the correct answer

1. Which statement about observing the sky is correct?

A. A useful observation needs only a colourful drawing.
B. Direction does not matter when describing the sky.
C. It is safe to view the Sun through ordinary sunglasses.
D. A scientific sky record includes date, time, direction and observing conditions.

2. What does the word “dark adaptation” mean in this lesson?

A. the amount of sky covered by clouds
B. the eyes becoming more sensitive in low light
C. a special certified filter made for safe Sun observation
D. to write, draw or save information

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A notebook is closed on a table.
C. Two sketches of the Moon are more useful when each includes the date, time and direction.
D. A pencil is sharpened.

4. Which word means “the eyes becoming more sensitive in low light”?

A. dark adaptation

B. cloud cover
C. record
D. solar filter

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Observe the same part of the sky for ten minutes. List changes such as moving clouds, aircraft, satellites or stars becoming visible.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Observing the sky can be understood without observation, evidence or comparison.
B. Every object or event connected with observing the sky behaves in exactly the same way.
C. The main idea of observing the sky is unrelated to science or everyday experience.
D. Safe, useful observations include time, place, direction and conditions.

B. Answer in one or two sentences

1. In one sentence, explain observing the sky.
2. Write two important facts about observing the sky.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. How can another student repeat your observation if you forgot to write the direction?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Your eyes can take about 20 to 30 minutes to become well adapted to darkness. A simple cardboard tube can help block side lights and make it easier to focus on one small patch of sky.

UNIT 1 · CHAPTER 8

Stars Visible at Night

Learning outcomes

- Explain stars visible at night using simple scientific language.
- Use the words star and twinkle correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Stars are enormous hot balls of gas that produce their own light. At night, many stars become visible because the sky is no longer brightened by direct sunlight.

Understanding the idea

Stars are present in the daytime sky too, but scattered sunlight usually hides them from our eyes.

Some stars look brighter because they are nearer, larger, hotter or naturally more luminous.

Stars seem to twinkle because their light passes through moving layers of Earth’s atmosphere.

The stars visible at a given time depend on the observer’s location, season and time of night.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

Notice how the terms star, twinkle, luminous appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

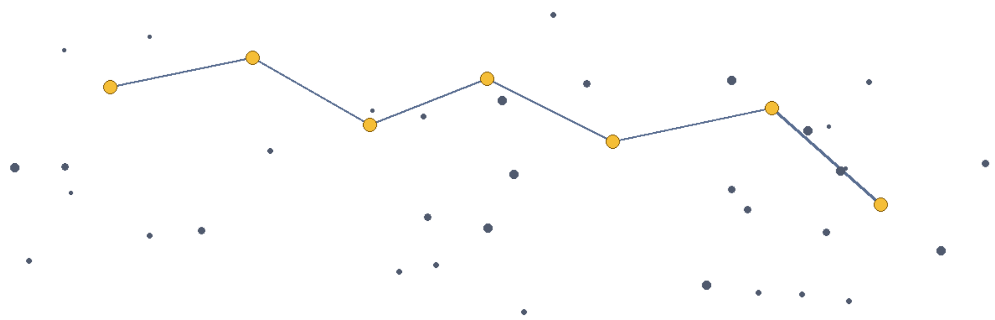


Figure 1.8 A simple star pattern. The lines are imagined guides; the stars may be at very different distances.

Example

A bright star near the horizon may twinkle more strongly because its light travels through more atmosphere.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

From a safe place, count stars inside a hand-sized square frame held at arm's length. Repeat in a darker place and compare.

Think and discuss

Does seeing more stars always mean that more stars are actually above that place? Explain.

What you have learnt

- Stars are present in the daytime sky too, but scattered sunlight usually hides them from our eyes.
- Some stars look brighter because they are nearer, larger, hotter or naturally more luminous.
- Stars produce light; their visibility depends on darkness, atmosphere and light pollution.

EXERCISES · STARS VISIBLE AT NIGHT

Exercises

A. Choose the correct answer

1. Which statement about stars visible at night is correct?

A. All stars are equally bright and equally distant.
B. Stars are holes in a dark roof.
C. Stars are easier to see at night because the sky is darker.
D. Stars are switched on only at night.

2. What does the word "star" mean in this lesson?

A. rapid changes in the apparent brightness or position of a star
B. a hot glowing ball of gas that produces energy and light
C. giving out light
D. how easily something can be seen

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A glass is filled with water.
C. A bright star near the horizon may twinkle more strongly because its light travels through more atmosphere.
D. A pencil is sharpened.

4. Which word means "a hot glowing ball of gas that produces energy and light"?

A. twinkle
B. star
C. luminous
D. visibility

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. From a safe place, count stars inside a hand-sized square frame held at arm's length. Repeat in a darker place and compare.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Stars visible at night can be understood without observation, evidence or comparison.
B. Stars produce light; their visibility depends on darkness, atmosphere and light pollution.
C. Every object or event connected with stars visible at night behaves in exactly the same way.
D. The main idea of stars visible at night is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain stars visible at night.
2. Write two important facts about stars visible at night.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Does seeing more stars always mean that more stars are actually above that place? Explain.
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The Sun is a star, but it looks much larger and brighter because it is far closer than all other stars. Most star names used in modern astronomy come from many cultures, including Arabic, Greek and Latin traditions.

UNIT 1 · CHAPTER 9

Introduction to Constellations

Learning outcomes

- Explain introduction to constellations using simple scientific language.
- Use the words constellation and asterism correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A constellation is an officially defined region of the sky, often recognised by a traditional star pattern. The stars in a pattern may look close together but can be at very different distances from Earth.

Understanding the idea

People in many cultures imagined shapes, animals and heroes by joining bright stars with imaginary lines.

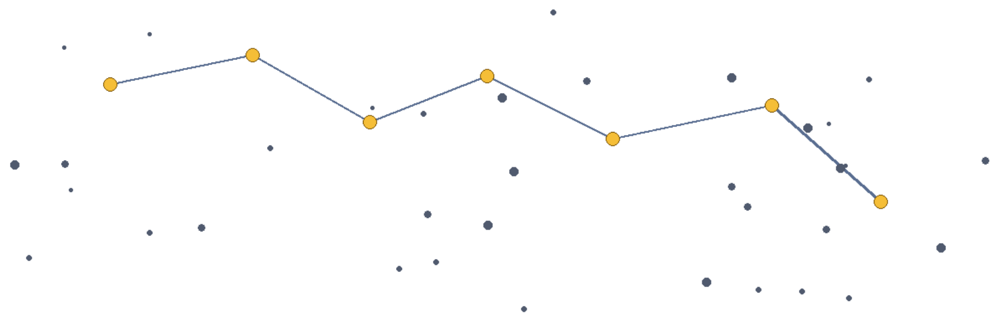
Modern astronomers divide the whole sky into 88 official constellations.

Constellations help people organise and describe areas of the sky, much like countries divide a map.

A familiar pattern called an asterism can be part of a larger constellation; the Big Dipper is an example.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

Do not learn constellation, asterism, pattern separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.



Constellation lines are imagined patterns; the stars can be at very different distances.

Figure 1.9 A simple star pattern. The lines are imagined guides; the stars may be at very different distances.

Example

Orion is easy to recognise because three fairly bright stars form a short line called Orion's Belt.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Place seven dots on paper. Ask family members to invent different patterns and names using the same dots.

Think and discuss

Why do imaginary lines between stars not prove that the stars are physically joined?

What you have learnt

- People in many cultures imagined shapes, animals and heroes by joining bright stars with imaginary lines.
- Modern astronomers divide the whole sky into 88 official constellations.
- Constellation patterns are useful sky maps, not real dot-to-dot objects in space.

EXERCISES · INTRODUCTION TO CONSTELLATIONS

Exercises

A. Choose the correct answer

1. Which statement about introduction to constellations is correct?

A. Every constellation is visible all night from every place.
B. Constellations are groups of stars tied together in space.
C. Constellations organise the sky into named regions and recognised patterns.
D. There are only twelve constellations in the whole sky.

2. What does the word "constellation" mean in this lesson?

A. a well-known constellation visible from many parts of Earth
B. an arrangement that can be recognised
C. an officially named region of the sky
D. a familiar star pattern that is not necessarily a whole constellation

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A notebook is closed on a table.
C. Orion is easy to recognise because three fairly bright stars form a short line called Orion's Belt.

D. A pencil is sharpened.

4. Which word means “an officially named region of the sky”?

A. asterism
B. pattern
C. constellation
D. Orion

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Place seven dots on paper. Ask family members to invent different patterns and names using the same dots.

6. Which sentence gives the best summary of this lesson?

A. Introduction to constellations can be understood without observation, evidence or comparison.
B. Every object or event connected with introduction to constellations behaves in exactly the same way.
C. The main idea of introduction to constellations is unrelated to science or everyday experience.
D. Constellation patterns are useful sky maps, not real dot-to-dot objects in space.

B. Answer in one or two sentences

1. In one sentence, explain introduction to constellations.
2. Write two important facts about introduction to constellations.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why do imaginary lines between stars not prove that the stars are physically joined?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The same constellation can appear upside down or at a different angle when viewed from another place or season. Indian traditions include many nakshatras, which are sky divisions used historically in calendars and astronomy.

UNIT 1 · CHAPTER 10

Astronomy in Everyday Life

Learning outcomes

- Explain astronomy in everyday life using simple scientific language.
- Use the words calendar and navigation correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Astronomy affects everyday life through timekeeping, calendars, navigation, communication, weather observation and technologies first developed for space research.

Understanding the idea

A day is linked to Earth’s rotation, a month to the Moon’s cycle and a year to Earth’s revolution around the Sun.

Early travellers used the Sun and stars to help find direction before modern navigation systems.

Satellites support television, phone links, weather forecasts, maps, disaster monitoring and navigation.

Studying space has encouraged advances in cameras, sensors, materials, computing and medical monitoring.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

The vocabulary of this chapter - calendar, navigation, satellite image - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

Example

A farmer may use a weather forecast made from satellite images to plan irrigation or protect a crop.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

List five objects or services used in one day. Circle any that depend directly or indirectly on satellites or astronomy.

Think and discuss

Which would be harder without satellites: finding a route, predicting a cyclone or sending television signals? Explain.

What you have learnt

- A day is linked to Earth's rotation, a month to the Moon's cycle and a year to Earth's revolution around the Sun.
- Early travellers used the Sun and stars to help find direction before modern navigation systems.
- Astronomy and space technology support many ordinary services on Earth.

EXERCISES · ASTRONOMY IN EVERYDAY LIFE

Exercises

A. Choose the correct answer

1. Which statement about astronomy in everyday life is correct?

A. Calendars are based only on random numbers.
B. Satellites are used only to take pictures of planets.
C. Astronomy supports calendars, navigation and many satellite services.
D. Astronomy has no connection with life on Earth.

2. What does the word "calendar" mean in this lesson?

A. finding position and direction of travel
B. a picture or measurement of Earth made from orbit
C. a system for organising days, months and years
D. tools and methods created using scientific knowledge

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A pencil is sharpened.
C. A glass is filled with water.
D. A farmer may use a weather forecast made from satellite images to plan irrigation or protect a crop.

4. Which word means "a system for organising days, months and years"?

A. navigation
B. satellite image
C. technology
D. calendar

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. List five objects or services used in one day. Circle any that depend directly or indirectly on satellites or astronomy.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Astronomy in everyday life can be understood without observation, evidence or comparison.
B. Astronomy and space technology support many ordinary services on Earth.
C. Every object or event connected with astronomy in everyday life behaves in exactly the same way.
D. The main idea of astronomy in everyday life is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain astronomy in everyday life.
2. Write two important facts about astronomy in everyday life.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Which would be harder without satellites: finding a route, predicting a cyclone or sending television signals? Explain.
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The word "month" is related historically to the Moon because many early calendars followed lunar phases. Space-based clocks help navigation systems calculate position with great accuracy.

UNIT 1 · CHAPTER 11

Simple Astronomical Observations

Learning outcomes

- Explain simple astronomical observations using simple scientific language.
- Use the words gnomon and data correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Simple astronomical observations turn curiosity into evidence. By observing the same object repeatedly, young scientists can notice patterns in position, shape, brightness and time.

Understanding the idea

A shadow stick can show the changing apparent position of the Sun without looking at it directly.

Moon sketches made over several weeks reveal a repeating cycle of phases.

A constellation recorded at the same clock time over several months appears in different parts of the sky.

Observations become stronger when measurements are repeated and compared.

When reading this lesson, look for patterns in time, direction and appearance. A single observation is useful, but repeated observations are stronger evidence.

The important words in this chapter include gnomon, data, repeat. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

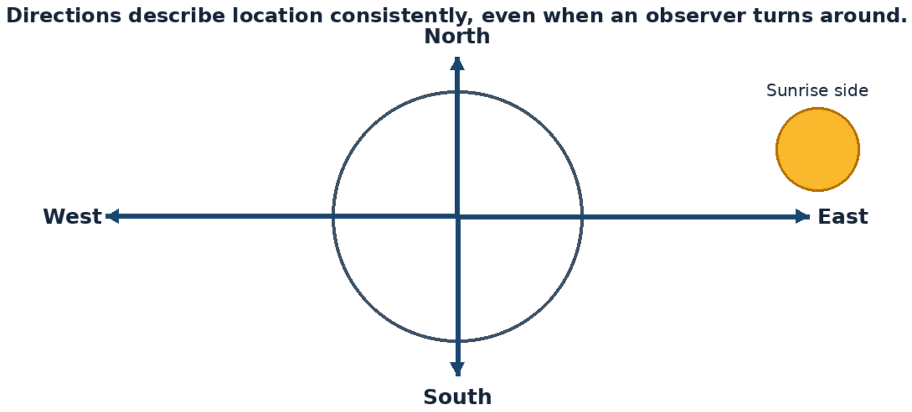


Figure 1.11 The four cardinal directions provide a fixed way to describe where an object appears.

Example

Marking the tip of a stick's shadow every hour creates a curved trail across the ground.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

With adult help, place a short stick upright in sunlight. Mark the shadow tip every hour without looking at the Sun.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

What pattern would you expect if the shadow activity were repeated on a different date?

What you have learnt

- A shadow stick can show the changing apparent position of the Sun without looking at it directly.
- Moon sketches made over several weeks reveal a repeating cycle of phases.
- Repeated, labelled observations reveal patterns that a single observation may miss.

EXERCISES · SIMPLE ASTRONOMICAL OBSERVATIONS

Exercises

A. Choose the correct answer

1. Which statement about simple astronomical observations is correct?

A. One observation is always enough to prove any idea.
B. Repeated and labelled observations help reveal sky patterns.
C. Sky observations do not need dates or times.
D. A shadow can be tracked safely by staring at the Sun.

2. What does the word "gnomon" mean in this lesson?

A. recorded facts or measurements
B. a regular or recognisable arrangement or change
C. the part of a sundial that casts a shadow
D. to do again in the same way

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A pencil is sharpened.
C. Marking the tip of a stick's shadow every hour creates a curved trail across the ground.
D. A glass is filled with water.

4. Which word means "the part of a sundial that casts a shadow"?

A. gnomon
B. data
C. repeat
D. pattern

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. With adult help, place a short stick upright in sunlight. Mark the shadow tip every hour without looking at the Sun.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Simple astronomical observations can be understood without observation, evidence or comparison.
B. Every object or event connected with simple astronomical observations behaves in exactly the same way.
C. The main idea of simple astronomical observations is unrelated to science or everyday experience.
D. Repeated, labelled observations reveal patterns that a single observation may miss.

B. Answer in one or two sentences

1. In one sentence, explain simple astronomical observations.
2. Write two important facts about simple astronomical observations.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. What pattern would you expect if the shadow activity were repeated on a different date?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Ancient observers built calendars by recording repeated sky patterns over many years. A gnomon is the upright part of a sundial that makes the shadow.

UNIT 1 REVIEW

Review · Fundamentals of Astronomy

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of what is astronomy? using one example.
2. Why might a village sky show more stars than the sky above a brightly lit city?
3. Write one correct statement and one common mistake about day and night.
4. Explain the main idea of sunrise and sunset using one example.
5. Why is “the Moon is on my left” not a reliable direction for another observer?
6. Write one correct statement and one common mistake about seasons.
7. Explain the main idea of observing the sky using one example.
8. Does seeing more stars always mean that more stars are actually above that place? Explain.

B. Compare and connect

1. Write one connection between “What is Astronomy?” and “Simple Astronomical Observations”.
2. Write one connection between “The Sky Around Us” and “Astronomy in Everyday Life”.

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 1 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about what is astronomy? is correct?

A. Astronomy is only the naming of stars.
B. Astronomy is the scientific study of space and objects beyond Earth.
C. Astronomy is a type of storytelling without evidence.
D. Astronomy studies only weather on Earth.

2. Which statement about the sky around us is correct?

A. The sky is a solid blue roof.
B. Stars disappear forever during daytime.
C. The visible sky changes with time, place, weather and artificial light.
D. Every person on Earth sees exactly the same sky at the same time.

3. Which statement about day and night is correct?

A. Earth becomes dark because the Sun switches off.
B. Day and night are caused by the Moon blocking the Sun every evening.
C. Day and night happen because clouds cover half of Earth.
D. Day and night are caused by Earth rotating on its axis.

4. Which statement about sunrise and sunset is correct?

A. The Sun appears to rise in the east and set in the west because Earth rotates.
B. Sunrise happens in the west everywhere.
C. Sunset occurs because the Sun becomes smaller.
D. The Sun circles Earth once every day.

5. Which statement about directions is correct?

A. North, south, east and west are the four cardinal directions.
B. A compass always points east-west.
C. Directions change whenever a person turns around.
D. Up, down, near and far are the four cardinal directions.

6. Which statement about seasons is correct?

A. Seasons are caused by the Moon changing shape.
B. Every place on Earth has the same season at the same time.
C. Seasons happen because Earth is much nearer the Sun in summer.
D. Earth's tilt and revolution cause the seasons.

7. Which statement about observing the sky is correct?

A. A useful observation needs only a colourful drawing.
B. Direction does not matter when describing the sky.
C. It is safe to view the Sun through ordinary sunglasses.
D. A scientific sky record includes date, time, direction and observing conditions.

8. Which statement about stars visible at night is correct?

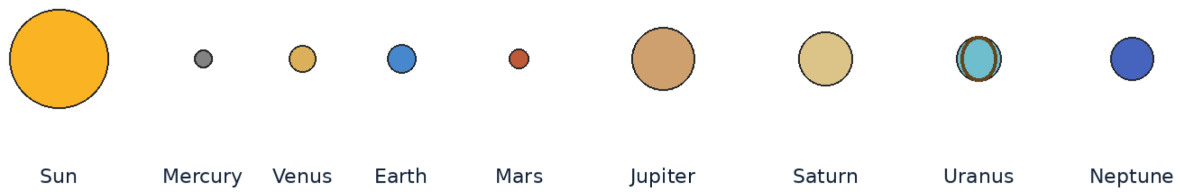
A. All stars are equally bright and equally distant.
B. Stars are holes in a dark roof.
C. Stars are easier to see at night because the sky is darker.
D. Stars are switched on only at night.

UNIT 2

The Solar System

The Solar System is a family of worlds held together by the Sun's gravity. This unit develops a clear picture of the Sun, planets, moons and smaller bodies without turning the subject into a list to memorise.

The planets are shown in order, not to scale.



Chapters in this unit

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 2 · CHAPTER 1

Our Sun

Learning outcomes

- Explain our sun using simple scientific language.
- Use the words solar system and fusion correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

The Sun is the star at the centre of our solar system. It is a huge sphere of very hot gas, mostly hydrogen and helium, and it supplies the light and heat that make life on Earth possible.

Understanding the idea

The Sun's gravity keeps planets, dwarf planets, asteroids and comets moving around it.

Energy is produced in the Sun's core when hydrogen nuclei join to form helium in a process called nuclear fusion.

Sunlight takes about eight minutes and twenty seconds to reach Earth.

The Sun is about 4.5 billion years old and is expected to continue shining for billions of years.

Compare objects by position, size, composition and motion. Remember that drawings of the Solar System are usually not made to one common scale.

The vocabulary of this chapter - solar system, fusion, core - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

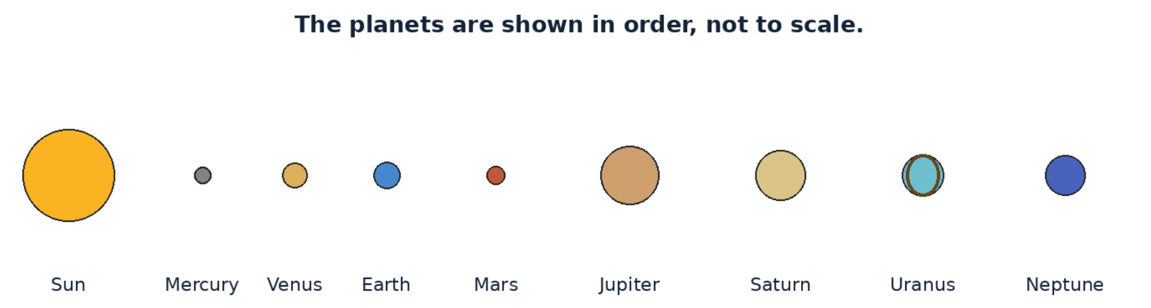


Figure 2.1 The Sun and eight planets shown in order. Sizes and distances are not to scale.

Example

Plants use sunlight to make food, and animals depend directly or indirectly on plants.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Place a ball in sunlight and observe the lit and dark halves. Do not look at the Sun.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

How can the Sun light Earth even though space between them is mostly empty?

What you have learnt

- The Sun’s gravity keeps planets, dwarf planets, asteroids and comets moving around it.
- Energy is produced in the Sun’s core when hydrogen nuclei join to form helium in a process called nuclear fusion.
- The Sun is a star whose light, heat and gravity shape the solar system.

Exercises

A. Choose the correct answer

1. Which statement about our sun is correct?
A. The Sun is the star at the centre of our solar system.
B. The Sun is made mainly of rock and ice.
C. The Sun shines by reflecting light from the Moon.
D. The Sun is a planet orbiting Earth.

2. What does the word “solar system” mean in this lesson?
A. the joining of small atomic nuclei that releases energy
B. the attraction between objects with mass
C. the Sun and all objects held in orbit around it
D. the central region of an object

3. Which observation or example best supports the lesson idea?
A. A pencil is sharpened.
B. A glass is filled with water.
C. Plants use sunlight to make food, and animals depend directly or indirectly on plants.
D. A notebook is closed on a table.

4. Which word means “the Sun and all objects held in orbit around it”?
A. fusion
B. core
C. gravity
D. solar system

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Guess once and do not record anything.
B. Place a ball in sunlight and observe the lit and dark halves. Do not look at the Sun.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?
A. The Sun is a star whose light, heat and gravity shape the solar system.
B. Our sun can be understood without observation, evidence or comparison.
C. Every object or event connected with our sun behaves in exactly the same way.
D. The main idea of our sun is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain our sun.
2. Write two important facts about our sun.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. How can the Sun light Earth even though space between them is mostly empty?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: More than one million Earths could fit inside the Sun by volume, although they could not be packed like solid balls without gaps. The Sun is classified as a yellow dwarf star, but its visible light is close to white when seen from space.

UNIT 2 · CHAPTER 2

The Eight Planets

Learning outcomes

- Explain the eight planets using simple scientific language.
- Use the words planet and rocky planet correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A planet is a large, nearly round body that orbits the Sun and has cleared most other objects from its orbital neighbourhood. Our solar system has eight planets.

Understanding the idea

Mercury, Venus, Earth and Mars are rocky inner planets with solid surfaces.

Jupiter and Saturn are gas giants, while Uranus and Neptune are often called ice giants.

Planets do not make visible light of their own; they are seen because they reflect sunlight.

Each planet has a different size, temperature, atmosphere, day length and number of moons.

Compare objects by position, size, composition and motion. Remember that drawings of the Solar System are usually not made to one common scale.

The important words in this chapter include planet, rocky planet, gas giant. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

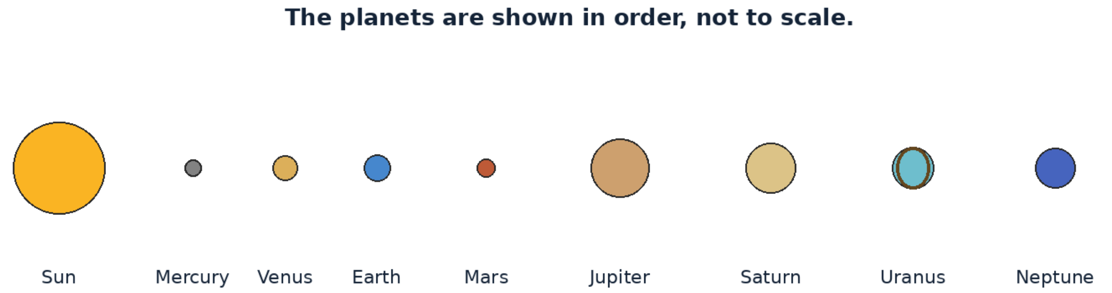


Figure 2.2 The Sun and eight planets shown in order. Sizes and distances are not to scale.

Example

Venus can look like a very bright “star” after sunset, but it is a planet reflecting sunlight.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Make eight planet cards. Sort them by rocky/giant, inner/outer and smaller/larger than Earth.

Think and discuss

Why is it misleading to draw the planets equally spaced in a classroom diagram?

What you have learnt

- Mercury, Venus, Earth and Mars are rocky inner planets with solid surfaces.
- Jupiter and Saturn are gas giants, while Uranus and Neptune are often called ice giants.
- There are eight planets: four rocky inner planets and four giant outer planets.

EXERCISES · THE EIGHT PLANETS

Exercises

A. Choose the correct answer

1. Which statement about the eight planets is correct?

A. Our solar system has eight planets orbiting the Sun.
B. Our solar system has nine official planets including Pluto.
C. Planets produce their own visible light like stars.
D. Every planet has a solid rocky surface.

2. What does the word “planet” mean in this lesson?

A. a very large planet made mostly of gases
B. a giant planet rich in water, ammonia and methane materials
C. a large nearly round body orbiting a star that has cleared its orbital neighbourhood
D. a planet made mainly of rock and metal with a solid surface

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A pencil is sharpened.
C. A glass is filled with water.
D. Venus can look like a very bright “star” after sunset, but it is a planet reflecting sunlight.

4. Which word means “a large nearly round body orbiting a star that has cleared its orbital neighbourhood”?

A. planet
B. rocky planet
C. gas giant
D. ice giant

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Make eight planet cards. Sort them by rocky/giant, inner/outer and smaller/larger than Earth.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. The eight planets can be understood without observation, evidence or comparison.
B. Every object or event connected with the eight planets behaves in exactly the same way.
C. There are eight planets: four rocky inner planets and four giant outer planets.
D. The main idea of the eight planets is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain the eight planets.
2. Write two important facts about the eight planets.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is it misleading to draw the planets equally spaced in a classroom diagram?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Jupiter is the largest planet, while Mercury is the smallest of the eight planets. A day on Venus is longer than its year when the terms are measured by its rotation and revolution periods.

UNIT 2 · CHAPTER 3

Dwarf Planets

Learning outcomes

- Explain dwarf planets using simple scientific language.
- Use the words dwarf planet and orbit correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A dwarf planet orbits the Sun and is nearly round, but it has not cleared other objects from the region around its orbit. It is not a moon.

Understanding the idea

The five officially recognised dwarf planets commonly listed are Ceres, Pluto, Haumea, Makemake and Eris.

Ceres lies in the main asteroid belt between Mars and Jupiter.

Pluto, Haumea, Makemake and Eris are icy worlds in the outer solar system.

Calling Pluto a dwarf planet does not make it unimportant; it gives scientists a clearer category for comparing worlds.

Compare objects by position, size, composition and motion. Remember that drawings of the Solar System are usually not made to one common scale.

Notice how the terms dwarf planet, orbit, neighbourhood appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

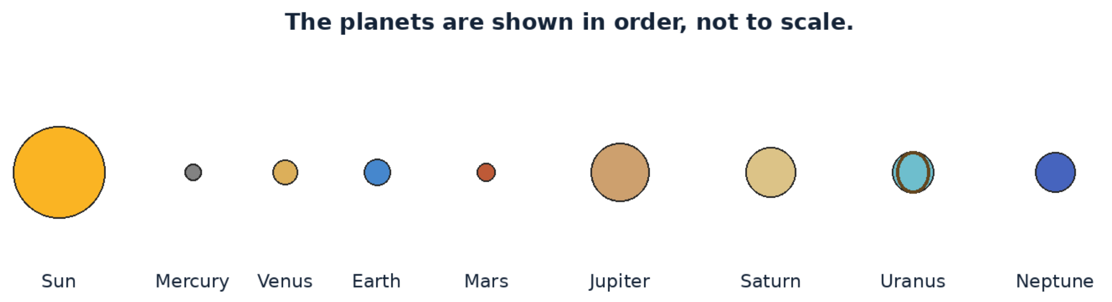


Figure 2.3 The Sun and eight planets shown in order. Sizes and distances are not to scale.

Example

A neighbourhood street can be clear or crowded. A planet has largely cleared its orbital “street,” while a dwarf planet shares its region with many objects.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Use hoops as orbital paths and small counters as nearby objects. Compare a mostly clear hoop with a crowded hoop.

Think and discuss

Which part of the planet definition does Pluto not satisfy?

What you have learnt

- The five officially recognised dwarf planets commonly listed are Ceres, Pluto, Haumea, Makemake and Eris.
- Ceres lies in the main asteroid belt between Mars and Jupiter.
- A dwarf planet is round and orbits the Sun, but it has not cleared its orbital neighbourhood.

EXERCISES · DWARF PLANETS

Exercises

A. Choose the correct answer

1. Which statement about dwarf planets is correct?

A. A dwarf planet is always a moon.
B. A dwarf planet orbits the Sun but has not cleared its orbital neighbourhood.
C. A dwarf planet does not orbit the Sun.
D. A dwarf planet must be smaller than every asteroid.

2. What does the word “dwarf planet” mean in this lesson?

A. the region near an object’s orbit
B. the curved path of one object around another
C. a nearly round body orbiting the Sun that has not cleared its orbital neighbourhood
D. a dwarf planet in the asteroid belt

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A glass is filled with water.
C. A neighbourhood street can be clear or crowded. A planet has largely cleared its orbital “street,” while a dwarf planet shares its region with many objects.
D. A notebook is closed on a table.

4. Which word means “a nearly round body orbiting the Sun that has not cleared its orbital neighbourhood”?

A. orbit
B. dwarf planet
C. neighbourhood
D. Ceres

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Use hoops as orbital paths and small counters as nearby objects. Compare a mostly clear hoop with a crowded hoop.

6. Which sentence gives the best summary of this lesson?

A. A dwarf planet is round and orbits the Sun, but it has not cleared its orbital neighbourhood.
B. Dwarf planets can be understood without observation, evidence or comparison.
C. Every object or event connected with dwarf planets behaves in exactly the same way.
D. The main idea of dwarf planets is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain dwarf planets.

2. Write two important facts about dwarf planets.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Which part of the planet definition does Pluto not satisfy?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Pluto has five known moons, including its largest moon, Charon. The dwarf planet Haumea spins rapidly and has an elongated shape rather than a perfect sphere.

UNIT 2 · CHAPTER 4

Order of the Planets

Learning outcomes

- Explain order of the planets using simple scientific language.
- Use the words inner planets and outer planets correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

The planets are arranged by increasing distance from the Sun: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

Understanding the idea

The first four planets are called inner planets because they are closer to the Sun and inside the asteroid belt.

The last four are outer planets and are much farther apart than most simple diagrams show.

Distance from the Sun affects a planet's orbital period, but temperature also depends on atmosphere and other properties.

A mnemonic can help recall order, but understanding the groups is more useful than memorising only a sentence.

Compare objects by position, size, composition and motion. Remember that drawings of the Solar System are usually not made to one common scale.

Do not learn inner planets, outer planets, orbital period separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

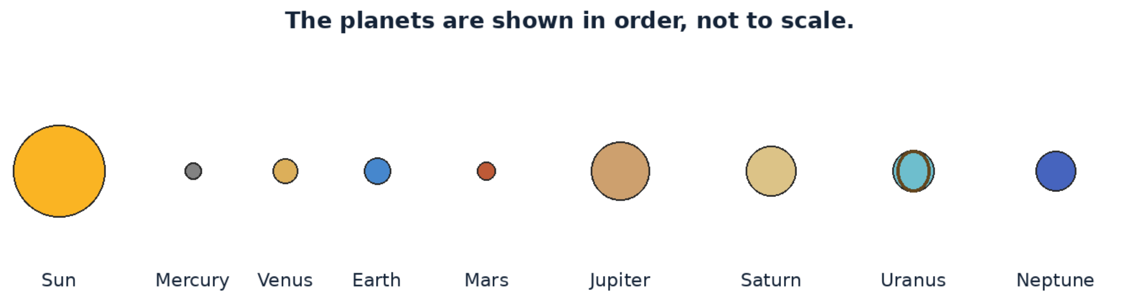


Figure 2.4 The Sun and eight planets shown in order. Sizes and distances are not to scale.

Example

Mars is the fourth planet, so it lies between Earth and Jupiter.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Write the first letters M V E M J S U N. Invent a respectful, memorable sentence using those initials.

Think and discuss

If a spacecraft travels outward from Earth, which planetary orbit does it reach next: Venus, Mars or Saturn?

What you have learnt

- The first four planets are called inner planets because they are closer to the Sun and inside the asteroid belt.
- The last four are outer planets and are much farther apart than most simple diagrams show.
- Planet order from the Sun is Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

EXERCISES · ORDER OF THE PLANETS

Exercises

A. Choose the correct answer

1. Which statement about order of the planets is correct?

A. Neptune is the closest planet to the Sun.
B. Earth is the third planet from the Sun, between Venus and Mars.
C. Mars is farther from the Sun than Jupiter.
D. Earth is between Jupiter and Saturn.

2. What does the word “inner planets” mean in this lesson?

A. Jupiter, Saturn, Uranus and Neptune
B. a region containing many rocky bodies between Mars and Jupiter
C. Mercury, Venus, Earth and Mars
D. the time an object takes to complete one orbit

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A glass is filled with water.
C. A notebook is closed on a table.
D. Mars is the fourth planet, so it lies between Earth and Jupiter.

4. Which word means “Mercury, Venus, Earth and Mars”?

A. outer planets
B. orbital period
C. inner planets
D. asteroid belt

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Write the first letters M V E M J S U N. Invent a respectful, memorable sentence using those initials.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Order of the planets can be understood without observation, evidence or comparison.
B. Every object or event connected with order of the planets behaves in exactly the same way.
C. Planet order from the Sun is Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

D. The main idea of order of the planets is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain order of the planets.
2. Write two important facts about order of the planets.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. If a spacecraft travels outward from Earth, which planetary orbit does it reach next: Venus, Mars or Saturn?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Neptune takes about 165 Earth years to complete one orbit around the Sun. Mercury completes an orbit in only about 88 Earth days.

UNIT 2 · CHAPTER 5

Natural Satellites

Learning outcomes

- Explain natural satellites using simple scientific language.
- Use the words natural satellite and moon correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A natural satellite is an object that orbits a planet, dwarf planet or other larger body without being built by people. Natural satellites are commonly called moons.

Understanding the idea

Earth has one natural satellite, the Moon.

Some planets have no moons, while the giant planets have many known moons.

Moons can be rocky, icy, cratered, volcanic or covered by hidden oceans.

A moon stays in orbit because its forward motion and gravity work together.

Compare objects by position, size, composition and motion. Remember that drawings of the Solar System are usually not made to one common scale.

The vocabulary of this chapter - natural satellite, moon, crater - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

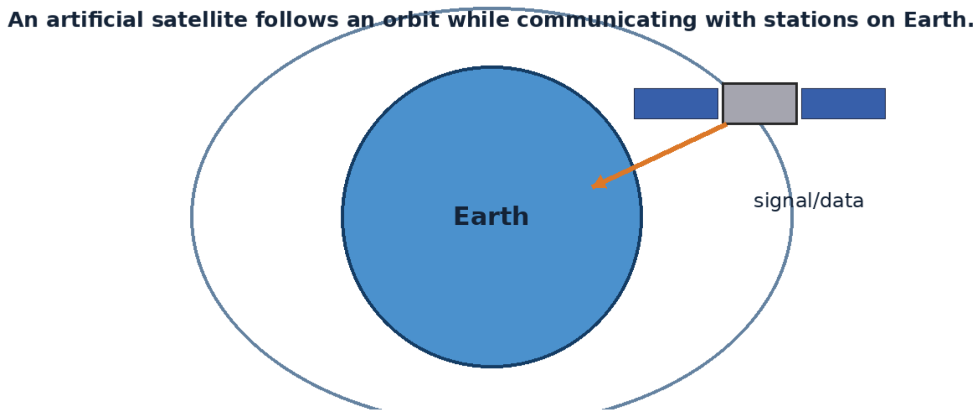


Figure 2.5 An artificial satellite follows an orbit and sends or receives information.

Example

The Moon keeps moving forward while Earth's gravity bends its path, so it continually falls around Earth instead of straight down.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Tie a soft ball securely to a string and, with an adult in a clear area, rotate it slowly to model inward pull and circular motion.

Think and discuss

Why does a moon not need an engine firing all the time to remain in orbit?

What you have learnt

- Earth has one natural satellite, the Moon.
- Some planets have no moons, while the giant planets have many known moons.
- Natural satellites are moons that orbit larger bodies because of gravity and motion.

EXERCISES · NATURAL SATELLITES

Exercises

A. Choose the correct answer

1. Which statement about natural satellites is correct?

A. A natural satellite is always built by engineers.
B. A natural satellite is a moon that orbits a larger body.
C. Moons orbit only the Sun and never planets.
D. Every planet has exactly one moon.

2. What does the word "natural satellite" mean in this lesson?

A. a bowl-shaped hollow often made by an impact
B. a naturally formed object that orbits a larger body
C. a repeated curved path around another object
D. a common name for a natural satellite

3. Which observation or example best supports the lesson idea?

A. The Moon keeps moving forward while Earth's gravity bends its path, so it continually falls around Earth instead of straight down.
B. A notebook is closed on a table.
C. A glass is filled with water.
D. A pencil is sharpened.

4. Which word means "a naturally formed object that orbits a larger body"?

A. moon
B. crater
C. orbit
D. natural satellite

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Tie a soft ball securely to a string and, with an adult in a clear area, rotate it slowly to model inward pull and circular motion.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Natural satellites are moons that orbit larger bodies because of gravity and motion.
B. Natural satellites can be understood without observation, evidence or comparison.
C. Every object or event connected with natural satellites behaves in exactly the same way.
D. The main idea of natural satellites is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain natural satellites.
2. Write two important facts about natural satellites.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why does a moon not need an engine firing all the time to remain in orbit?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Ganymede, a moon of Jupiter, is larger than the planet Mercury. Saturn's moon Titan has a thick atmosphere and lakes of liquid methane and ethane.

UNIT 2 · CHAPTER 6

Asteroids

Learning outcomes

- Explain asteroids using simple scientific language.
- Use the words asteroid and asteroid belt correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Asteroids are small rocky or metallic bodies that orbit the Sun. Most known asteroids are in the main asteroid belt between Mars and Jupiter, but others travel in different regions.

Understanding the idea

Asteroids are leftover building materials from the early solar system.

They range from tiny boulders to bodies hundreds of kilometres wide.

Some asteroids come near Earth's orbit and are carefully tracked by astronomers.

Space missions study asteroids to learn about solar-system history and possible resources.

Compare objects by position, size, composition and motion. Remember that drawings of the Solar System are usually not made to one common scale.

The important words in this chapter include asteroid, asteroid belt, near-Earth object. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

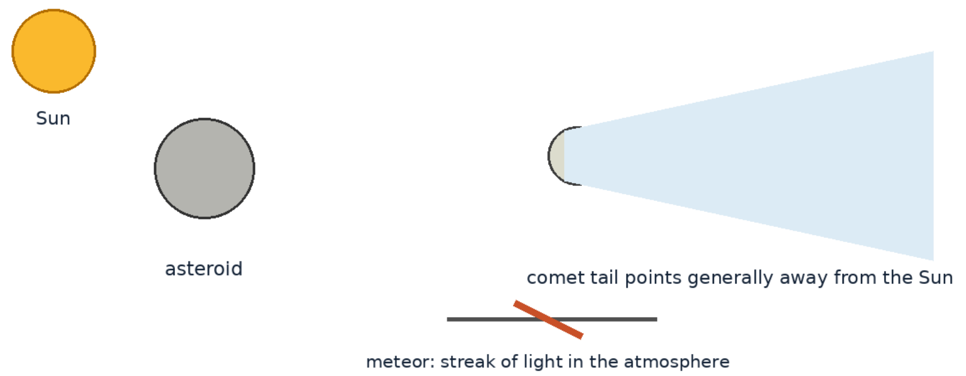


Figure 2.6 Asteroids, comets and meteors are different kinds of small Solar System objects and events.

Example

Ceres was first discovered as an asteroid and is now classified as a dwarf planet because it is nearly round.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Scatter 30 beans across a large floor area. Notice that a “belt” can contain many objects while still leaving large gaps.

Think and discuss

Why is a movie scene showing thousands of asteroids almost touching one another usually unrealistic?

What you have learnt

- Asteroids are leftover building materials from the early solar system.
- They range from tiny boulders to bodies hundreds of kilometres wide.
- Asteroids are rocky or metallic objects orbiting the Sun, mostly in widely spaced groups.

EXERCISES · ASTEROIDS

Exercises

A. Choose the correct answer

1. Which statement about asteroids is correct?

A. Asteroids are stars that have stopped shining.
B. All asteroids lie on Earth’s surface.
C. Most asteroids are rocky or metallic bodies orbiting the Sun.
D. The asteroid belt is a solid ring with no gaps.

2. What does the word “asteroid” mean in this lesson?

A. containing or made from metal
B. a small rocky or metallic body orbiting the Sun
C. the main region of asteroids between Mars and Jupiter
D. an asteroid or comet whose orbit brings it near Earth’s orbit

3. Which observation or example best supports the lesson idea?

A. Ceres was first discovered as an asteroid and is now classified as a dwarf planet because it is nearly round.
B. A notebook is closed on a table.
C. A pencil is sharpened.
D. A glass is filled with water.

4. Which word means “a small rocky or metallic body orbiting the Sun”?

A. asteroid
B. asteroid belt
C. near-Earth object
D. metallic

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Scatter 30 beans across a large floor area. Notice that a “belt” can contain many objects while still leaving large gaps.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Asteroids can be understood without observation, evidence or comparison.
B. Every object or event connected with asteroids behaves in exactly the same way.
C. Asteroids are rocky or metallic objects orbiting the Sun, mostly in widely spaced groups.
D. The main idea of asteroids is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain asteroids.
2. Write two important facts about asteroids.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is a movie scene showing thousands of asteroids almost touching one another usually unrealistic?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Spacecraft can usually pass through the asteroid belt because the objects are separated by enormous distances. Some asteroids have their own tiny moons.

UNIT 2 · CHAPTER 7

Comets

Learning outcomes

- Explain comets using simple scientific language.
- Use the words comet and nucleus correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Comets are small bodies made of ice, dust and rock that orbit the Sun. When a comet approaches the Sun, some ice turns to gas and carries dust outward, forming a glowing coma and one or more tails.

Understanding the idea

A comet’s solid centre is called its nucleus.

The coma is the cloud of gas and dust around the warmed nucleus.

Comet tails point generally away from the Sun because of sunlight and the solar wind.

Many comets spend most of their time far from the Sun and become active only during a close approach.

Compare objects by position, size, composition and motion. Remember that drawings of the Solar System are usually not made to one common scale.

Notice how the terms comet, nucleus, coma appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

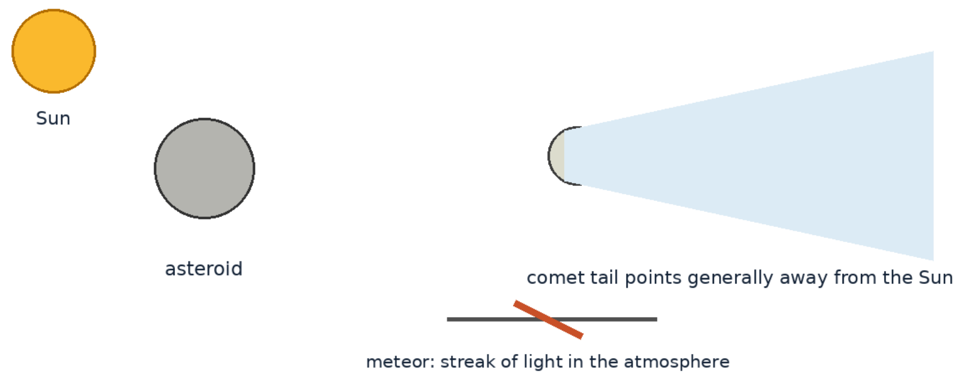


Figure 2.7 Asteroids, comets and meteors are different kinds of small Solar System objects and events.

Example

A comet moving away from the Sun can still have its tail pointing away from the Sun, so the tail may appear in front of its direction of travel.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Place a desk fan as the “solar wind” and hold strips of paper behind a ball. Move the ball around the fan and note that strips point away from it.

Think and discuss

Why does a comet not always show a bright tail when it is far from the Sun?

What you have learnt

- A comet’s solid centre is called its nucleus.
- The coma is the cloud of gas and dust around the warmed nucleus.
- Comet activity increases near the Sun, and its tails point away from the Sun.

EXERCISES · COMETS

Exercises

A. Choose the correct answer

1. Which statement about comets is correct?

A. A comet is a burning star falling through Earth’s atmosphere.
B. Comets are made only of liquid water.
C. A comet’s tail always points behind its motion.
D. A comet is an icy body that can form a coma and tails near the Sun.

2. What does the word “comet” mean in this lesson?

A. the solid central part of a comet
B. an icy, dusty body orbiting the Sun
C. the cloud around an active comet nucleus
D. a stream of charged particles flowing from the Sun

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A glass is filled with water.
C. A pencil is sharpened.
D. A comet moving away from the Sun can still have its tail pointing away from the Sun, so the tail may appear in front of its direction of travel.

4. Which word means "an icy, dusty body orbiting the Sun"?

A. nucleus
B. comet
C. coma
D. solar wind

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Place a desk fan as the "solar wind" and hold strips of paper behind a ball. Move the ball around the fan and note that strips point away from it.

6. Which sentence gives the best summary of this lesson?

A. Comet activity increases near the Sun, and its tails point away from the Sun.
B. Comets can be understood without observation, evidence or comparison.
C. Every object or event connected with comets behaves in exactly the same way.
D. The main idea of comets is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain comets.
2. Write two important facts about comets.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why does a comet not always show a bright tail when it is far from the Sun?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Comet Halley returns to the inner solar system roughly every 76 years. A comet can develop two main tails: a curved dust tail and a straighter ion tail.

UNIT 2 · CHAPTER 8

Meteors and Meteorites

Learning outcomes

- Explain meteors and meteorites using simple scientific language.
- Use the words meteoroid and meteor correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A meteoroid is a small rock or metal object in space. A meteor is the bright streak seen when a meteoroid heats the air while entering an atmosphere. A meteorite is a piece that reaches the ground.

Understanding the idea

Most meteors are caused by very small particles and burn or break apart high in the atmosphere.

A meteor is often called a "shooting star," but it is not a star.

Meteor showers occur when Earth passes through streams of debris, often left by comets.

Scientists study meteorites because some preserve ancient material from the early solar system.

Compare objects by position, size, composition and motion. Remember that drawings of the Solar System are usually not made to one common scale.

Do not learn meteoroid, meteor, meteorite separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

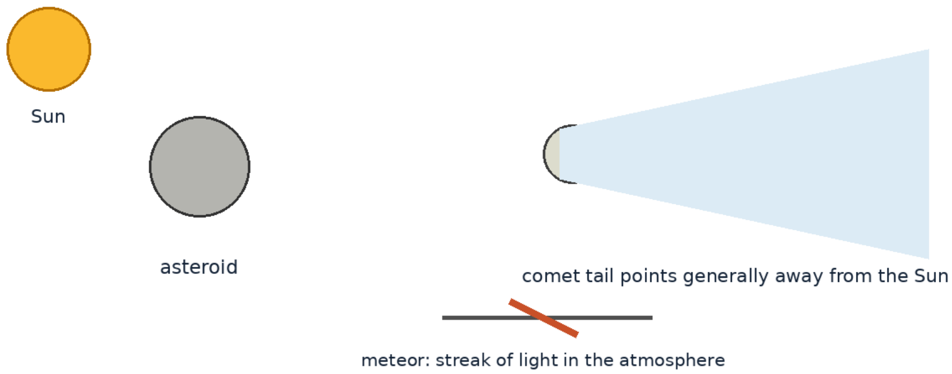


Figure 2.8 Asteroids, comets and meteors are different kinds of small Solar System objects and events.

Example

The same object can be called a meteoroid in space, a meteor while glowing in the atmosphere and a meteorite if it lands.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Write the three words on cards and place them beside drawings of space, sky and ground.

Think and discuss

A bright streak vanishes before reaching the ground. Which word describes what you saw, and why?

What you have learnt

- Most meteors are caused by very small particles and burn or break apart high in the atmosphere.
- A meteor is often called a “shooting star,” but it is not a star.
- Location and stage determine the words meteoroid, meteor and meteorite.

EXERCISES · METEORS AND METEORITES

Exercises

A. Choose the correct answer

1. Which statement about meteors and meteorites is correct?

A. A meteor is the light seen when a meteoroid enters an atmosphere.
B. A meteorite is a cloud around a comet.
C. A meteor is another name for a planet.
D. A meteoroid is a star visible only at night.

2. What does the word “meteoroid” mean in this lesson?

A. material from a meteoroid that reaches the ground
B. many meteors seen when Earth crosses a debris stream
C. a small natural rock or metal object moving in space
D. the streak of light produced during atmospheric entry

3. Which observation or example best supports the lesson idea?

A. The same object can be called a meteoroid in space, a meteor while glowing in the atmosphere and a meteorite if it lands.
B. A notebook is closed on a table.
C. A glass is filled with water.

D. A pencil is sharpened.

4. Which word means "a small natural rock or metal object moving in space"?

A. meteor
B. meteorite
C. meteoroid
D. meteor shower

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Write the three words on cards and place them beside drawings of space, sky and ground.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Meteors and meteorites can be understood without observation, evidence or comparison.
B. Every object or event connected with meteors and meteorites behaves in exactly the same way.
C. Location and stage determine the words meteoroid, meteor and meteorite.
D. The main idea of meteors and meteorites is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain meteors and meteorites.
2. Write two important facts about meteors and meteorites.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. A bright streak vanishes before reaching the ground. Which word describes what you saw, and why?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Many meteorites are attracted to magnets because they contain iron, but not every magnetic rock is a meteorite. During a strong meteor shower, many streaks appear to come from one region of the sky because of perspective.

UNIT 2 · CHAPTER 9

Characteristics of the Planets

Learning outcomes

- Explain characteristics of the planets using simple scientific language.
- Use the words characteristic and atmosphere correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Planetary characteristics are features used to compare worlds, such as size, mass, distance from the Sun, atmosphere, surface, temperature, rotation, rings and moons.

Understanding the idea

Venus is the hottest planet because its thick carbon-dioxide atmosphere traps heat strongly.

Mars is cold and dusty, with a thin atmosphere and evidence that liquid water flowed there long ago.

Jupiter has the greatest mass of any planet and a long-lived giant storm called the Great Red Spot.

Saturn has bright rings; Uranus rotates with an extreme tilt; Neptune has very fast winds.

Compare objects by position, size, composition and motion. Remember that drawings of the Solar System are usually not made to one common scale.

The vocabulary of this chapter - characteristic, atmosphere, ring system - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

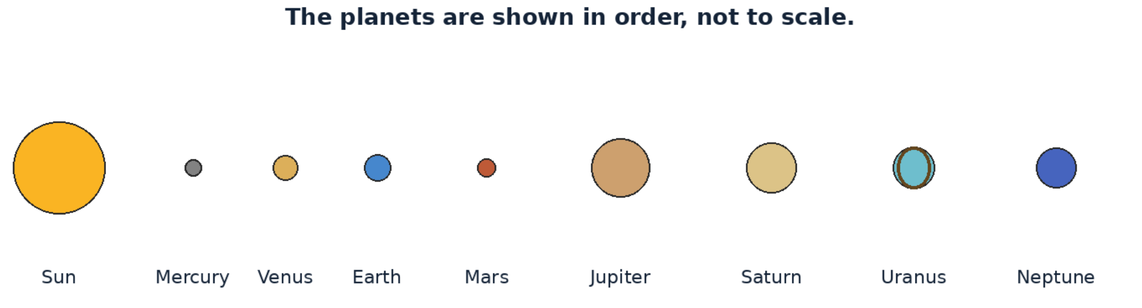


Figure 2.9 The Sun and eight planets shown in order. Sizes and distances are not to scale.

Example

Mercury is closest to the Sun, but Venus is hotter because atmosphere matters as well as distance.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Choose three planets and make "identity cards" using five characteristics. Ask a partner to guess each planet.

Think and discuss

Which two characteristics would best help distinguish Saturn from Jupiter?

What you have learnt

- Venus is the hottest planet because its thick carbon-dioxide atmosphere traps heat strongly.
- Mars is cold and dusty, with a thin atmosphere and evidence that liquid water flowed there long ago.
- Planets can be identified by combinations of characteristics, not by one feature alone.

EXERCISES · CHARACTERISTICS OF THE PLANETS

Exercises

A. Choose the correct answer

1. Which statement about characteristics of the planets is correct?

A. Every planet has the same atmosphere.
B. Only Saturn has any rings.
C. Distance from the Sun is the only feature that matters.
D. Planet characteristics include size, atmosphere, surface, rings, moons and motion.

2. What does the word "characteristic" mean in this lesson?

A. many particles orbiting a planet in flat bands
B. the gases surrounding a world
C. a feature used to describe or identify something
D. the time a world takes to spin once

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A notebook is closed on a table.

C. A glass is filled with water.
D. Mercury is closest to the Sun, but Venus is hotter because atmosphere matters as well as distance.

4. Which word means “a feature used to describe or identify something”?
A. atmosphere
B. ring system
C. rotation period
D. characteristic

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Guess once and do not record anything.
B. Choose three planets and make “identity cards” using five characteristics. Ask a partner to guess each planet.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?
A. Planets can be identified by combinations of characteristics, not by one feature alone.
B. Characteristics of the planets can be understood without observation, evidence or comparison.
C. Every object or event connected with characteristics of the planets behaves in exactly the same way.
D. The main idea of characteristics of the planets is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain characteristics of the planets.
2. Write two important facts about characteristics of the planets.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Which two characteristics would best help distinguish Saturn from Jupiter?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: All four giant planets have ring systems, although Saturn’s are the easiest to see. Earth is the only world currently known to have stable oceans of liquid water on its surface and life.

UNIT 2 REVIEW

Review · The Solar System

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of our sun using one example.
2. Why is it misleading to draw the planets equally spaced in a classroom diagram?
3. Write one correct statement and one common mistake about dwarf planets.
4. Explain the main idea of order of the planets using one example.
5. Why does a moon not need an engine firing all the time to remain in orbit?
6. Write one correct statement and one common mistake about asteroids.
7. Explain the main idea of comets using one example.
8. A bright streak vanishes before reaching the ground. Which word describes what you saw, and why?

B. Compare and connect

1. Write one connection between “Our Sun” and “Characteristics of the Planets”.
2. Write one connection between “The Eight Planets” and “Meteors and Meteorites”.

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 2 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about our sun is correct?

A. The Sun is the star at the centre of our solar system.
B. The Sun is made mainly of rock and ice.
C. The Sun shines by reflecting light from the Moon.
D. The Sun is a planet orbiting Earth.

2. Which statement about the eight planets is correct?

A. Our solar system has eight planets orbiting the Sun.
B. Our solar system has nine official planets including Pluto.
C. Planets produce their own visible light like stars.
D. Every planet has a solid rocky surface.

3. Which statement about dwarf planets is correct?

A. A dwarf planet is always a moon.
B. A dwarf planet orbits the Sun but has not cleared its orbital neighbourhood.
C. A dwarf planet does not orbit the Sun.
D. A dwarf planet must be smaller than every asteroid.

4. Which statement about order of the planets is correct?

A. Neptune is the closest planet to the Sun.
B. Earth is the third planet from the Sun, between Venus and Mars.
C. Mars is farther from the Sun than Jupiter.
D. Earth is between Jupiter and Saturn.

5. Which statement about natural satellites is correct?

A. A natural satellite is always built by engineers.
B. A natural satellite is a moon that orbits a larger body.
C. Moons orbit only the Sun and never planets.
D. Every planet has exactly one moon.

6. Which statement about asteroids is correct?

A. Asteroids are stars that have stopped shining.
B. All asteroids lie on Earth's surface.
C. Most asteroids are rocky or metallic bodies orbiting the Sun.
D. The asteroid belt is a solid ring with no gaps.

7. Which statement about comets is correct?

A. A comet is a burning star falling through Earth's atmosphere.
B. Comets are made only of liquid water.
C. A comet's tail always points behind its motion.
D. A comet is an icy body that can form a coma and tails near the Sun.

8. Which statement about meteors and meteorites is correct?

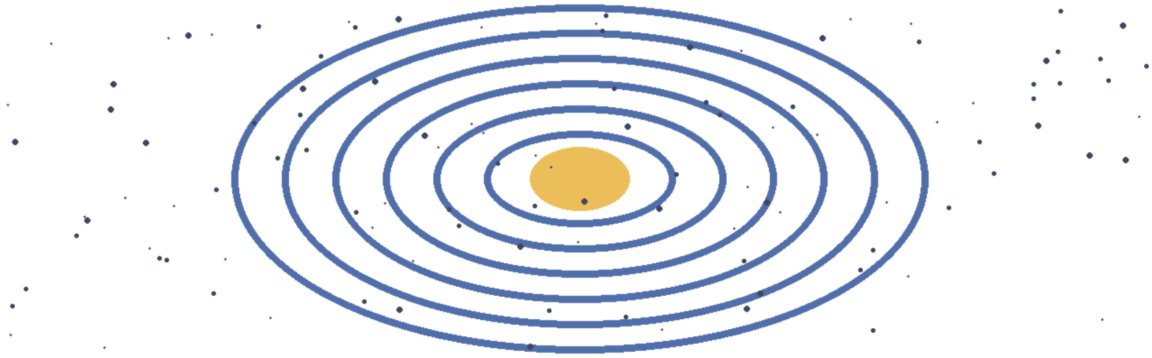
A. A meteor is the light seen when a meteoroid enters an atmosphere.
B. A meteorite is a cloud around a comet.
C. A meteor is another name for a planet.
D. A meteoroid is a star visible only at night.

UNIT 3

Stars, Galaxies & the Universe

Stars and galaxies take us beyond the Solar System. Learners compare brightness, distance and scale while understanding that the Universe is much larger than anything seen in everyday life.

A galaxy is a vast system of stars, gas, dust and dark matter held together by gravity.



Chapters in this unit

1. What are Stars?
2. Brightness of Stars
3. The Nearest Stars
4. The Milky Way Galaxy
5. Other Galaxies
6. Understanding the Universe

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 3 · CHAPTER 1

What are Stars?

Learning outcomes

- Explain what are stars using simple scientific language.
- Use the words plasma and fusion correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Stars are enormous spheres of hot plasma held together by gravity. They produce energy through nuclear fusion and release that energy as light, heat and other radiation.

Understanding the idea

A star forms when gravity pulls gas and dust together until the centre becomes hot and dense enough for fusion.

A star's colour gives a clue to surface temperature: blue-white stars are generally hotter than red stars.

Stars change during long life cycles, and their later stages depend strongly on mass.

The Sun is an ordinary star in size compared with many stars, but it is special to us because it is nearby.

Separate how bright an object looks from how much light it actually gives. Distance and scale are central to this unit.

The important words in this chapter include plasma, fusion, gravity. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

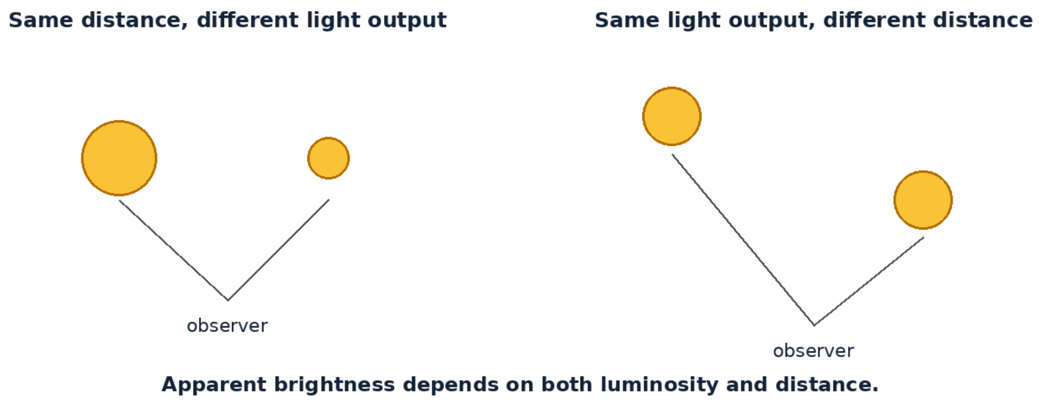


Figure 3.1 A star's apparent brightness depends on its light output and its distance from the observer.

Example

A candle flame looks brighter when close, just as the nearby Sun looks far brighter than distant stars.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Compare red and blue paper circles labelled cooler and hotter. Add a yellow circle for the Sun between them.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Why can a small nearby star look brighter than a much larger distant star?

What you have learnt

- A star forms when gravity pulls gas and dust together until the centre becomes hot and dense enough for fusion.
- A star's colour gives a clue to surface temperature: blue-white stars are generally hotter than red stars.
- Stars make their own energy and light through fusion.

Exercises

A. Choose the correct answer

1. Which statement about what are stars? is correct?

A. All stars have the same temperature and size.
B. Stars are small points attached to the sky.
C. Stars are hot glowing spheres that produce energy through fusion.
D. Stars shine only by reflecting sunlight.

2. What does the word "plasma" mean in this lesson?

A. a very hot state of matter containing charged particles
B. the stages through which an object changes over time
C. attraction between masses
D. joining atomic nuclei and releasing energy

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A candle flame looks brighter when close, just as the nearby Sun looks far brighter than distant stars.
C. A notebook is closed on a table.
D. A glass is filled with water.

4. Which word means "a very hot state of matter containing charged particles"?

A. plasma
B. fusion
C. gravity
D. life cycle

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Compare red and blue paper circles labelled cooler and hotter. Add a yellow circle for the Sun between them.

6. Which sentence gives the best summary of this lesson?

A. What are stars can be understood without observation, evidence or comparison.
B. Stars make their own energy and light through fusion.
C. Every object or event connected with what are stars behaves in exactly the same way.
D. The main idea of what are stars is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain what are stars?
2. Write two important facts about what are stars
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can a small nearby star look brighter than a much larger distant star?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Stars are not five-pointed shapes; the points in drawings represent the way light spreads in eyes or instruments. Some stars exist in pairs or larger systems and orbit one another.

UNIT 3 · CHAPTER 2

Brightness of Stars

Learning outcomes

- Explain brightness of stars using simple scientific language.
- Use the words apparent brightness and luminosity correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

The brightness of a star as seen from Earth depends on its true energy output and its distance from us. Astronomers separate apparent brightness from intrinsic brightness.

Understanding the idea

Apparent brightness describes how bright an object looks from the observer's location.

Luminosity is the total amount of energy a star gives out each second.

Distance can make a powerful star look faint or a modest star look bright.

Dust between stars can absorb and scatter light, making a star appear dimmer and redder.

Separate how bright an object looks from how much light it actually gives. Distance and scale are central to this unit.

Notice how the terms apparent brightness, luminosity, distance appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

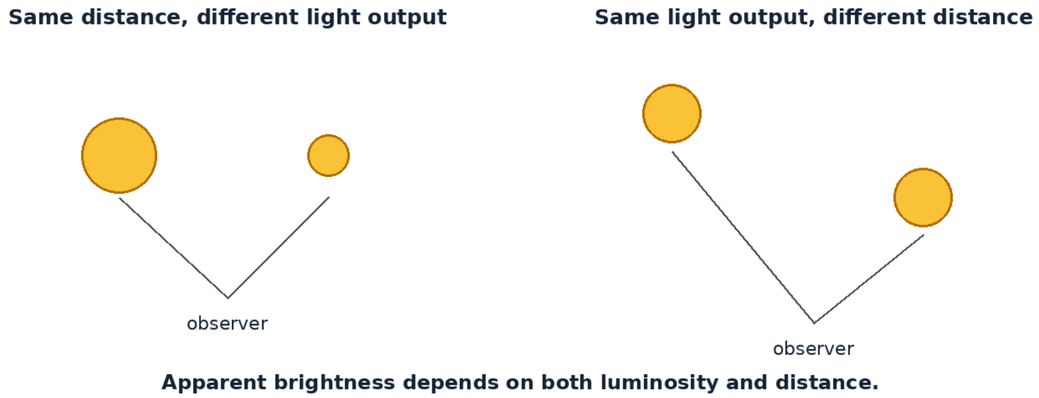


Figure 3.2 A star's apparent brightness depends on its light output and its distance from the observer.

Example

Two identical torches look different in brightness when one is nearby and the other is far away.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Place two identical lights at different distances. Compare how bright they look while keeping safety rules.

Think and discuss

Can apparent brightness alone tell us which star produces more energy? Why not?

What you have learnt

- Apparent brightness describes how bright an object looks from the observer's location.
- Luminosity is the total amount of energy a star gives out each second.
- How bright a star looks is not the same as how much energy it truly produces.

EXERCISES · BRIGHTNESS OF STARS

Exercises

A. Choose the correct answer

1. Which statement about brightness of stars is correct?

A. The brightest-looking star must always be the largest.
B. Distance never affects how bright a star looks.
C. A star's apparent brightness depends on luminosity, distance and intervening material.
D. All stars give out exactly the same amount of energy.

2. What does the word "apparent brightness" mean in this lesson?

A. how bright an object looks from a particular place
B. a number system astronomers use to describe brightness
C. the space between two points
D. the total energy a star gives out per second

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A notebook is closed on a table.
C. A glass is filled with water.
D. Two identical torches look different in brightness when one is nearby and the other is far away.

4. Which word means "how bright an object looks from a particular place"?

A. luminosity
B. apparent brightness
C. distance
D. magnitude

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Place two identical lights at different distances. Compare how bright they look while keeping safety rules.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Brightness of stars can be understood without observation, evidence or comparison.
B. Every object or event connected with brightness of stars behaves in exactly the same way.
C. The main idea of brightness of stars is unrelated to science or everyday experience.
D. How bright a star looks is not the same as how much energy it truly produces.

B. Answer in one or two sentences

1. In one sentence, explain brightness of stars.
2. Write two important facts about brightness of stars.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Can apparent brightness alone tell us which star produces more energy? Why not?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Astronomers use a magnitude scale in which smaller or more negative numbers indicate brighter objects. The brightest-looking star in Earth's night sky is Sirius, but it is not the most luminous star known.

UNIT 3 · CHAPTER 3

The Nearest Stars

Learning outcomes

- Explain the nearest stars using simple scientific language.
- Use the words light-year and Proxima Centauri correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

The Sun is the nearest star to Earth. The next nearest known star is Proxima Centauri, part of the Alpha Centauri star system, a little over four light-years away.

Understanding the idea

A light-year is a unit of distance: the distance light travels in one year.

Sunlight reaches Earth in minutes, but light from the next nearest stars takes more than four years.

Alpha Centauri A and B form a bright pair, while Proxima Centauri is a smaller red dwarf nearby in the same system.

Even our nearest stellar neighbours are so far away that present spacecraft would take thousands of years to reach them.

Separate how bright an object looks from how much light it actually gives. Distance and scale are central to this unit.

Do not learn light-year, Proxima Centauri, star system separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

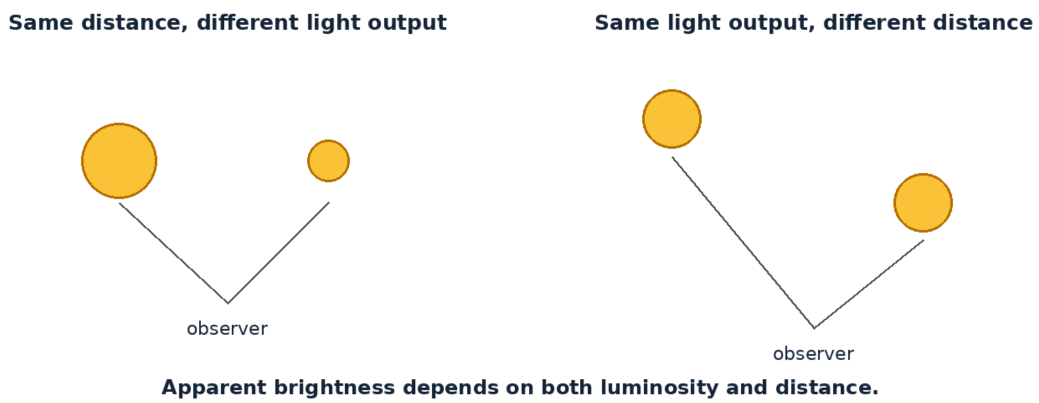


Figure 3.3 A star's apparent brightness depends on its light output and its distance from the observer.

Example

If Proxima Centauri suddenly changed today, Earth would not receive the new light for more than four years.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Walk ten steps for a model Earth-Sun distance, then discuss why the next star would lie far beyond the playground on the same scale.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Why is a light-year a distance and not a length of time in astronomy?

What you have learnt

- A light-year is a unit of distance: the distance light travels in one year.
- Sunlight reaches Earth in minutes, but light from the next nearest stars takes more than four years.
- The Sun is our nearest star; every other star is enormously farther away.

EXERCISES · THE NEAREST STARS

Exercises

A. Choose the correct answer

1. Which statement about the nearest stars is correct?

A. A light-year measures how long a school year lasts.
B. Proxima Centauri is a planet in our solar system.
C. Polaris is closer than the Sun.
D. The Sun is the nearest star to Earth.

2. What does the word “light-year” mean in this lesson?

A. the distance light travels in one year
B. the nearest known star beyond the Sun
C. two or more stars related by gravity
D. a small, relatively cool and faint type of star

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A pencil is sharpened.
C. If Proxima Centauri suddenly changed today, Earth would not receive the new light for more than four years.
D. A glass is filled with water.

4. Which word means “the distance light travels in one year”?

A. Proxima Centauri
B. star system
C. light-year
D. red dwarf

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Walk ten steps for a model Earth-Sun distance, then discuss why the next star would lie far beyond the playground on the same scale.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. The nearest stars can be understood without observation, evidence or comparison.
B. The Sun is our nearest star; every other star is enormously farther away.
C. Every object or event connected with the nearest stars behaves in exactly the same way.
D. The main idea of the nearest stars is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain the nearest stars.

2. Write two important facts about the nearest stars.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is a light-year a distance and not a length of time in astronomy?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: One light-year is about 9.46 trillion kilometres. Proxima Centauri has at least one confirmed planet in its system, although conditions there are very different from Earth.

UNIT 3 · CHAPTER 4

The Milky Way Galaxy

Learning outcomes

- Explain the milky way galaxy using simple scientific language.
- Use the words galaxy and Milky Way correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

The Milky Way is the galaxy that contains the Sun, Earth and our solar system. It is a barred spiral galaxy made of stars, gas, dust and dark matter.

Understanding the idea

The Milky Way contains hundreds of billions of stars, although the exact number is uncertain.

Our solar system lies in a smaller spiral feature called the Orion Spur, far from the galactic centre.

From a dark location, the Milky Way can appear as a pale band because we are looking through the galaxy's crowded disc.

The Sun takes roughly 230 million years to complete one orbit around the Milky Way's centre.

Separate how bright an object looks from how much light it actually gives. Distance and scale are central to this unit.

The vocabulary of this chapter - galaxy, Milky Way, spiral arm - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

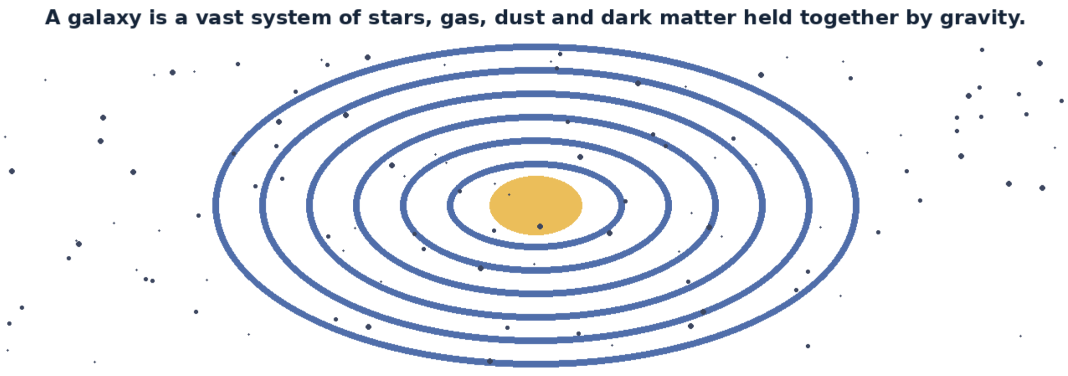


Figure 3.4 A galaxy contains enormous numbers of stars, together with gas, dust and dark matter.

Example

A person inside a large forest sees trees in every direction; similarly, we view the Milky Way from inside it rather than from above.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Sprinkle rice in a thin circular layer on dark paper. View it from above and then edge-on to model a galaxy disc.

Think and discuss

Why can no spacecraft currently take a complete photograph of the Milky Way from outside?

What you have learnt

- The Milky Way contains hundreds of billions of stars, although the exact number is uncertain.
- Our solar system lies in a smaller spiral feature called the Orion Spur, far from the galactic centre.
- We live inside the Milky Way, a vast barred spiral galaxy.

EXERCISES · THE MILKY WAY GALAXY

Exercises

A. Choose the correct answer

1. Which statement about the milky way galaxy is correct?

A. The Milky Way is another name for the solar system.
B. The solar system is located inside the Milky Way galaxy.
C. The Milky Way contains only eight stars.
D. Earth lies at the exact centre of the Milky Way.

2. What does the word “galaxy” mean in this lesson?

A. the galaxy containing our solar system
B. the central region of a galaxy
C. a curved region of stars and gas in a spiral galaxy
D. a huge gravitational system of stars, gas, dust and dark matter

3. Which observation or example best supports the lesson idea?

A. A person inside a large forest sees trees in every direction; similarly, we view the Milky Way from inside it rather than from above.
B. A glass is filled with water.
C. A notebook is closed on a table.
D. A pencil is sharpened.

4. Which word means “a huge gravitational system of stars, gas, dust and dark matter”?

A. Milky Way
B. spiral arm
C. galactic centre
D. galaxy

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Sprinkle rice in a thin circular layer on dark paper. View it from above and then edge-on to model a galaxy disc.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. The milky way galaxy can be understood without observation, evidence or comparison.
B. Every object or event connected with the milky way galaxy behaves in exactly the same way.
C. The main idea of the milky way galaxy is unrelated to science or everyday experience.
D. We live inside the Milky Way, a vast barred spiral galaxy.

B. Answer in one or two sentences

1. In one sentence, explain the milky way galaxy.
2. Write two important facts about the milky way galaxy.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can no spacecraft currently take a complete photograph of the Milky Way from outside?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The centre of the Milky Way contains a supermassive black hole called Sagittarius A*. Nearly every star visible without a telescope belongs to the Milky Way.

UNIT 3 · CHAPTER 5

Other Galaxies

Learning outcomes

- Explain other galaxies using simple scientific language.
- Use the words spiral galaxy and elliptical galaxy correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

The universe contains many galaxies beyond the Milky Way. Galaxies come in several broad shapes, including spiral, elliptical and irregular.

Understanding the idea

The Andromeda Galaxy is a large spiral galaxy and the nearest major galaxy to the Milky Way.

The Large and Small Magellanic Clouds are irregular satellite galaxies visible mainly from the Southern Hemisphere.

Galaxies can contain millions, billions or even trillions of stars.

Gravity gathers galaxies into groups and clusters, and those clusters form even larger cosmic structures.

Separate how bright an object looks from how much light it actually gives. Distance and scale are central to this unit.

The important words in this chapter include spiral galaxy, elliptical galaxy, irregular galaxy. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

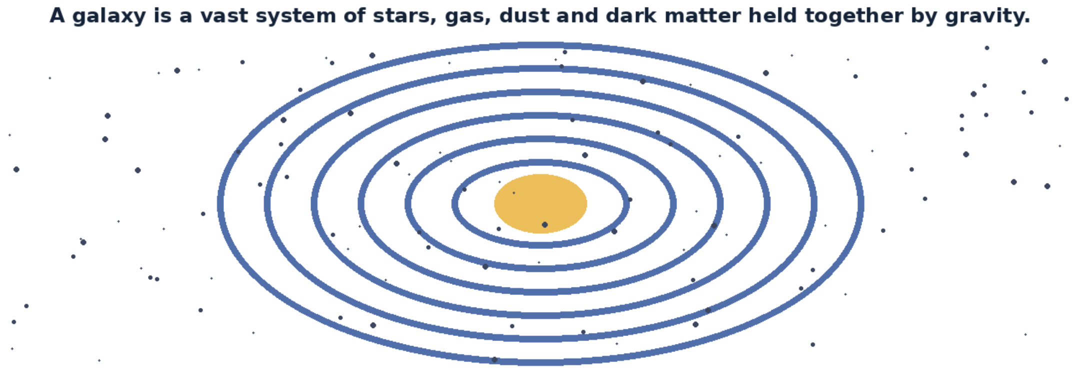


Figure 3.5 A galaxy contains enormous numbers of stars, together with gas, dust and dark matter.

Example

A telescope image may show one large nearby galaxy and many tiny-looking galaxies much farther away in the background.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Sort cut-out galaxy pictures by shape. Explain that real classifications can be more complex than three boxes.

Think and discuss

Why does a very distant galaxy look smaller even if it is physically large?

What you have learnt

- The Andromeda Galaxy is a large spiral galaxy and the nearest major galaxy to the Milky Way.
- The Large and Small Magellanic Clouds are irregular satellite galaxies visible mainly from the Southern Hemisphere.
- The Milky Way is one of many galaxies in a universe organised by gravity.

Exercises

A. Choose the correct answer

1. Which statement about other galaxies is correct?

A. A galaxy contains only one star and one planet.
B. Galaxies exist in many shapes and contain enormous numbers of stars.
C. Galaxies are found only inside the solar system.
D. Every galaxy is identical to the Milky Way.

2. What does the word “spiral galaxy” mean in this lesson?

A. a rounded or oval galaxy with little visible spiral structure
B. a group of galaxies held together by gravity
C. a galaxy with a disc and curved arms
D. a galaxy without a regular spiral or elliptical shape

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A pencil is sharpened.
C. A notebook is closed on a table.
D. A telescope image may show one large nearby galaxy and many tiny-looking galaxies much farther away in the background.

4. Which word means “a galaxy with a disc and curved arms”?

A. spiral galaxy
B. elliptical galaxy
C. irregular galaxy
D. cluster

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Sort cut-out galaxy pictures by shape. Explain that real classifications can be more complex than three boxes.

6. Which sentence gives the best summary of this lesson?

A. Other galaxies can be understood without observation, evidence or comparison.
B. The Milky Way is one of many galaxies in a universe organised by gravity.
C. Every object or event connected with other galaxies behaves in exactly the same way.
D. The main idea of other galaxies is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain other galaxies.
2. Write two important facts about other galaxies.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why does a very distant galaxy look smaller even if it is physically large?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The Milky Way and Andromeda are moving toward each other and are expected to interact billions of years from now. Looking at a distant galaxy means seeing it as it was in the past because its light needed time to reach us.

UNIT 3 · CHAPTER 6

Understanding the Universe

Learning outcomes

- Explain understanding the universe using simple scientific language.
- Use the words universe and expansion correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

The universe includes all space, time, matter and energy. It contains galaxies, stars, planets, radiation and structures on scales far larger than everyday experience.

Understanding the idea

Evidence shows that the universe has been expanding from a much hotter, denser early state for about 13.8 billion years.

The observable universe is the region whose light has had time to reach us; the whole universe may be larger.

Scientists build models of the universe using observations such as galaxy motion, background radiation and element amounts.

Many questions remain open, including the nature of dark matter and dark energy.

Separate how bright an object looks from how much light it actually gives. Distance and scale are central to this unit.

Notice how the terms universe, expansion, observable universe appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

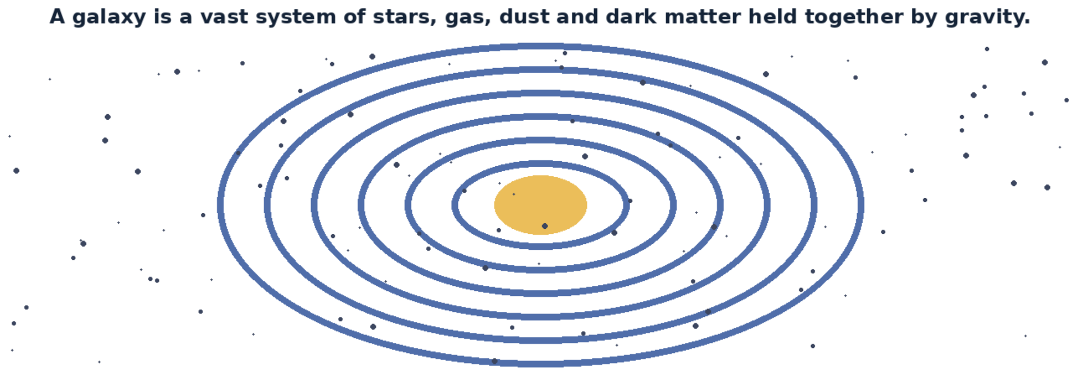


Figure 3.6 A galaxy contains enormous numbers of stars, together with gas, dust and dark matter.

Example

Like dots on an inflating balloon surface, distant galaxies can become farther apart as space expands, although the balloon is only a model.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Draw five nested circles showing increasingly large cosmic scales. Write one object found at each scale.

Think and discuss

Why do scientists say a distant galaxy is seen in the past?

What you have learnt

- Evidence shows that the universe has been expanding from a much hotter, denser early state for about 13.8 billion years.
- The observable universe is the region whose light has had time to reach us; the whole universe may be larger.
- The universe is everything that exists physically, studied through evidence and models.

EXERCISES · UNDERSTANDING THE UNIVERSE

Exercises

A. Choose the correct answer

1. Which statement about understanding the universe is correct?

A. The universe is a small empty shell around Earth.
B. The universe contains only the Milky Way.
C. The universe includes all space, time, matter and energy.
D. Scientists understand every part of the universe completely.

2. What does the word “universe” mean in this lesson?

A. a simplified representation used to explain or test an idea
B. the region whose signals have had time to reach us
C. all space, time, matter and energy
D. the increase of distances between widely separated galaxies as space changes

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. Like dots on an inflating balloon surface, distant galaxies can become farther apart as space expands, although the balloon is only a model.
C. A pencil is sharpened.

D. A notebook is closed on a table.

4. Which word means “all space, time, matter and energy”?

A. expansion
B. universe
C. observable universe
D. model

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Draw five nested circles showing increasingly large cosmic scales. Write one object found at each scale.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Understanding the universe can be understood without observation, evidence or comparison.
B. Every object or event connected with understanding the universe behaves in exactly the same way.
C. The main idea of understanding the universe is unrelated to science or everyday experience.
D. The universe is everything that exists physically, studied through evidence and models.

B. Answer in one or two sentences

1. In one sentence, explain understanding the universe.
2. Write two important facts about understanding the universe.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why do scientists say a distant galaxy is seen in the past?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: There is no known ordinary edge or central point in space that we can travel to and call “the centre of the universe.” Astronomy lets us look back in time because light from distant objects began its journey long ago.

UNIT 3 REVIEW

Review · Stars, Galaxies & the Universe

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of what are stars? using one example.
2. Can apparent brightness alone tell us which star produces more energy? Why not?
3. Write one correct statement and one common mistake about the nearest stars.
4. Explain the main idea of the milky way galaxy using one example.
5. Why does a very distant galaxy look smaller even if it is physically large?
6. Write one correct statement and one common mistake about understanding the universe.
7. Create one new question from this unit and write its answer.

B. Compare and connect

1. Write one connection between “What are Stars?” and “Understanding the Universe”.
2. Write one connection between “Brightness of Stars” and “Other Galaxies”.

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 3 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about what are stars? is correct?
- A. All stars have the same temperature and size.
- B. Stars are small points attached to the sky.
- C. Stars are hot glowing spheres that produce energy through fusion.
- D. Stars shine only by reflecting sunlight.

2. Which statement about brightness of stars is correct?
- A. The brightest-looking star must always be the largest.
- B. Distance never affects how bright a star looks.
- C. A star's apparent brightness depends on luminosity, distance and intervening material.
- D. All stars give out exactly the same amount of energy.

3. Which statement about the nearest stars is correct?
- A. A light-year measures how long a school year lasts.
- B. Proxima Centauri is a planet in our solar system.
- C. Polaris is closer than the Sun.
- D. The Sun is the nearest star to Earth.

4. Which statement about the milky way galaxy is correct?
- A. The Milky Way is another name for the solar system.
- B. The solar system is located inside the Milky Way galaxy.
- C. The Milky Way contains only eight stars.
- D. Earth lies at the exact centre of the Milky Way.

5. Which statement about other galaxies is correct?
- A. A galaxy contains only one star and one planet.
- B. Galaxies exist in many shapes and contain enormous numbers of stars.
- C. Galaxies are found only inside the solar system.
- D. Every galaxy is identical to the Milky Way.

6. Which statement about understanding the universe is correct?
- A. The universe is a small empty shell around Earth.
- B. The universe contains only the Milky Way.
- C. The universe includes all space, time, matter and energy.
- D. Scientists understand every part of the universe completely.

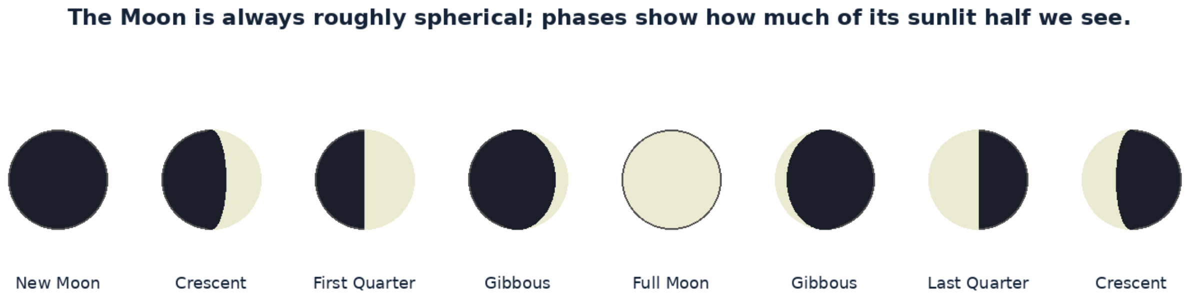
7. Which word means "a very hot state of matter containing charged particles"?
- A. plasma
- B. fusion
- C. gravity
- D. life cycle

8. Which word means "how bright an object looks from a particular place"?
- A. luminosity
- B. apparent brightness
- C. distance
- D. magnitude

UNIT 4

Earth & Moon

Earth and the Moon form a connected system. Their motions explain familiar cycles such as day and night, seasons, Moon phases, eclipses and tides.



Chapters in this unit

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 4 · CHAPTER 1

Earth: Our Home Planet

Learning outcomes

- Explain earth: our home planet using simple scientific language.
- Use the words Earth and atmosphere correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Earth is the third planet from the Sun and the only world currently known to support life. It has liquid-water oceans, a protective atmosphere, active geology and a suitable range of temperatures.

Understanding the idea

Earth is nearly spherical, with a slightly wider equator than pole-to-pole distance.

About 71 percent of Earth's surface is covered by ocean and about 29 percent by land.

The atmosphere contains mostly nitrogen and oxygen and helps regulate temperature and block harmful radiation.

Earth's magnetic field helps deflect many charged particles from the Sun.

Use models carefully. A lamp, ball and globe can show relationships, but a model is smaller and simpler than the real system.

Notice how the terms Earth, atmosphere, magnetic field appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

Example

A thin layer of air makes breathing and weather possible, even though it looks almost invisible from the ground.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Make a list of five conditions Earth provides for life: water, air, energy, suitable temperature and nutrients.

Think and discuss

Why is Earth's atmosphere important even though it is thin compared with the planet's size?

What you have learnt

- Earth is nearly spherical, with a slightly wider equator than pole-to-pole distance.
- About 71 percent of Earth's surface is covered by ocean and about 29 percent by land.
- Earth supports life through a connected system of air, water, land and energy from the Sun.

EXERCISES · EARTH: OUR HOME PLANET

Exercises

A. Choose the correct answer

1. Which statement about earth: our home planet is correct?

A. Earth is the largest planet in the solar system.

B. Earth is the third planet from the Sun and the only known world with life.
C. Earth has no atmosphere or liquid water.
D. Earth is the closest planet to the Sun.

2. What does the word "Earth" mean in this lesson?
A. suitable for living things
B. the third planet from the Sun and our home world
C. a region where magnetic forces act
D. the gases surrounding a planet

3. Which observation or example best supports the lesson idea?
A. A notebook is closed on a table.
B. A glass is filled with water.
C. A thin layer of air makes breathing and weather possible, even though it looks almost invisible from the ground.
D. A pencil is sharpened.

4. Which word means "the third planet from the Sun and our home world"?
A. atmosphere
B. Earth
C. magnetic field
D. habitable

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Guess once and do not record anything.
B. Make a list of five conditions Earth provides for life: water, air, energy, suitable temperature and nutrients.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?
A. Earth: our home planet can be understood without observation, evidence or comparison.
B. Every object or event connected with earth: our home planet behaves in exactly the same way.
C. Earth supports life through a connected system of air, water, land and energy from the Sun.
D. The main idea of earth: our home planet is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain earth: our home planet.
2. Write two important facts about earth: our home planet.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is Earth's atmosphere important even though it is thin compared with the planet's size?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Earth is sometimes called the "blue planet" because oceans dominate its appearance from space. The deepest ocean trenches and highest mountains are tiny compared with Earth's total diameter.

UNIT 4 · CHAPTER 2

Continents and Oceans

Learning outcomes

- Explain continents and oceans using simple scientific language.
- Use the words continent and ocean correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Earth's surface is divided into large land areas called continents and vast connected bodies of salt water called oceans. Maps help us represent these features on a flat surface.

Understanding the idea

The seven commonly taught continents are Asia, Africa, North America, South America, Antarctica, Europe and Australia.

The five named oceans are the Pacific, Atlantic, Indian, Southern and Arctic Oceans.

All oceans are connected, forming one global ocean even though different regions have different names.

Continents move very slowly because Earth's outer shell is broken into tectonic plates.

Use models carefully. A lamp, ball and globe can show relationships, but a model is smaller and simpler than the real system.

Do not learn continent, ocean, map separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

Example

India lies in Asia and is bordered by the Indian Ocean region to the south.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

On an outline map, label India, Asia, the Indian Ocean and one continent in each hemisphere.

Think and discuss

Why can a flat map never show a spherical Earth without some distortion?

What you have learnt

- The seven commonly taught continents are Asia, Africa, North America, South America, Antarctica, Europe and Australia.
- The five named oceans are the Pacific, Atlantic, Indian, Southern and Arctic Oceans.
- Continents are major land regions; oceans are connected parts of one global ocean.

EXERCISES · CONTINENTS AND OCEANS

Exercises

A. Choose the correct answer

1. Which statement about continents and oceans is correct?

A. The Arctic is a continent and Antarctica is an ocean.
B. Earth's oceans are completely separate from one another.
C. Earth is commonly described as having seven continents and five oceans.
D. India is located in South America.

2. What does the word "continent" mean in this lesson?

A. a vast body of salt water
B. one of Earth's very large land regions
C. a representation of a place or surface
D. a slowly moving piece of Earth's rigid outer layer

3. Which observation or example best supports the lesson idea?

A. India lies in Asia and is bordered by the Indian Ocean region to the south.

B. A notebook is closed on a table.
C. A pencil is sharpened.
D. A glass is filled with water.

4. Which word means “one of Earth’s very large land regions”?
A. ocean
B. map
C. continent
D. tectonic plate

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. On an outline map, label India, Asia, the Indian Ocean and one continent in each hemisphere.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?
A. Continents are major land regions; oceans are connected parts of one global ocean.
B. Continents and oceans can be understood without observation, evidence or comparison.
C. Every object or event connected with continents and oceans behaves in exactly the same way.
D. The main idea of continents and oceans is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain continents and oceans.
2. Write two important facts about continents and oceans.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can a flat map never show a spherical Earth without some distortion?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The Pacific Ocean is larger than all Earth’s land areas combined. Antarctica is a continent covered by a thick ice sheet, while the Arctic is mainly an ocean surrounded by land.

UNIT 4 · CHAPTER 3

Earth’s Rotation

Learning outcomes

- Explain earth’s rotation using simple scientific language.
- Use the words rotation and axis correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Earth rotates from west to east around an imaginary axis. One rotation takes about 24 hours and creates the repeating daily pattern of day and night.

Understanding the idea

The axis passes through the geographic North and South Poles.

Earth’s eastward rotation makes the Sun, Moon and stars appear to move westward across the sky.

Rotation also helps explain time zones because different longitudes face the Sun at different times.

Earth rotates continuously, but we do not feel constant motion because we and the surrounding air move with it.

Use models carefully. A lamp, ball and globe can show relationships, but a model is smaller and simpler than the real system.

The vocabulary of this chapter - rotation, axis, time zone - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

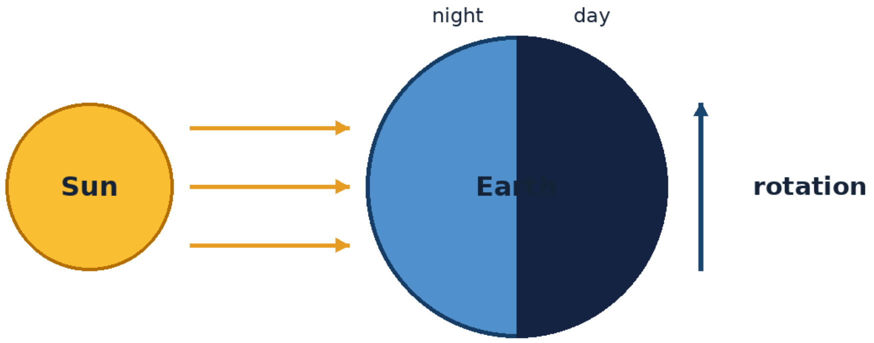


Figure 4.3 Earth's rotation carries places into sunlight and then into darkness.

Example

A passenger in a smoothly moving train may not feel motion until looking outside; similarly, steady rotation is not felt directly.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Spin a globe slowly from west to east while a lamp remains fixed. Watch labelled cities enter daylight.

Think and discuss

Why do stars appear to move across the sky if their positions do not change greatly during one night?

What you have learnt

- The axis passes through the geographic North and South Poles.
- Earth's eastward rotation makes the Sun, Moon and stars appear to move westward across the sky.
- Earth's rotation is a west-to-east spin completed in about 24 hours.

EXERCISES · EARTH'S ROTATION

Exercises

A. Choose the correct answer

1. Which statement about earth's rotation is correct?

A. Rotation causes the Moon's phases.
B. Earth rotates from west to east once in about 24 hours.
C. Earth rotates once every 365 years.
D. Earth rotates from east to west.

2. What does the word "rotation" mean in this lesson?

A. a region using the same standard clock time
B. an imaginary line through a rotating object
C. spinning around an axis
D. imaginary north-south lines used to locate places east or west

3. Which observation or example best supports the lesson idea?

A. A passenger in a smoothly moving train may not feel motion until looking outside; similarly, steady rotation is not felt directly.
B. A glass is filled with water.
C. A pencil is sharpened.
D. A notebook is closed on a table.

4. Which word means “spinning around an axis”?
A. axis
B. time zone
C. longitude
D. rotation

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Spin a globe slowly from west to east while a lamp remains fixed. Watch labelled cities enter daylight.

6. Which sentence gives the best summary of this lesson?
A. Earth’s rotation can be understood without observation, evidence or comparison.
B. Every object or event connected with earth’s rotation behaves in exactly the same way.
C. Earth’s rotation is a west-to-east spin completed in about 24 hours.
D. The main idea of earth’s rotation is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain earth’s rotation.
2. Write two important facts about earth’s rotation.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why do stars appear to move across the sky if their positions do not change greatly during one night?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: At the equator, Earth’s surface moves due to rotation at roughly 1,670 kilometres per hour. The direction of rotation is counterclockwise when Earth is viewed from above the North Pole.

UNIT 4 · CHAPTER 4

Earth’s Revolution

Learning outcomes

- Explain earth’s revolution using simple scientific language.
- Use the words revolution and orbit correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Earth revolves around the Sun along a slightly oval path called an orbit. One revolution takes about 365.25 days and forms the basis of a year.

Understanding the idea

Earth’s orbit is nearly circular, so its distance from the Sun changes only modestly.

The extra quarter day each year is managed by adding a leap day to most years divisible by four, with calendar exceptions.

Earth moves around the Sun while rotating on its axis at the same time.

Revolution combined with axial tilt produces the yearly cycle of seasons.

Use models carefully. A lamp, ball and globe can show relationships, but a model is smaller and simpler than the real system.

The important words in this chapter include revolution, orbit, year. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

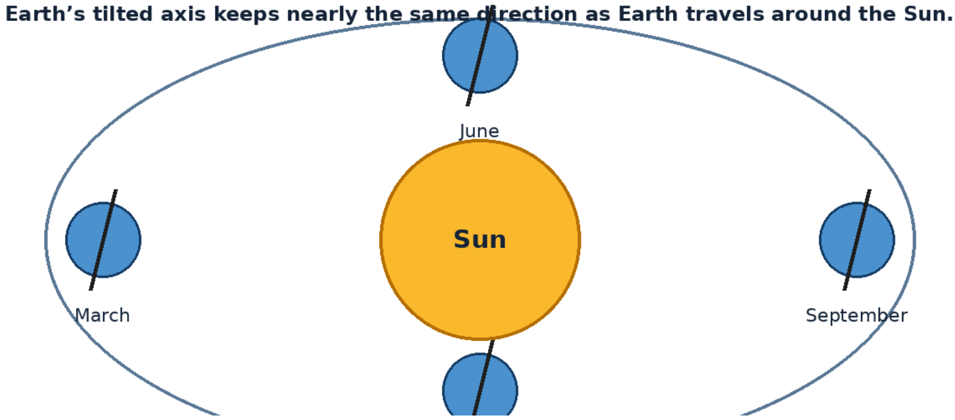


Figure 4.4 Earth travels around the Sun while its tilted axis keeps nearly the same direction in space.

Example

After four ordinary years, four quarter-days total roughly one day, which is why a leap day is sometimes added.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Walk in a circle around a chair while turning a small ball. Identify which motion is revolution and which is rotation.

Think and discuss

Can Earth rotate many times during one revolution? Use day and year lengths to explain.

What you have learnt

- Earth's orbit is nearly circular, so its distance from the Sun changes only modestly.
- The extra quarter day each year is managed by adding a leap day to most years divisible by four, with calendar exceptions.
- Revolution is Earth's yearly journey around the Sun; rotation is its daily spin.

EXERCISES · EARTH'S REVOLUTION

Exercises

A. Choose the correct answer

1. Which statement about earth's revolution is correct?

A. Earth completes one revolution every 24 hours.
B. Earth's orbit is centred on the Moon.
C. Revolution means spinning on an axis.
D. Earth completes one revolution around the Sun in about 365.25 days.

2. What does the word "revolution" mean in this lesson?

A. Earth's average distance from the Sun
B. the motion of one object travelling around another
C. the time Earth takes to orbit the Sun

D. the curved path followed around another object

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.

B. A pencil is sharpened.

C. A glass is filled with water.

D. After four ordinary years, four quarter-days total roughly one day, which is why a leap day is sometimes added.

4. Which word means “the motion of one object travelling around another”?

A. revolution

B. orbit

C. year

D. astronomical unit

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Walk in a circle around a chair while turning a small ball. Identify which motion is revolution and which is rotation.

B. Guess once and do not record anything.

C. Change several conditions at the same time and compare nothing.

D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Revolution is Earth’s yearly journey around the Sun; rotation is its daily spin.

B. Earth’s revolution can be understood without observation, evidence or comparison.

C. Every object or event connected with earth’s revolution behaves in exactly the same way.

D. The main idea of earth’s revolution is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain earth’s revolution.
2. Write two important facts about earth’s revolution.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Can Earth rotate many times during one revolution? Use day and year lengths to explain.
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Earth travels nearly 940 million kilometres during one orbit around the Sun. The average Earth-Sun distance is called one astronomical unit, or AU.

UNIT 4 · CHAPTER 5

Day and Night Revisited

Learning outcomes

- Explain day and night revisited using simple scientific language.
- Use the words terminator and local time correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Day and night can be understood more deeply by combining Earth’s shape, rotation and sunlight. The boundary between daylight and darkness moves across the globe as Earth turns.

Understanding the idea

The day-night boundary is called the terminator.

Sunrise occurs when a location rotates from the dark side into the illuminated side.

Sunset occurs when a location rotates from the illuminated side into darkness.

Day length changes through the year because Earth's axis is tilted.

Use models carefully. A lamp, ball and globe can show relationships, but a model is smaller and simpler than the real system.

Notice how the terms terminator, local time, midday appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

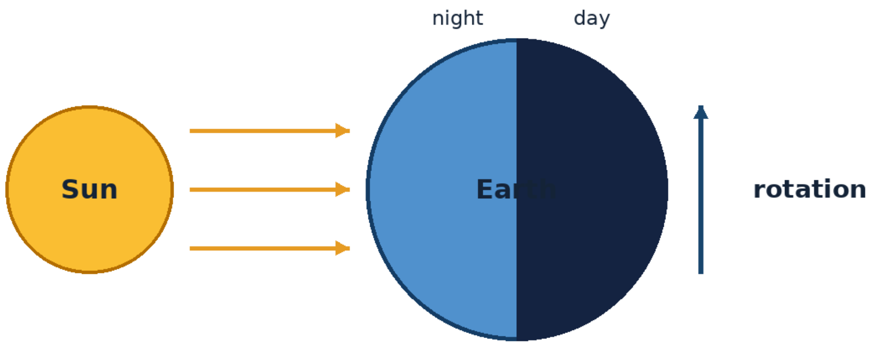


Figure 4.5 Earth's rotation carries places into sunlight and then into darkness.

Example

Cities at different longitudes cross the terminator at different times, producing different local times.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Mark four points around a ball: sunrise, midday, sunset and midnight. Rotate the ball under a lamp.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Why is it incorrect to say the dark half of Earth is always the same half?

What you have learnt

- The day-night boundary is called the terminator.
- Sunrise occurs when a location rotates from the dark side into the illuminated side.
- As Earth rotates, every location moves through daylight and darkness.

EXERCISES · DAY AND NIGHT REVISITED

Exercises

A. Choose the correct answer

1. Which statement about day and night revisited is correct?

A. The moving day-night boundary crosses Earth as the planet rotates.
B. Every city experiences noon at the same moment.
C. The same half of Earth is permanently dark.
D. Sunrise is caused by the Sun moving around Earth.

2. What does the word "terminator" mean in this lesson?

A. the middle of the night
B. the middle part of the day when the Sun is highest
C. the boundary between the lit and dark halves of a world

D. clock time for a particular location

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. Cities at different longitudes cross the terminator at different times, producing different local times.
C. A notebook is closed on a table.
D. A pencil is sharpened.

4. Which word means "the boundary between the lit and dark halves of a world"?

A. local time
B. terminator
C. midday
D. midnight

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Mark four points around a ball: sunrise, midday, sunset and midnight. Rotate the ball under a lamp.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Day and night revisited can be understood without observation, evidence or comparison.
B. Every object or event connected with day and night revisited behaves in exactly the same way.
C. As Earth rotates, every location moves through daylight and darkness.
D. The main idea of day and night revisited is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain day and night revisited.
2. Write two important facts about day and night revisited.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is it incorrect to say the dark half of Earth is always the same half?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: From space, the day-night boundary often appears curved because Earth is round. Near the poles, daylight or darkness can last for many weeks or months during parts of the year.

UNIT 4 · CHAPTER 6

Seasons Revisited

Learning outcomes

- Explain seasons revisited using simple scientific language.
- Use the words solstice and equinox correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

The seasonal cycle becomes clearer when we track sunlight angle, day length and opposite hemispheres. Earth's axis keeps nearly the same direction in space as the planet travels around the Sun.

Understanding the idea

Solstices occur when one hemisphere is tilted most toward or away from the Sun.

Equinoxes occur when neither hemisphere is tilted strongly toward the Sun and day and night are nearly equal in length worldwide.

Seasonal temperature changes lag behind daylight changes because land and oceans take time to warm or cool.

Local climates, monsoons, altitude and distance from oceans affect how seasons feel.

Use models carefully. A lamp, ball and globe can show relationships, but a model is smaller and simpler than the real system.

Do not learn solstice, equinox, climate separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

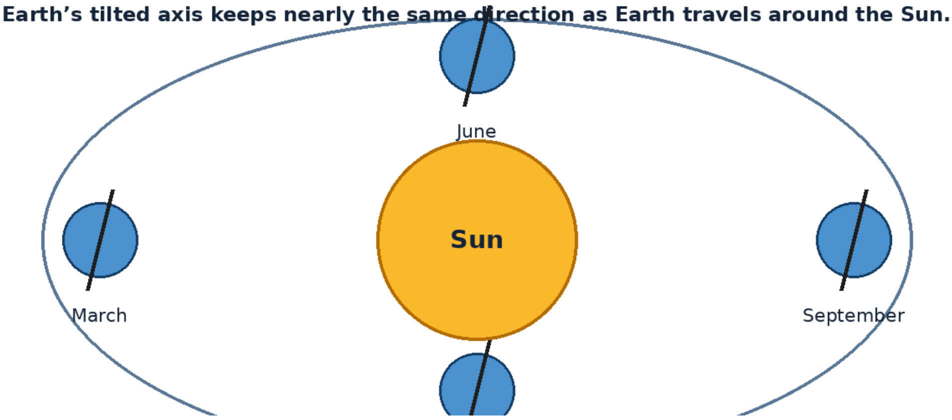


Figure 4.6 Earth travels around the Sun while its tilted axis keeps nearly the same direction in space.

Example

India's monsoon seasons are influenced by large-scale wind and heating patterns, so local seasonal experience is more complex than four equal seasons.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Measure the length of a fixed object's noon shadow once a week. Compare the pattern over several months.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Why can the warmest days occur after the longest day of the year?

What you have learnt

- Solstices occur when one hemisphere is tilted most toward or away from the Sun.
- Equinoxes occur when neither hemisphere is tilted strongly toward the Sun and day and night are nearly equal in length worldwide.
- Seasons involve changing sunlight angle and day length, modified by local climate.

EXERCISES · SEASONS REVISITED

Exercises

A. Choose the correct answer

1. Which statement about seasons revisited is correct?

A. A solstice happens every day at noon.
B. Solstices and equinoxes mark important points in Earth's seasonal cycle.
C. An equinox means the Sun stops shining.
D. Seasons are identical in every climate.

2. What does the word "solstice" mean in this lesson?

A. a time when neither hemisphere is strongly tilted toward the Sun
B. a seasonal wind system often linked with major rainfall changes
C. the usual long-term weather pattern of a place
D. a time when a hemisphere is most tilted toward or away from the Sun

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A pencil is sharpened.
C. India's monsoon seasons are influenced by large-scale wind and heating patterns, so local seasonal experience is more complex than four equal seasons.
D. A notebook is closed on a table.

4. Which word means "a time when a hemisphere is most tilted toward or away from the Sun"?

A. equinox
B. climate
C. solstice
D. monsoon

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Measure the length of a fixed object's noon shadow once a week. Compare the pattern over several months.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Seasons involve changing sunlight angle and day length, modified by local climate.
B. Seasons revisited can be understood without observation, evidence or comparison.
C. Every object or event connected with seasons revisited behaves in exactly the same way.
D. The main idea of seasons revisited is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain seasons revisited.
2. Write two important facts about seasons revisited.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can the warmest days occur after the longest day of the year?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: At an equinox, the Sun is directly overhead at the equator at local noon. The word equinox comes from words meaning "equal night," though day and night are not exactly twelve hours everywhere.

UNIT 4 · CHAPTER 7

Phases of the Moon

Learning outcomes

- Explain phases of the moon using simple scientific language.
- Use the words phase and waxing correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Moon phases are the changing shapes of the Moon's sunlit half that we see from Earth as the Moon orbits our planet. The Moon does not make its own visible light.

Understanding the idea

At all times, sunlight illuminates roughly half of the Moon, except during a lunar eclipse.

The main sequence is new Moon, waxing crescent, first quarter, waxing gibbous, full Moon, waning gibbous, third quarter and waning crescent.

The phase cycle takes about 29.5 days from one new Moon to the next.

Phases are not caused by Earth's shadow; Earth's shadow causes a lunar eclipse.

Use models carefully. A lamp, ball and globe can show relationships, but a model is smaller and simpler than the real system.

The vocabulary of this chapter - phase, waxing, waning - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

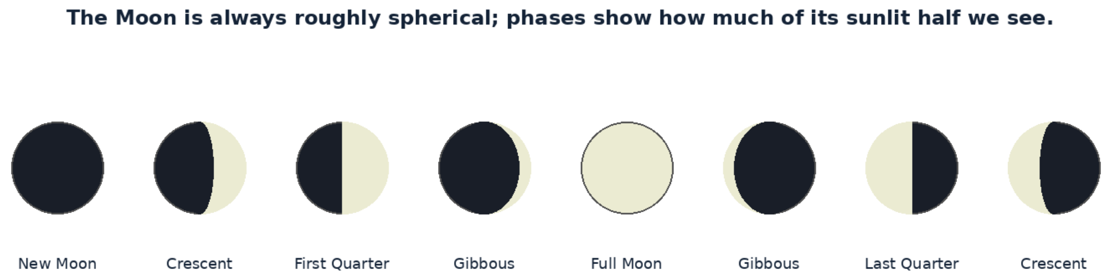


Figure 4.7 Moon phases show the changing portion of the Moon's sunlit half visible from Earth.

Example

At first quarter, we see half of the Moon's disc illuminated even though one quarter of the phase cycle has passed.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Use a lamp as the Sun and a ball as the Moon. Stand as Earth and move the ball around your head to view phases.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Why is there not a lunar eclipse at every full Moon?

What you have learnt

- At all times, sunlight illuminates roughly half of the Moon, except during a lunar eclipse.
- The main sequence is new Moon, waxing crescent, first quarter, waxing gibbous, full Moon, waning gibbous, third quarter and waning crescent.
- Moon phases are views of the Moon's sunlit half as it orbits Earth.

EXERCISES · PHASES OF THE MOON

Exercises

A. Choose the correct answer

1. Which statement about phases of the moon is correct?

A. The Moon makes more light during a full Moon.
B. Moon phases happen because clouds cover the Moon.
C. Earth's shadow causes every Moon phase.
D. Moon phases are caused by changing viewing geometry as the Moon orbits Earth.

2. What does the word "phase" mean in this lesson?

A. the illuminated part appears to shrink
B. a thin curved illuminated shape
C. the illuminated part appears to grow
D. the apparent shape of the Moon's illuminated part

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. At first quarter, we see half of the Moon's disc illuminated even though one quarter of the phase cycle has passed.
C. A pencil is sharpened.
D. A notebook is closed on a table.

4. Which word means "the apparent shape of the Moon's illuminated part"?

A. waxing
B. waning
C. crescent
D. phase

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Use a lamp as the Sun and a ball as the Moon. Stand as Earth and move the ball around your head to view phases.

6. Which sentence gives the best summary of this lesson?

A. Phases of the moon can be understood without observation, evidence or comparison.
B. Every object or event connected with phases of the moon behaves in exactly the same way.
C. Moon phases are views of the Moon's sunlit half as it orbits Earth.
D. The main idea of phases of the moon is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain phases of the moon.
2. Write two important facts about phases of the moon.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is there not a lunar eclipse at every full Moon?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: A full Moon rises near sunset and sets near sunrise. The line between the Moon's lit and dark parts is called the terminator, where shadows reveal surface relief.

UNIT 4 · CHAPTER 8

Solar and Lunar Eclipses

Learning outcomes

- Explain solar and lunar eclipses using simple scientific language.
- Use the words eclipse and solar eclipse correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

An eclipse occurs when one space object moves into the shadow of another or blocks our view of it. A solar eclipse involves the Moon blocking the Sun, while a lunar eclipse involves the Moon entering Earth's shadow.

Understanding the idea

A solar eclipse occurs near new Moon when the Sun, Moon and Earth align closely.

A lunar eclipse occurs near full Moon when the Sun, Earth and Moon align closely.

Eclipses do not occur every month because the Moon's orbit is tilted relative to Earth's orbit around the Sun.

Viewing the Sun requires certified eclipse glasses or approved indirect methods; ordinary sunglasses are unsafe.

Use models carefully. A lamp, ball and globe can show relationships, but a model is smaller and simpler than the real system.

The important words in this chapter include eclipse, solar eclipse, lunar eclipse. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

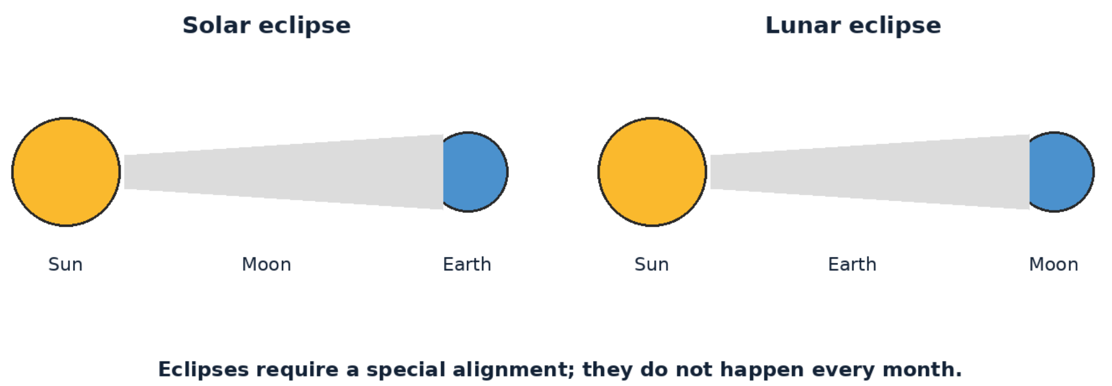


Figure 4.8 Solar and lunar eclipses occur when the Sun, Earth and Moon line up in particular ways.

Example

During a total lunar eclipse, the Moon can look reddish because Earth's atmosphere bends and filters sunlight into the shadow.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Model eclipses with a lamp and two balls. Never use the real Sun for this activity.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Why can more people usually see a lunar eclipse than a total solar eclipse?

What you have learnt

- A solar eclipse occurs near new Moon when the Sun, Moon and Earth align closely.
- A lunar eclipse occurs near full Moon when the Sun, Earth and Moon align closely.
- Solar and lunar eclipses require special alignments; safe Sun viewing is essential.

Exercises

A. Choose the correct answer

1. Which statement about solar and lunar eclipses is correct?

A. Ordinary sunglasses make direct Sun viewing safe.
B. A solar eclipse happens when the Moon passes between the Sun and Earth.
C. A lunar eclipse happens when Venus blocks the Moon.
D. A solar eclipse occurs at every full Moon.

2. What does the word "eclipse" mean in this lesson?

A. the Moon blocks some or all of the Sun from view
B. a nearly straight arrangement of objects
C. the Moon passes through Earth's shadow
D. an event in which one object blocks light or enters a shadow

3. Which observation or example best supports the lesson idea?

A. During a total lunar eclipse, the Moon can look reddish because Earth's atmosphere bends and filters sunlight into the shadow.
B. A pencil is sharpened.
C. A glass is filled with water.
D. A notebook is closed on a table.

4. Which word means "an event in which one object blocks light or enters a shadow"?

A. eclipse
B. solar eclipse
C. lunar eclipse
D. alignment

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Model eclipses with a lamp and two balls. Never use the real Sun for this activity.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Solar and lunar eclipses require special alignments; safe Sun viewing is essential.
B. Solar and lunar eclipses can be understood without observation, evidence or comparison.
C. Every object or event connected with solar and lunar eclipses behaves in exactly the same way.
D. The main idea of solar and lunar eclipses is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain solar and lunar eclipses.
2. Write two important facts about solar and lunar eclipses.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can more people usually see a lunar eclipse than a total solar eclipse?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: A total solar eclipse is visible only along a narrow path, while a lunar eclipse can be seen from much of Earth's night side. The reddish lunar-eclipse colour is related to the same scattering that makes many sunsets red.

UNIT 4 · CHAPTER 9

Tides

Learning outcomes

- Explain tides using simple scientific language.
- Use the words tide and high tide correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Tides are regular rises and falls of sea level caused mainly by the Moon's gravity, with an additional effect from the Sun. Earth's rotation carries coastlines through regions of higher and lower water level.

Understanding the idea

The Moon pulls more strongly on the side of Earth closer to it than on the far side, helping create two tidal bulges.

Many coasts experience about two high tides and two low tides each day, but local geography can change the pattern.

Spring tides have a larger range and occur near new and full Moon when Sun, Earth and Moon are aligned.

Neap tides have a smaller range and occur near first and third quarter phases.

Use models carefully. A lamp, ball and globe can show relationships, but a model is smaller and simpler than the real system.

Notice how the terms tide, high tide, spring tide appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

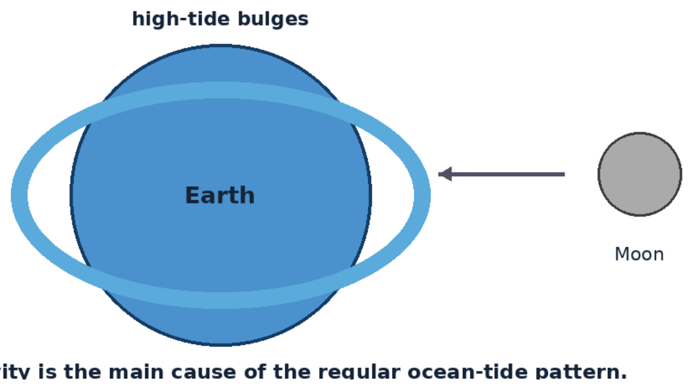


Figure 4.9 The Moon's gravity is the main cause of the regular pattern of ocean tides.

Example

A narrow bay may have a much larger tidal range than an open coast because the shape of the shoreline affects water movement.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Use a tide table for an Indian coastal city and circle high and low tide times for one day.

Think and discuss

Why can two beaches have different tide heights even on the same day?

What you have learnt

- The Moon pulls more strongly on the side of Earth closer to it than on the far side, helping create two tidal bulges.
- Many coasts experience about two high tides and two low tides each day, but local geography can change the pattern.
- Tides result mainly from the Moon's gravity, modified by the Sun and local coastline shape.

Exercises

A. Choose the correct answer

1. Which statement about tides is correct?

A. The Moon's gravity is the main cause of ocean tides on Earth.
B. Spring tides happen only during the spring season.
C. The Sun has no effect on tides.
D. Tides are caused only by ocean winds.

2. What does the word "tide" mean in this lesson?

A. the regular rise and fall of sea level
B. a tide with a smaller-than-usual range near quarter Moon
C. a time when sea level is relatively high
D. a tide with a larger-than-usual range near new or full Moon

3. Which observation or example best supports the lesson idea?

A. A narrow bay may have a much larger tidal range than an open coast because the shape of the shoreline affects water movement.
B. A notebook is closed on a table.
C. A glass is filled with water.
D. A pencil is sharpened.

4. Which word means "the regular rise and fall of sea level"?

A. high tide
B. tide
C. spring tide
D. neap tide

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Use a tide table for an Indian coastal city and circle high and low tide times for one day.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Tides can be understood without observation, evidence or comparison.
B. Every object or event connected with tides behaves in exactly the same way.
C. Tides result mainly from the Moon's gravity, modified by the Sun and local coastline shape.
D. The main idea of tides is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain tides.
2. Write two important facts about tides.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can two beaches have different tide heights even on the same day?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The word “spring” in spring tide means “to rise up” and is not connected with the season of spring. The Sun is far more massive than the Moon, but the Moon has a stronger tide-raising effect on Earth because it is much closer.

UNIT 4 REVIEW

Review · Earth & Moon

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of earth: our home planet using one example.
2. Why can a flat map never show a spherical Earth without some distortion?
3. Write one correct statement and one common mistake about earth’s rotation.
4. Explain the main idea of earth’s revolution using one example.
5. Why is it incorrect to say the dark half of Earth is always the same half?
6. Write one correct statement and one common mistake about seasons revisited.
7. Explain the main idea of phases of the moon using one example.
8. Why can more people usually see a lunar eclipse than a total solar eclipse?

B. Compare and connect

1. Write one connection between “Earth: Our Home Planet” and “Tides”.
2. Write one connection between “Continents and Oceans” and “Solar and Lunar Eclipses”.

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 4 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about earth: our home planet is correct?
   A. Earth is the largest planet in the solar system.
   B. Earth is the third planet from the Sun and the only known world with life.
   C. Earth has no atmosphere or liquid water.
   D. Earth is the closest planet to the Sun.

2. Which statement about continents and oceans is correct?
   A. The Arctic is a continent and Antarctica is an ocean.
   B. Earth’s oceans are completely separate from one another.
   C. Earth is commonly described as having seven continents and five oceans.
   D. India is located in South America.

3. Which statement about earth’s rotation is correct?
   A. Rotation causes the Moon’s phases.
   B. Earth rotates from west to east once in about 24 hours.
   C. Earth rotates once every 365 years.
   D. Earth rotates from east to west.

4. Which statement about earth’s revolution is correct?
   A. Earth completes one revolution every 24 hours.
   B. Earth’s orbit is centred on the Moon.
   C. Revolution means spinning on an axis.
   D. Earth completes one revolution around the Sun in about 365.25 days.

5. Which statement about day and night revisited is correct?
   A. The moving day-night boundary crosses Earth as the planet rotates.

B. Every city experiences noon at the same moment.
C. The same half of Earth is permanently dark.
D. Sunrise is caused by the Sun moving around Earth.

6. Which statement about seasons revisited is correct?

A. A solstice happens every day at noon.
B. Solstices and equinoxes mark important points in Earth’s seasonal cycle.
C. An equinox means the Sun stops shining.
D. Seasons are identical in every climate.

7. Which statement about phases of the moon is correct?

A. The Moon makes more light during a full Moon.
B. Moon phases happen because clouds cover the Moon.
C. Earth’s shadow causes every Moon phase.
D. Moon phases are caused by changing viewing geometry as the Moon orbits Earth.

8. Which statement about solar and lunar eclipses is correct?

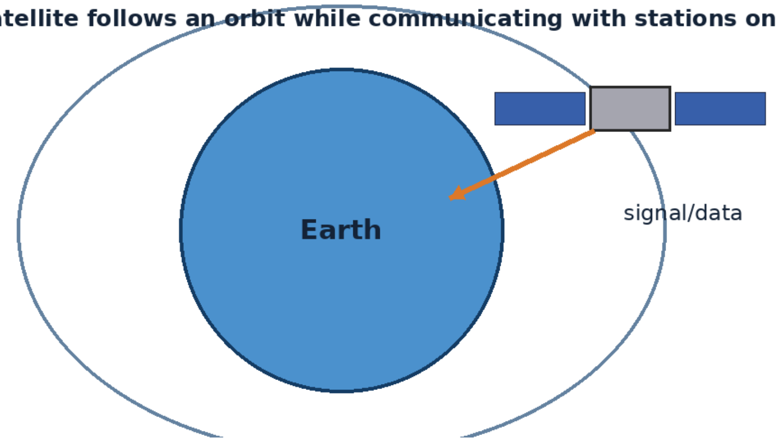
A. Ordinary sunglasses make direct Sun viewing safe.
B. A solar eclipse happens when the Moon passes between the Sun and Earth.
C. A lunar eclipse happens when Venus blocks the Moon.
D. A solar eclipse occurs at every full Moon.

UNIT 5

Satellites & Space Technology

Satellites are machines that extend human sight and communication. This unit explains how artificial satellites support weather forecasts, navigation, television, disaster response and scientific observation.

An artificial satellite follows an orbit while communicating with stations on Earth.



Chapters in this unit

1. **Natural and Artificial Satellites**
2. **Communication Satellites**
3. **Weather Satellites**
4. **GPS and Navigation**
5. **Satellites in Everyday Life**

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 5 · CHAPTER 1

Natural and Artificial Satellites

Learning outcomes

- Explain natural and artificial satellites using simple scientific language.
- Use the words satellite and artificial satellite correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A satellite is an object that moves around a larger object. Natural satellites form in nature, while artificial satellites are designed and launched by people for specific purposes.

Understanding the idea

The Moon is Earth’s natural satellite; communication, navigation and weather spacecraft are artificial satellites.

Artificial satellites carry a main body called a bus and useful instruments called payloads.

Satellites need power, communication, temperature control, guidance and a suitable orbit.

A satellite remains in orbit because gravity bends its forward-moving path around Earth.

Follow the path of information: a satellite observes or receives a signal, equipment processes it, and people use the result on Earth.

Do not learn satellite, artificial satellite, payload separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

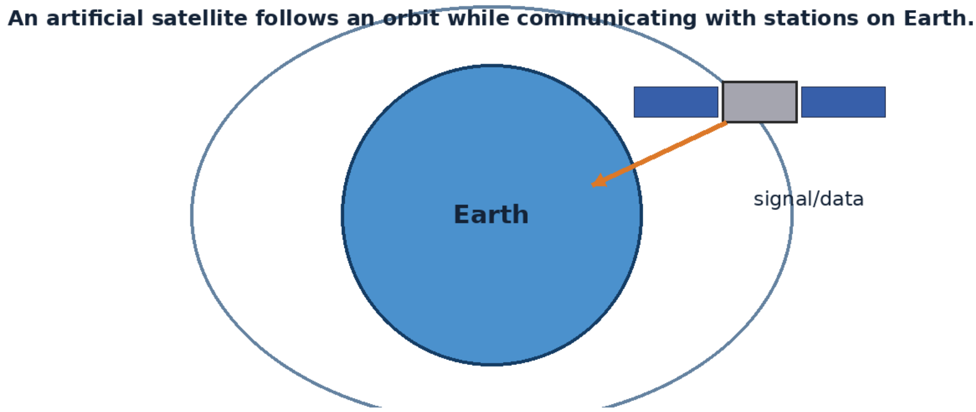


Figure 5.1 An artificial satellite follows an orbit and sends or receives information.

Example

A weather satellite’s payload may include cameras and sensors, while its bus supplies power and controls pointing.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Build a paper satellite and label which parts provide power, communicate and collect information.

Think and discuss

Why must an artificial satellite carry both a payload and supporting systems?

What you have learnt

- The Moon is Earth's natural satellite; communication, navigation and weather spacecraft are artificial satellites.
- Artificial satellites carry a main body called a bus and useful instruments called payloads.
- Natural satellites are moons; artificial satellites are human-made machines placed in orbit.

EXERCISES · NATURAL AND ARTIFICIAL SATELLITES

Exercises

A. Choose the correct answer

1. Which statement about natural and artificial satellites is correct?

A. Artificial satellites are human-made spacecraft placed in orbit for specific tasks.
B. Artificial satellites can work without power or communication.
C. A satellite stays up because gravity disappears in space.
D. Every satellite is a natural moon.

2. What does the word "satellite" mean in this lesson?

A. an object that moves around a larger body
B. the supporting structure and systems of a spacecraft
C. the instruments or equipment that perform a mission
D. a human-made object placed in orbit

3. Which observation or example best supports the lesson idea?

A. A weather satellite's payload may include cameras and sensors, while its bus supplies power and controls pointing.
B. A glass is filled with water.
C. A pencil is sharpened.
D. A notebook is closed on a table.

4. Which word means "an object that moves around a larger body"?

A. artificial satellite
B. payload
C. satellite
D. satellite bus

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Build a paper satellite and label which parts provide power, communicate and collect information.

6. Which sentence gives the best summary of this lesson?

A. Natural and artificial satellites can be understood without observation, evidence or comparison.
B. Every object or event connected with natural and artificial satellites behaves in exactly the same way.
C. The main idea of natural and artificial satellites is unrelated to science or everyday experience.
D. Natural satellites are moons; artificial satellites are human-made machines placed in orbit.

B. Answer in one or two sentences

1. In one sentence, explain natural and artificial satellites.
2. Write two important facts about natural and artificial satellites.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why must an artificial satellite carry both a payload and supporting systems?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The first artificial satellite, Sputnik 1, was launched in 1957. Some tiny satellites called CubeSats can be built from standard box-shaped units.

UNIT 5 · CHAPTER 2

Communication Satellites

Learning outcomes

- Explain communication satellites using simple scientific language.
- Use the words radio wave and uplink correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Communication satellites receive signals from one location, process or strengthen them and send them to another location. They help connect places separated by oceans, mountains or long distances.

Understanding the idea

Signals travel between ground antennas and satellite antennas using radio waves.

Many communication satellites use geostationary orbit, appearing almost fixed above the equator.

Satellites support television, telephone, internet links, emergency communication and remote education.

A communication link includes an uplink from Earth and a downlink back to Earth.

Follow the path of information: a satellite observes or receives a signal, equipment processes it, and people use the result on Earth.

The vocabulary of this chapter - radio wave, uplink, downlink - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

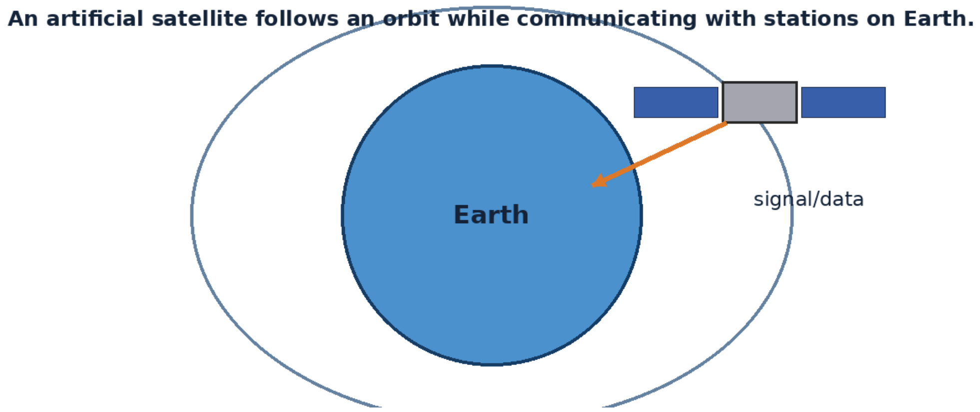


Figure 5.2 An artificial satellite follows an orbit and sends or receives information.

Example

A live television event can be sent from one city to a satellite and then down to viewers across a wide region.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Use three students as transmitter, satellite and receiver. Pass a written message through the "satellite" and discuss accuracy.

Think and discuss

Why is a satellite especially useful for connecting remote islands or mountain communities?

What you have learnt

- Signals travel between ground antennas and satellite antennas using radio waves.
- Many communication satellites use geostationary orbit, appearing almost fixed above the equator.
- Communication satellites relay radio signals over large areas.

EXERCISES · COMMUNICATION SATELLITES

Exercises

A. Choose the correct answer

1. Which statement about communication satellites is correct?

A. Communication satellites carry sound through ordinary air across space.
B. A downlink travels from Earth up to a satellite.
C. Communication satellites relay signals between distant ground locations.
D. Geostationary satellites hover without orbiting.

2. What does the word "radio wave" mean in this lesson?

A. a type of electromagnetic wave used for communication
B. an orbit where a satellite appears fixed above one point on the equator
C. a signal sent from a satellite to Earth
D. a signal sent from Earth to a satellite

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A pencil is sharpened.
C. A notebook is closed on a table.
D. A live television event can be sent from one city to a satellite and then down to viewers across a wide region.

4. Which word means "a type of electromagnetic wave used for communication"?

A. uplink
B. downlink
C. geostationary orbit
D. radio wave

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Use three students as transmitter, satellite and receiver. Pass a written message through the "satellite" and discuss accuracy.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Communication satellites can be understood without observation, evidence or comparison.
B. Communication satellites relay radio signals over large areas.
C. Every object or event connected with communication satellites behaves in exactly the same way.
D. The main idea of communication satellites is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain communication satellites.
2. Write two important facts about communication satellites.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is a satellite especially useful for connecting remote islands or mountain communities?

2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: A geostationary satellite orbits once in about 24 hours in the same direction as Earth rotates. Satellites can restore communication after disasters damage ground networks.

UNIT 5 · CHAPTER 3

Weather Satellites

Learning outcomes

- Explain weather satellites using simple scientific language.
- Use the words meteorologist and geostationary correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Weather satellites observe clouds, water vapour, temperatures, storms and other features from space. Their measurements help meteorologists understand current weather and improve forecasts.

Understanding the idea

Geostationary weather satellites watch the same broad region continuously.

Polar-orbiting satellites pass over different parts of Earth and can collect detailed global measurements.

Satellite images help track cyclones, monsoon clouds, dust, smoke, fog and rainfall systems.

Forecasts combine satellite data with ground observations, weather balloons, radar and computer models.

Follow the path of information: a satellite observes or receives a signal, equipment processes it, and people use the result on Earth.

The important words in this chapter include meteorologist, geostationary, polar orbit. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

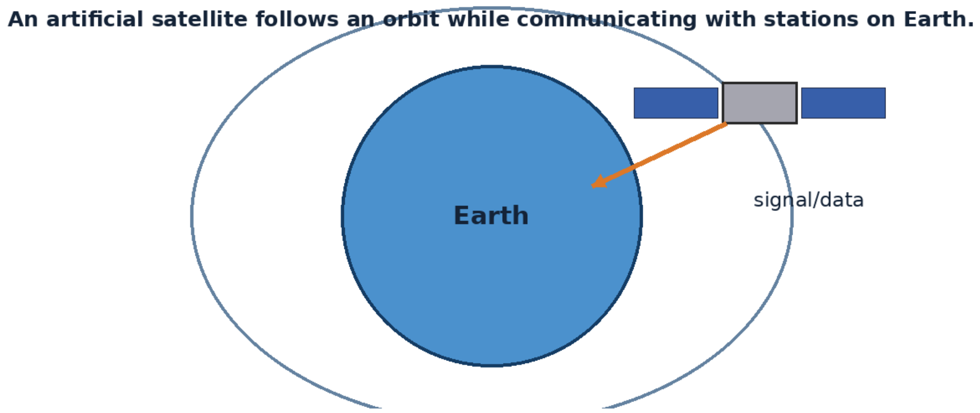


Figure 5.3 An artificial satellite follows an orbit and sends or receives information.

Example

Repeated satellite images show a cyclone's movement and changing cloud structure, helping authorities issue warnings.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Compare two weather images taken several hours apart. Draw an arrow showing the direction a cloud system moved.

Think and discuss

Why is one satellite image less useful than a sequence of images for tracking a storm?

What you have learnt

- Geostationary weather satellites watch the same broad region continuously.
- Polar-orbiting satellites pass over different parts of Earth and can collect detailed global measurements.
- Weather satellites provide repeated, wide-area observations used with other data to make forecasts.

Exercises

A. Choose the correct answer

1. Which statement about weather satellites is correct?

A. Weather satellites help monitor clouds and storms from space.
B. Forecasts use only one photograph and no other data.
C. Weather satellites create all weather on Earth.
D. A polar satellite remains fixed over one city.

2. What does the word “meteorologist” mean in this lesson?

A. appearing fixed over one region of Earth
B. a scientist who studies weather and the atmosphere
C. a scientific prediction of future weather
D. an orbit that passes near both poles

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A pencil is sharpened.
C. Repeated satellite images show a cyclone’s movement and changing cloud structure, helping authorities issue warnings.
D. A glass is filled with water.

4. Which word means “a scientist who studies weather and the atmosphere”?

A. meteorologist
B. geostationary
C. polar orbit
D. forecast

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Compare two weather images taken several hours apart. Draw an arrow showing the direction a cloud system moved.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Weather satellites can be understood without observation, evidence or comparison.
B. Every object or event connected with weather satellites behaves in exactly the same way.
C. The main idea of weather satellites is unrelated to science or everyday experience.
D. Weather satellites provide repeated, wide-area observations used with other data to make forecasts.

B. Answer in one or two sentences

1. In one sentence, explain weather satellites.

2. Write two important facts about weather satellites.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is one satellite image less useful than a sequence of images for tracking a storm?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: India's INSAT series has provided important meteorological observations and warning support. Some weather satellites measure infrared energy, allowing cloud observations at night.

UNIT 5 · CHAPTER 4

GPS and Navigation

Learning outcomes

- Explain gps and navigation using simple scientific language.
- Use the words navigation and GPS correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Satellite navigation helps a receiver find its position by measuring precisely timed signals from several satellites. GPS is one global system; India also operates the regional NavIC system.

Understanding the idea

Navigation satellites carry very accurate clocks and broadcast their position and time.

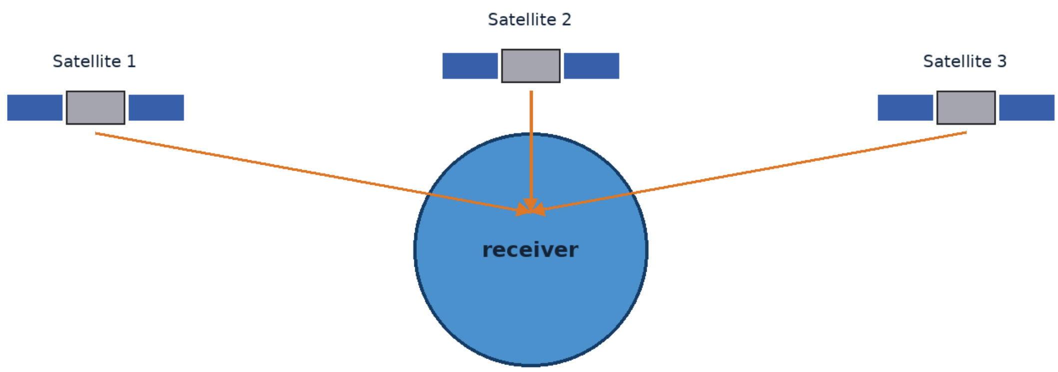
A receiver compares signal arrival times to estimate distance from multiple satellites.

Using at least four suitable satellite signals allows a receiver to solve for three-dimensional position and clock error.

Navigation supports maps, transport, farming, emergency response, timing and scientific measurements.

Follow the path of information: a satellite observes or receives a signal, equipment processes it, and people use the result on Earth.

Notice how the terms navigation, GPS, NavIC appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.



A receiver compares timed signals from several satellites to calculate position.

Figure 5.4 A navigation receiver compares timed signals from several satellites to calculate position.

Example

A phone map combines satellite positioning with digital maps and often with mobile-network or Wi-Fi information.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Hide an object and give a partner distances from three classroom landmarks. Discuss how multiple distances narrow the location.

Think and discuss

Why is one satellite signal not enough to identify a unique position on Earth?

What you have learnt

- Navigation satellites carry very accurate clocks and broadcast their position and time.
- A receiver compares signal arrival times to estimate distance from multiple satellites.
- Navigation receivers calculate position from precisely timed signals sent by multiple satellites.

EXERCISES · GPS AND NAVIGATION

Exercises

A. Choose the correct answer

1. Which statement about gps and navigation is correct?

A. A navigation receiver uses timed signals from several satellites to calculate position.
B. A phone finds position because one satellite takes a photograph of the user.
C. Navigation needs no accurate timing.
D. GPS works using sound waves through space.

2. What does the word "navigation" mean in this lesson?

A. finding position and choosing a route
B. a device that detects and uses incoming signals
C. the United States global satellite navigation system
D. India's regional satellite navigation system

3. Which observation or example best supports the lesson idea?

A. A phone map combines satellite positioning with digital maps and often with mobile-network or Wi-Fi information.
B. A pencil is sharpened.
C. A glass is filled with water.
D. A notebook is closed on a table.

4. Which word means "finding position and choosing a route"?

A. GPS
B. navigation
C. NavIC
D. receiver

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Hide an object and give a partner distances from three classroom landmarks. Discuss how multiple distances narrow the location.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Gps and navigation can be understood without observation, evidence or comparison.
B. Navigation receivers calculate position from precisely timed signals sent by multiple satellites.
C. Every object or event connected with gps and navigation behaves in exactly the same way.
D. The main idea of gps and navigation is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain gps and navigation.
2. Write two important facts about gps and navigation.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is one satellite signal not enough to identify a unique position on Earth?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Satellite navigation is mainly a one-way service: the receiver listens to satellite signals and does not need to tell every satellite where it is. NavIC was developed by ISRO to provide positioning services over India and a surrounding region.

UNIT 5 · CHAPTER 5

Satellites in Everyday Life

Learning outcomes

- Explain satellites in everyday life using simple scientific language.
- Use the words remote sensing and Earth observation correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Satellites quietly support many daily activities. They help communicate, navigate, observe Earth, monitor disasters, study climate and provide accurate timing.

Understanding the idea

Earth-observation satellites map farms, forests, water, cities, coasts and disaster-affected areas.

Communication satellites support television, remote links and services in difficult terrain.

Navigation satellites guide vehicles and also provide timing used by networks and power systems.

Scientific satellites study Earth, the Sun and the universe, turning space measurements into useful knowledge.

Follow the path of information: a satellite observes or receives a signal, equipment processes it, and people use the result on Earth.

Do not learn remote sensing, Earth observation, disaster monitoring separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

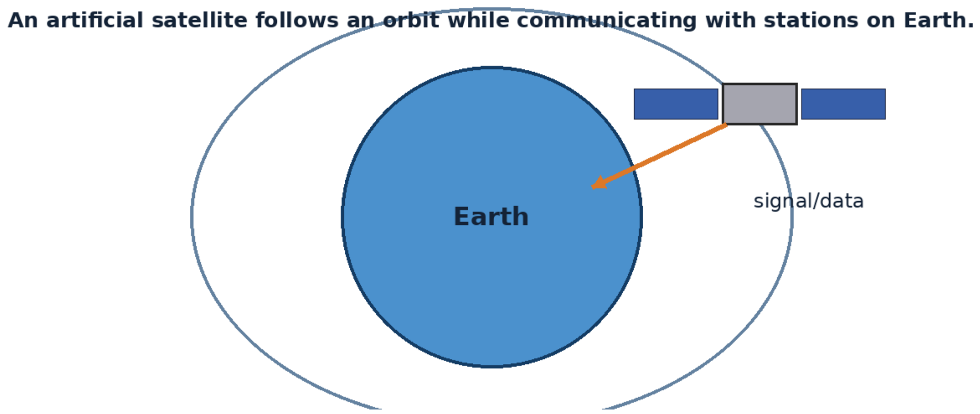


Figure 5.5 An artificial satellite follows an orbit and sends or receives information.

Example

During flooding, satellite images can help map water-covered areas even where roads are blocked.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Keep a one-day “satellite diary.” Note each time you use a map, forecast, broadcast or service that may depend on satellite timing.

Think and discuss

Which satellite service would be most important during a cyclone warning, and why?

What you have learnt

- Earth-observation satellites map farms, forests, water, cities, coasts and disaster-affected areas.
- Communication satellites support television, remote links and services in difficult terrain.
- Satellites provide information and connections that support life on Earth.

EXERCISES · SATELLITES IN EVERYDAY LIFE

Exercises

A. Choose the correct answer

1. Which statement about satellites in everyday life is correct?

A. Satellite data cannot help people on Earth.
B. Satellites support communication, navigation, weather, mapping and disaster response.
C. Satellites are useful only to astronauts.
D. Every satellite has exactly the same mission.

2. What does the word “remote sensing” mean in this lesson?

A. precise measurement and distribution of time
B. observing dangerous events to support warnings and response
C. measuring and imaging Earth from aircraft or spacecraft
D. collecting information about an object or area from a distance

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. During flooding, satellite images can help map water-covered areas even where roads are blocked.
C. A notebook is closed on a table.
D. A pencil is sharpened.

4. Which word means “collecting information about an object or area from a distance”?

A. Earth observation
B. disaster monitoring
C. remote sensing
D. timing

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Keep a one-day “satellite diary.” Note each time you use a map, forecast, broadcast or service that may depend on satellite timing.

6. Which sentence gives the best summary of this lesson?

A. Satellites in everyday life can be understood without observation, evidence or comparison.
B. Every object or event connected with satellites in everyday life behaves in exactly the same way.

C. The main idea of satellites in everyday life is unrelated to science or everyday experience.
D. Satellites provide information and connections that support life on Earth.

B. Answer in one or two sentences

1. In one sentence, explain satellites in everyday life.
2. Write two important facts about satellites in everyday life.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Which satellite service would be most important during a cyclone warning, and why?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Satellite measurements can show slow changes such as glacier movement, shoreline change and forest loss. Remote sensing means learning about a place without touching it directly, often by measuring reflected or emitted energy.

UNIT 5 REVIEW

Review · Satellites & Space Technology

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of natural and artificial satellites using one example.
2. Why is a satellite especially useful for connecting remote islands or mountain communities?
3. Write one correct statement and one common mistake about weather satellites.
4. Explain the main idea of gps and navigation using one example.
5. Which satellite service would be most important during a cyclone warning, and why?
6. Create one new question from this unit and write its answer.

B. Compare and connect

1. Write one connection between “Natural and Artificial Satellites” and “Satellites in Everyday Life”.
2. Write one connection between “Communication Satellites” and “GPS and Navigation”.

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 5 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about natural and artificial satellites is correct?
A. Artificial satellites are human-made spacecraft placed in orbit for specific tasks.
B. Artificial satellites can work without power or communication.
C. A satellite stays up because gravity disappears in space.
D. Every satellite is a natural moon.

2. Which statement about communication satellites is correct?
A. Communication satellites carry sound through ordinary air across space.
B. A downlink travels from Earth up to a satellite.
C. Communication satellites relay signals between distant ground locations.
D. Geostationary satellites hover without orbiting.

3. Which statement about weather satellites is correct?
A. Weather satellites help monitor clouds and storms from space.
B. Forecasts use only one photograph and no other data.

C. Weather satellites create all weather on Earth.
D. A polar satellite remains fixed over one city.

4. Which statement about gps and navigation is correct?

A. A navigation receiver uses timed signals from several satellites to calculate position.
B. A phone finds position because one satellite takes a photograph of the user.
C. Navigation needs no accurate timing.
D. GPS works using sound waves through space.

5. Which statement about satellites in everyday life is correct?

A. Satellite data cannot help people on Earth.
B. Satellites support communication, navigation, weather, mapping and disaster response.
C. Satellites are useful only to astronauts.
D. Every satellite has exactly the same mission.

6. Which word means “an object that moves around a larger body”?

A. artificial satellite
B. payload
C. satellite
D. satellite bus

7. Which word means “a type of electromagnetic wave used for communication”?

A. uplink
B. downlink
C. geostationary orbit
D. radio wave

8. Which word means “a scientist who studies weather and the atmosphere”?

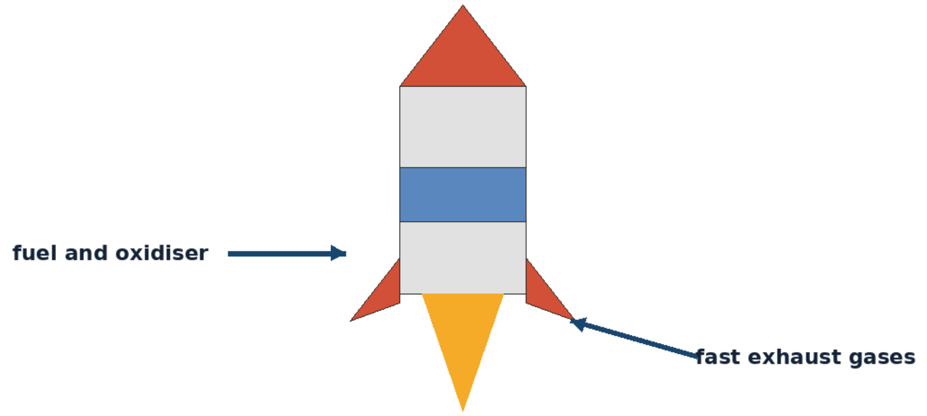
A. meteorologist
B. geostationary
C. polar orbit
D. forecast

UNIT 6

Rockets & Launch Vehicles

A rocket is a carefully designed moving system, not simply a tube filled with fuel. This unit introduces thrust, stages, launch sites, astronauts and the idea of reusing launch hardware.

A rocket moves forward as high-speed gases are pushed backward.



Chapters in this unit

1. **What is a Rocket?**
2. **How Rockets Reach Space**
3. **Launch Pads**
4. **Astronauts**
5. **Reusable Rockets**

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 6 · CHAPTER 1

What is a Rocket?

Learning outcomes

- Explain what a Rocket is using simple scientific language.
- Use the words rocket and thrust correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A rocket is a vehicle or engine that produces thrust by pushing exhaust in one direction so the rocket accelerates in the opposite direction. Rockets can work in space because they carry both fuel and an oxidiser.

Understanding the idea

A rocket engine turns stored chemical energy into fast-moving hot exhaust.

The backward exhaust and forward rocket motion demonstrate Newton's third law of motion.

A launch vehicle is a complete rocket system used to carry a payload into space.

Rockets must be strong, light, controllable and able to survive vibration, heating and pressure changes.

Look for cause and effect. Expelled gases produce thrust, stages reduce unnecessary mass, and guidance keeps the vehicle on the planned path.

The vocabulary of this chapter - rocket, thrust, exhaust - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

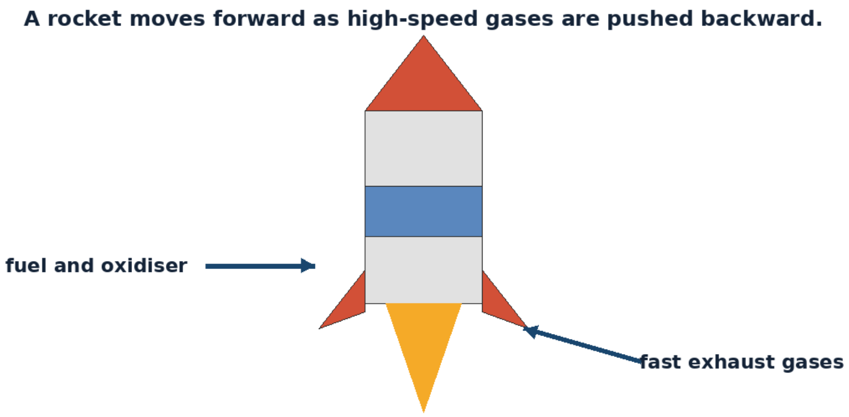


Figure 6.1 A rocket gains thrust by pushing high-speed exhaust gases backward.

Example

An untied balloon rushes forward as air escapes backward, although a real rocket works in a far more controlled and powerful way.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Inflate a balloon, release it safely and observe motion opposite the escaping air. Do not attach hard or sharp objects.

Think and discuss

Why can a rocket continue to fire above the atmosphere where there is almost no air?

What you have learnt

- A rocket engine turns stored chemical energy into fast-moving hot exhaust.
- The backward exhaust and forward rocket motion demonstrate Newton’s third law of motion.
- Rockets move forward by expelling mass backward at high speed.

Exercises

A. Choose the correct answer

1. Which statement about what is a rocket? is correct?

A. A rocket is pulled upward by the Moon.
B. A rocket works only by pushing against air.
C. Rockets need wings to flap in space.
D. A rocket produces thrust by pushing exhaust backward.

2. What does the word “rocket” mean in this lesson?

A. a chemical that allows fuel to burn without using outside air
B. the pushing force that accelerates a rocket
C. a vehicle or engine that produces thrust by expelling material
D. hot gases expelled by an engine

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A pencil is sharpened.
C. A notebook is closed on a table.
D. An untied balloon rushes forward as air escapes backward, although a real rocket works in a far more controlled and powerful way.

4. Which word means “a vehicle or engine that produces thrust by expelling material”?

A. thrust
B. exhaust
C. oxidiser
D. rocket

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Inflate a balloon, release it safely and observe motion opposite the escaping air. Do not attach hard or sharp objects.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Rockets move forward by expelling mass backward at high speed.
B. A rocket can be understood without observation, evidence or comparison.
C. Every object or event connected with a rocket behaves in exactly the same way.
D. The main idea of a rocket is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain what is a rocket?
2. Write two important facts about a rocket
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can a rocket continue to fire above the atmosphere where there is almost no air?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Rockets do not push against the ground or air; they move by accelerating exhaust, so they work in a vacuum. The word rocket can describe a small engine, a missile or a large space launch vehicle, depending on context.

UNIT 6 · CHAPTER 2

How Rockets Reach Space

Learning outcomes

- Explain how rockets reach space using simple scientific language.
- Use the words orbit and drag correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Reaching space requires more than going high. A spacecraft must gain enough sideways speed for gravity to curve its path around Earth, creating orbit.

Understanding the idea

At liftoff, thrust must exceed the rocket's weight to begin accelerating upward.

Rockets usually tilt gradually to build horizontal speed as they climb.

Atmospheric drag decreases with altitude, but gravity continues to act.

For low Earth orbit, a spacecraft travels at roughly 7.8 kilometres per second, depending on altitude.

Look for cause and effect. Expelled gases produce thrust, stages reduce unnecessary mass, and guidance keeps the vehicle on the planned path.

The important words in this chapter include orbit, drag, velocity. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

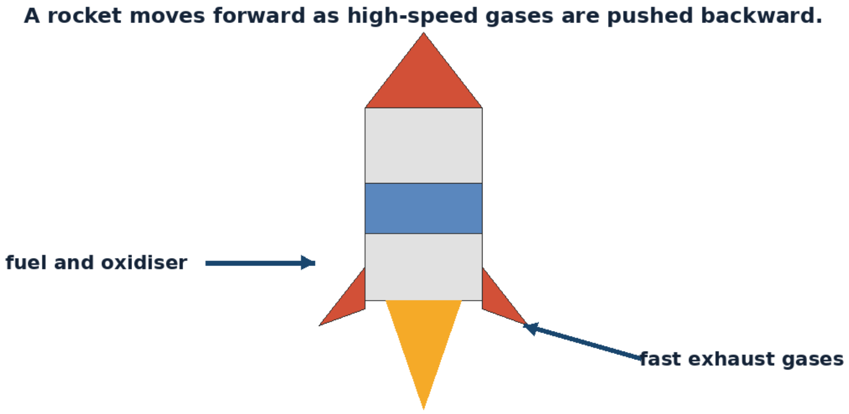


Figure 6.2 A rocket gains thrust by pushing high-speed exhaust gases backward.

Example

Throwing a ball faster makes it travel farther before falling. An orbiting spacecraft is moving so fast sideways that Earth curves away beneath it.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Roll a marble at different speeds along a curved bowl to discuss how speed changes its path. This is only an analogy.

Think and discuss

Why does an orbiting spacecraft not simply fly straight away from Earth?

What you have learnt

- At liftoff, thrust must exceed the rocket's weight to begin accelerating upward.
- Rockets usually tilt gradually to build horizontal speed as they climb.
- Orbit requires high sideways speed while gravity continually bends the path.

Exercises

A. Choose the correct answer

1. Which statement about how rockets reach space is correct?

A. A rocket must point straight upward for the whole flight.
B. A spacecraft reaches orbit by gaining very high sideways speed.
C. Gravity stops completely at the edge of space.
D. Orbit is achieved only by rising above clouds.

2. What does the word "orbit" mean in this lesson?

A. a curved path around another body
B. resistance caused by moving through air or another fluid
C. speed in a particular direction
D. reaching space without completing an orbit

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. Throwing a ball faster makes it travel farther before falling. An orbiting spacecraft is moving so fast sideways that Earth curves away beneath it.
C. A glass is filled with water.
D. A pencil is sharpened.

4. Which word means "a curved path around another body"?

A. orbit
B. drag
C. velocity
D. suborbital

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Roll a marble at different speeds along a curved bowl to discuss how speed changes its path. This is only an analogy.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. How rockets reach space can be understood without observation, evidence or comparison.
B. Every object or event connected with how rockets reach space behaves in exactly the same way.
C. Orbit requires high sideways speed while gravity continually bends the path.
D. The main idea of how rockets reach space is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain how rockets reach space.
2. Write two important facts about how rockets reach space.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why does an orbiting spacecraft not simply fly straight away from Earth?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Most of a launch vehicle's energy is used to gain speed rather than simply to gain height. An object can briefly enter space on a suborbital path and fall back without completing an orbit.

UNIT 6 · CHAPTER 3

Launch Pads

Learning outcomes

- Explain launch pads using simple scientific language.
- Use the words launch pad and flame trench correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A launch pad is a specially designed site where a rocket is assembled, checked, fuelled and launched. It includes structures and systems for safety, support and control.

Understanding the idea

Launch towers provide access, cables, pipes and sometimes support arms before liftoff.

Flame trenches and water-deluge systems manage heat, sound and exhaust energy.

Tracking stations and mission control monitor the rocket and can respond to problems.

Launch sites are chosen using safety, geography, weather, access and desired orbit.

Look for cause and effect. Expelled gases produce thrust, stages reduce unnecessary mass, and guidance keeps the vehicle on the planned path.

Notice how the terms launch pad, flame trench, mission control appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

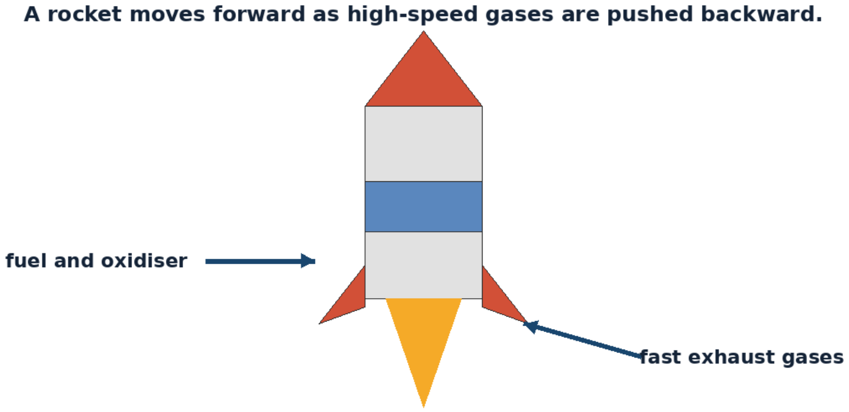


Figure 6.3 A rocket gains thrust by pushing high-speed exhaust gases backward.

Example

Sriharikota on India's east coast allows many launches over the Bay of Bengal, reducing risk to populated areas.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Design a paper launch complex map showing safe viewing area, control centre, road, pad and no-entry zone.

Think and discuss

Why are launch pads surrounded by large restricted areas?

What you have learnt

- Launch towers provide access, cables, pipes and sometimes support arms before liftoff.
- Flame trenches and water-deluge systems manage heat, sound and exhaust energy.
- A launch pad is a complete safety and support system, not just a flat platform.

EXERCISES · LAUNCH PADS

Exercises

A. Choose the correct answer

1. Which statement about launch pads is correct?

A. A launch pad is only a painted circle on the ground.
B. Rockets need no checks before launch.
C. Launch sites are placed inside crowded city centres.
D. Launch pads include systems for support, safety, exhaust control and monitoring.

2. What does the word "launch pad" mean in this lesson?

A. a channel that directs hot exhaust away
B. the team and facility that monitor and manage a mission
C. the prepared site from which a rocket launches
D. an area kept clear to reduce risk

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. Sriharikota on India's east coast allows many launches over the Bay of Bengal, reducing risk to populated areas.
C. A pencil is sharpened.
D. A glass is filled with water.

4. Which word means "the prepared site from which a rocket launches"?

A. flame trench
B. launch pad
C. mission control
D. safe zone

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Design a paper launch complex map showing safe viewing area, control centre, road, pad and no-entry zone.

6. Which sentence gives the best summary of this lesson?

A. A launch pad is a complete safety and support system, not just a flat platform.
B. Launch pads can be understood without observation, evidence or comparison.
C. Every object or event connected with launch pads behaves in exactly the same way.
D. The main idea of launch pads is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain launch pads.
2. Write two important facts about launch pads.

3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why are launch pads surrounded by large restricted areas?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Water used at a launch pad can reduce damaging sound energy as well as absorb heat. Launching eastward near the equator can gain some helpful speed from Earth's rotation.

UNIT 6 · CHAPTER 4

Astronauts

Learning outcomes

- Explain astronauts using simple scientific language.
- Use the words astronaut and microgravity correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Astronauts are people trained to travel or work in space. Their work can include operating spacecraft, conducting experiments, repairing equipment and communicating science to people on Earth.

Understanding the idea

Astronaut training includes science, engineering, physical fitness, emergency practice, teamwork and spacecraft systems.

In microgravity, astronauts float because they and their spacecraft are falling around Earth together.

Life-support systems provide air, safe pressure, temperature control, water and waste management.

Astronauts exercise regularly to reduce muscle and bone loss during long missions.

Look for cause and effect. Expelled gases produce thrust, stages reduce unnecessary mass, and guidance keeps the vehicle on the planned path.

Do not learn astronaut, microgravity, life support separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

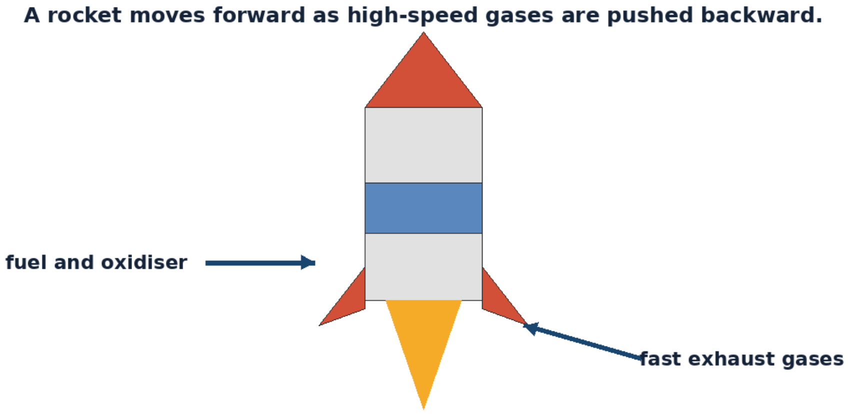


Figure 6.4 A rocket gains thrust by pushing high-speed exhaust gases backward.

Example

On the International Space Station, an astronaut may perform an experiment, exercise, photograph Earth and speak with mission control in one day.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Try completing a simple puzzle while wearing thick gloves to understand why spacesuit work requires practice.

Think and discuss

Why is teamwork as important as physical fitness for an astronaut?

What you have learnt

- Astronaut training includes science, engineering, physical fitness, emergency practice, teamwork and spacecraft systems.
- In microgravity, astronauts float because they and their spacecraft are falling around Earth together.
- Astronauts are trained professionals supported by spacecraft and life-support systems.

EXERCISES · ASTRONAUTS

Exercises

A. Choose the correct answer

1. Which statement about astronauts is correct?
A. Astronauts float because there is no gravity anywhere in orbit.
B. Astronauts need no exercise in space.
C. Astronauts train in science, systems, fitness, emergencies and teamwork.
D. A spacesuit is only clothing for warmth.

2. What does the word “astronaut” mean in this lesson?
A. a person trained to travel or work in space
B. a condition of continuous free fall that makes objects seem nearly weightless
C. systems that provide conditions needed for survival
D. work performed outside a spacecraft in space

3. Which observation or example best supports the lesson idea?
A. A glass is filled with water.
B. A notebook is closed on a table.
C. A pencil is sharpened.
D. On the International Space Station, an astronaut may perform an experiment, exercise, photograph Earth and speak with mission control in one day.

4. Which word means “a person trained to travel or work in space”?
A. microgravity
B. life support
C. astronaut
D. spacewalk

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Try completing a simple puzzle while wearing thick gloves to understand why spacesuit work requires practice.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?
A. Astronauts can be understood without observation, evidence or comparison.
B. Every object or event connected with astronauts behaves in exactly the same way.
C. Astronauts are trained professionals supported by spacecraft and life-support systems.

D. The main idea of astronauts is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain astronauts.
2. Write two important facts about astronauts.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is teamwork as important as physical fitness for an astronaut?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Astronauts aboard the ISS can experience about sixteen sunrises and sunsets in one Earth day because the station orbits quickly. Spacewalk suits are small personal spacecraft with pressure, oxygen, cooling and communication systems.

UNIT 6 · CHAPTER 5

Reusable Rockets

Learning outcomes

- Explain reusable rockets using simple scientific language.
- Use the words reusable and booster correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A reusable rocket is designed so that one or more major parts can be recovered and flown again. Reuse can lower cost and reduce manufacturing demand, but it requires careful engineering and inspection.

Understanding the idea

Some boosters return vertically using engines, while other reusable vehicles may glide or use parachutes.

Recovery requires guidance, heat protection, landing systems and extra fuel or equipment.

Not every part of a reusable system is reused on every mission.

Engineers compare the cost, reliability and performance of reuse with building new hardware.

Look for cause and effect. Expelled gases produce thrust, stages reduce unnecessary mass, and guidance keeps the vehicle on the planned path.

The vocabulary of this chapter - reusable, booster, recovery - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

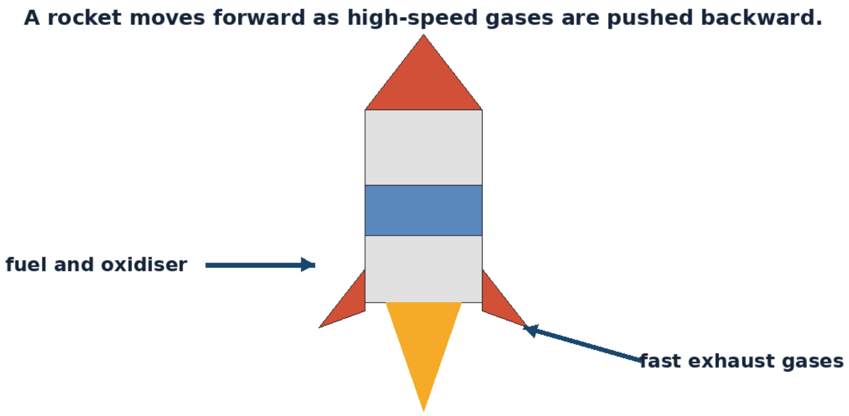


Figure 6.5 A rocket gains thrust by pushing high-speed exhaust gases backward.

Example

A reusable booster may separate after its work, turn around, restart engines and land while the upper stage continues to orbit.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Compare a reusable water bottle with a disposable bottle. List benefits and extra care needed, then relate the idea cautiously to rockets.

Think and discuss

Why might a reusable rocket carry extra equipment that a fully expendable rocket does not need?

What you have learnt

- Some boosters return vertically using engines, while other reusable vehicles may glide or use parachutes.
- Recovery requires guidance, heat protection, landing systems and extra fuel or equipment.
- Rocket reuse can reduce replacement needs but adds recovery and maintenance challenges.

EXERCISES · REUSABLE ROCKETS

Exercises

A. Choose the correct answer

1. Which statement about reusable rockets is correct?

A. Every reusable rocket part returns automatically.
B. Reuse removes the need for safety checks.
C. Reusable rockets recover and fly major hardware again after inspection.
D. Reusable rockets need no fuel.

2. What does the word "reusable" mean in this lesson?

A. designed to be used more than once
B. a rocket stage that adds thrust, especially early in flight
C. bringing hardware back safely
D. to inspect, repair and prepare for reuse

3. Which observation or example best supports the lesson idea?

A. A reusable booster may separate after its work, turn around, restart engines and land while the upper stage continues to orbit.
B. A notebook is closed on a table.
C. A glass is filled with water.
D. A pencil is sharpened.

4. Which word means "designed to be used more than once"?

A. booster
B. recovery
C. refurbish
D. reusable

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Compare a reusable water bottle with a disposable bottle. List benefits and extra care needed, then relate the idea cautiously to rockets.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Rocket reuse can reduce replacement needs but adds recovery and maintenance challenges.
B. Reusable rockets can be understood without observation, evidence or comparison.
C. Every object or event connected with reusable rockets behaves in exactly the same way.
D. The main idea of reusable rockets is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain reusable rockets.
2. Write two important facts about reusable rockets.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why might a reusable rocket carry extra equipment that a fully expendable rocket does not need?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: ISRO has tested technologies related to reusable launch vehicles, including winged experimental vehicles. A reusable rocket must still be inspected and refurbished; reuse does not mean "no maintenance."

UNIT 6 REVIEW

Review · Rockets & Launch Vehicles

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of what is a rocket? using one example.
2. Why does an orbiting spacecraft not simply fly straight away from Earth?
3. Write one correct statement and one common mistake about launch pads.
4. Explain the main idea of astronauts using one example.
5. Why might a reusable rocket carry extra equipment that a fully expendable rocket does not need?
6. Create one new question from this unit and write its answer.

B. Compare and connect

1. Write one connection between "What is a Rocket?" and "Reusable Rockets".
2. Write one connection between "How Rockets Reach Space" and "Astronauts".

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 6 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about what is a rocket? is correct?
A. A rocket is pulled upward by the Moon.
B. A rocket works only by pushing against air.
C. Rockets need wings to flap in space.
D. A rocket produces thrust by pushing exhaust backward.

2. Which statement about how rockets reach space is correct?
A. A rocket must point straight upward for the whole flight.
B. A spacecraft reaches orbit by gaining very high sideways speed.
C. Gravity stops completely at the edge of space.
D. Orbit is achieved only by rising above clouds.

3. Which statement about launch pads is correct?

A. A launch pad is only a painted circle on the ground.
B. Rockets need no checks before launch.
C. Launch sites are placed inside crowded city centres.
D. Launch pads include systems for support, safety, exhaust control and monitoring.

4. Which statement about astronauts is correct?

A. Astronauts float because there is no gravity anywhere in orbit.
B. Astronauts need no exercise in space.
C. Astronauts train in science, systems, fitness, emergencies and teamwork.
D. A spacesuit is only clothing for warmth.

5. Which statement about reusable rockets is correct?

A. Every reusable rocket part returns automatically.
B. Reuse removes the need for safety checks.
C. Reusable rockets recover and fly major hardware again after inspection.
D. Reusable rockets need no fuel.

6. Which word means "a vehicle or engine that produces thrust by expelling material"?

A. thrust
B. exhaust
C. oxidiser
D. rocket

7. Which word means "a curved path around another body"?

A. orbit
B. drag
C. velocity
D. suborbital

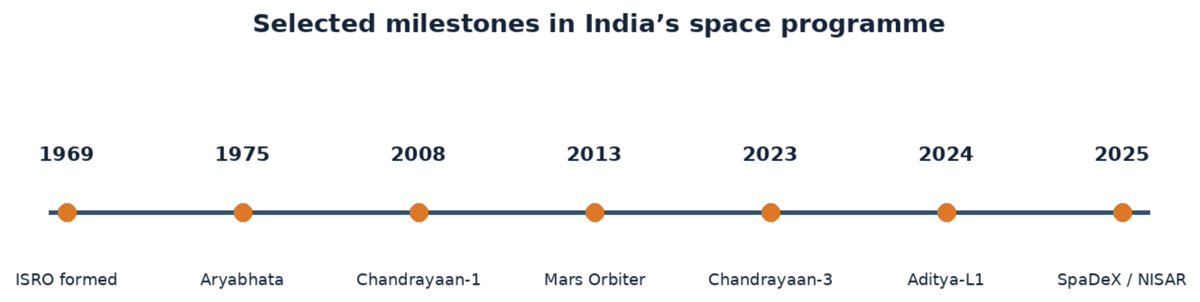
8. Which word means "the prepared site from which a rocket launches"?

A. flame trench
B. launch pad
C. mission control
D. safe zone

UNIT 7

ISRO & India's Space Missions

India's space programme links scientific research with national needs. This unit follows ISRO, important missions and the people whose work helped India become a major spacefaring nation.



Chapters in this unit

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 7 · CHAPTER 1

Introduction to ISRO

Learning outcomes

- Explain introduction to isro using simple scientific language.
- Use the words ISRO and space agency correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

The Indian Space Research Organisation, or ISRO, is India's national space agency. It develops satellites, launch vehicles, scientific missions and space-based services for national development and exploration.

Understanding the idea

ISRO was established in 1969, building on earlier Indian space research efforts led by scientists such as Vikram Sarabhai.

Its work includes communication, navigation, weather, Earth observation, planetary science and launch technology.

Major centres across India specialise in rockets, satellites, applications, tracking, propulsion and research.

ISRO works with universities, industry, other government organisations and international space agencies.

Read mission accounts as sequences: purpose, launch, journey, operations and results. A planned mission is not the same as a completed mission.

The important words in this chapter include ISRO, space agency, ground station. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

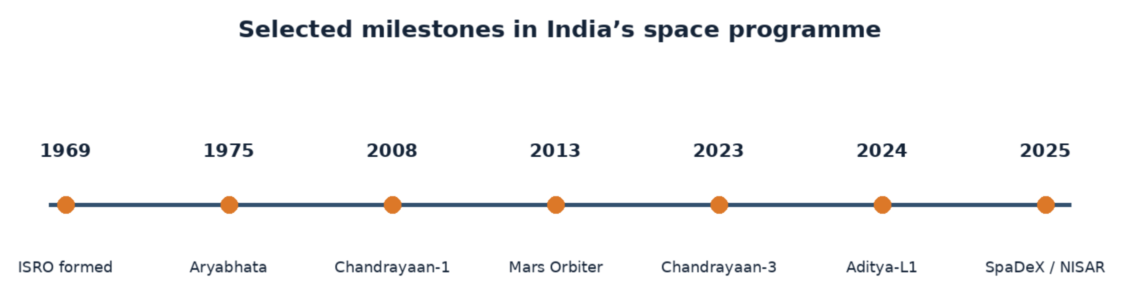


Figure 7.1 Selected milestones in India's space programme, arranged as a simple timeline.

Example

A cyclone warning may use an ISRO weather satellite, while a farmer may use an Earth-observation product and a driver may use NavIC.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Create an "ISRO mission wheel" with spokes for communication, weather, navigation, Earth observation, science and launchers.

Think and discuss

Why does a national space programme need both spacecraft and ground systems?

What you have learnt

- ISRO was established in 1969, building on earlier Indian space research efforts led by scientists such as Vikram Sarabhai.
- Its work includes communication, navigation, weather, Earth observation, planetary science and launch technology.
- ISRO combines science, engineering and public services through India's space programme.

Exercises

A. Choose the correct answer

1. Which statement about introduction to isro is correct?

A. ISRO studies only the Moon.
B. ISRO is India's national space agency, established in 1969.
C. ISRO is a weather station with no spacecraft.
D. ISRO was created to operate ordinary ships at sea.

2. What does the word "ISRO" mean in this lesson?

A. a research rocket that makes a suborbital flight
B. an organisation that plans and carries out space activities
C. the Indian Space Research Organisation
D. a facility that communicates with spacecraft

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A pencil is sharpened.
C. A notebook is closed on a table.
D. A cyclone warning may use an ISRO weather satellite, while a farmer may use an Earth-observation product and a driver may use NavIC.

4. Which word means "the Indian Space Research Organisation"?

A. ISRO
B. space agency
C. ground station
D. sounding rocket

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Create an "ISRO mission wheel" with spokes for communication, weather, navigation, Earth observation, science and launchers.

6. Which sentence gives the best summary of this lesson?

A. Introduction to isro can be understood without observation, evidence or comparison.
B. ISRO combines science, engineering and public services through India's space programme.
C. Every object or event connected with introduction to isro behaves in exactly the same way.
D. The main idea of introduction to isro is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain introduction to isro.
2. Write two important facts about introduction to isro.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why does a national space programme need both spacecraft and ground systems?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: India's first sounding rocket was launched from Thumba in Kerala in 1963. The Satish Dhawan Space Centre at Sriharikota is India's principal orbital launch site.

UNIT 7 · CHAPTER 2

Chandrayaan Missions

Learning outcomes

- Explain chandrayaan missions using simple scientific language.
- Use the words Chandrayaan and soft landing correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Chandrayaan is India's series of lunar missions. The missions have studied the Moon from orbit and on the surface while developing India's deep-space and landing capabilities.

Understanding the idea

Chandrayaan-1 launched in 2008 and made important mineral and water-related measurements of the Moon.

Chandrayaan-2 launched in 2019; its orbiter continued valuable lunar science, although the Vikram lander did not achieve a soft landing.

Chandrayaan-3 launched in 2023 and successfully soft-landed the Vikram lander on 23 August 2023 in the lunar southern high-latitude region.

The Pragyan rover explored near the landing site and performed in-situ measurements.

Read mission accounts as sequences: purpose, launch, journey, operations and results. A planned mission is not the same as a completed mission.

Notice how the terms Chandrayaan, soft landing, lander appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

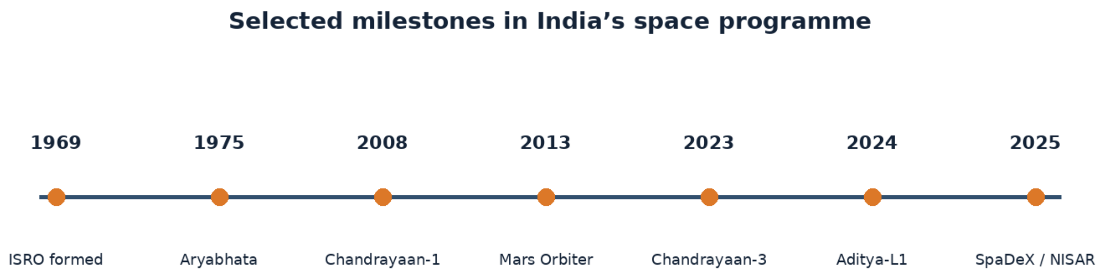


Figure 7.2 Selected milestones in India's space programme, arranged as a simple timeline.

Example

A mission can have several parts with different outcomes: Chandrayaan-2's orbiter succeeded even though its landing attempt did not.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Make three mission cards and sort each objective into orbiting, landing, roving or measuring.

Think and discuss

Why is a soft landing harder than simply reaching lunar orbit?

What you have learnt

- Chandrayaan-1 launched in 2008 and made important mineral and water-related measurements of the Moon.
- Chandrayaan-2 launched in 2019; its orbiter continued valuable lunar science, although the Vikram lander did not achieve a soft landing.
- Chandrayaan missions developed India's lunar science, orbiting, landing and roving experience.

EXERCISES · CHANDRAYAAN MISSIONS

Exercises

A. Choose the correct answer

1. Which statement about chandrayaan missions is correct?
A. Chandrayaan-2 had no successful orbiter.
B. A soft landing means crashing at high speed.
C. Chandrayaan-3 was India's first satellite of any kind.
D. Chandrayaan-3 achieved a successful lunar soft landing in August 2023.

2. What does the word "Chandrayaan" mean in this lesson?
A. a spacecraft designed to reach and operate on a surface
B. the name of India's lunar mission series
C. a controlled landing slow enough to avoid destructive impact
D. a mobile robotic vehicle that travels on another world

3. Which observation or example best supports the lesson idea?
A. A glass is filled with water.
B. A mission can have several parts with different outcomes: Chandrayaan-2's orbiter succeeded even though its landing attempt did not.
C. A pencil is sharpened.
D. A notebook is closed on a table.

4. Which word means "the name of India's lunar mission series"?
A. soft landing
B. Chandrayaan
C. lander
D. rover

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Make three mission cards and sort each objective into orbiting, landing, roving or measuring.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?
A. Chandrayaan missions can be understood without observation, evidence or comparison.
B. Every object or event connected with chandrayaan missions behaves in exactly the same way.
C. The main idea of chandrayaan missions is unrelated to science or everyday experience.
D. Chandrayaan missions developed India's lunar science, orbiting, landing and roving experience.

B. Answer in one or two sentences

1. In one sentence, explain chandrayaan missions.
2. Write two important facts about chandrayaan missions.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is a soft landing harder than simply reaching lunar orbit?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: India became the fourth country to achieve a soft landing on the Moon with Chandrayaan-3. India observes 23 August as National Space Day, linked to the Chandrayaan-3 landing.

UNIT 7 · CHAPTER 3

Mangalyaan

Learning outcomes

- Explain mangalyaan using simple scientific language.
- Use the words Mangalyaan and interplanetary correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Mangalyaan, officially the Mars Orbiter Mission, was India’s first interplanetary mission. It demonstrated deep-space technology and studied Mars from orbit.

Understanding the idea

The spacecraft launched on 5 November 2013 and entered Mars orbit on 24 September 2014.

India became the first country to reach Mars orbit successfully on its first attempt.

The mission tested navigation, communication, autonomy and operations across enormous distances.

Mangalyaan operated for about eight years, far beyond its planned mission duration.

Read mission accounts as sequences: purpose, launch, journey, operations and results. A planned mission is not the same as a completed mission.

Do not learn Mangalyaan, interplanetary, autonomy separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

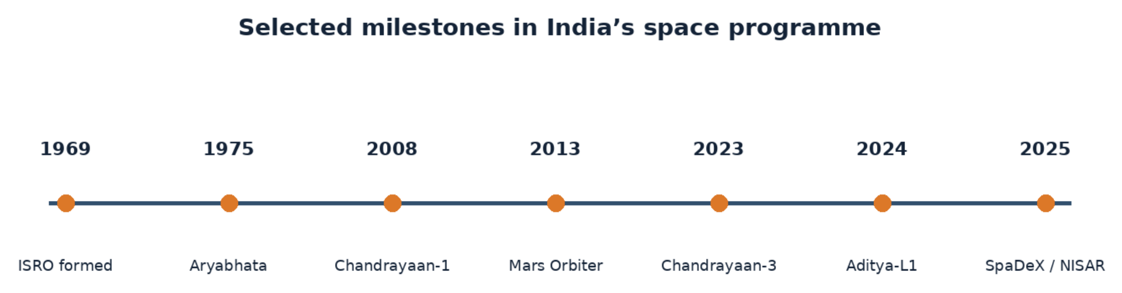


Figure 7.3 Selected milestones in India’s space programme, arranged as a simple timeline.

Example

A radio message between Earth and Mars can take many minutes, so a spacecraft must handle some situations without immediate instructions.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Send a partner a sequence of instructions for a maze, but allow no corrections after starting. Discuss why spacecraft need careful planning.

Think and discuss

Why can mission controllers not steer a Mars spacecraft as if it were a remote-control car?

What you have learnt

- The spacecraft launched on 5 November 2013 and entered Mars orbit on 24 September 2014.
- India became the first country to reach Mars orbit successfully on its first attempt.
- Mangalyaan proved India could navigate, communicate and operate a spacecraft at Mars.

EXERCISES · MANGALYAAN

Exercises

A. Choose the correct answer

1. Which statement about mangalyaan is correct?

A. Mangalyaan entered Mars orbit in September 2014 on India's first attempt.
B. Mangalyaan landed astronauts on Mars.
C. Signals between Earth and Mars travel instantly.
D. Mangalyaan orbited the Moon rather than Mars.

2. What does the word "Mangalyaan" mean in this lesson?

A. the ability to perform some tasks without immediate human control
B. the travel time of a signal between distant places
C. India's Mars Orbiter Mission
D. travelling or operating between planets

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A notebook is closed on a table.
C. A radio message between Earth and Mars can take many minutes, so a spacecraft must handle some situations without immediate instructions.
D. A pencil is sharpened.

4. Which word means "India's Mars Orbiter Mission"?

A. interplanetary
B. autonomy
C. Mangalyaan
D. time delay

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Send a partner a sequence of instructions for a maze, but allow no corrections after starting. Discuss why spacecraft need careful planning.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Mangalyaan can be understood without observation, evidence or comparison.
B. Mangalyaan proved India could navigate, communicate and operate a spacecraft at Mars.
C. Every object or event connected with mangalyaan behaves in exactly the same way.
D. The main idea of mangalyaan is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain mangalyaan.
2. Write two important facts about mangalyaan.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can mission controllers not steer a Mars spacecraft as if it were a remote-control car?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Mangalyaan captured full-disc colour images of Mars that were used for science and public outreach. The word Mangalyaan combines “Mangal,” meaning Mars, and “yaan,” meaning vehicle or craft.

UNIT 7 · CHAPTER 4

Aditya-L1

Learning outcomes

- Explain aditya-l1 using simple scientific language.
- Use the words Aditya-L1 and L1 correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Aditya-L1 is India’s first dedicated space-based solar observatory. It studies the Sun and space weather from a halo orbit around the Sun-Earth L1 region, about 1.5 million kilometres from Earth toward the Sun.

Understanding the idea

The spacecraft launched on 2 September 2023 and entered its planned halo orbit on 6 January 2024.

From near L1, Aditya-L1 can observe the Sun continuously with few interruptions by Earth.

Its instruments study the solar atmosphere, radiation, particles and magnetic fields.

Understanding eruptions from the Sun can improve knowledge of space weather that affects satellites and technology.

Read mission accounts as sequences: purpose, launch, journey, operations and results. A planned mission is not the same as a completed mission.

The vocabulary of this chapter - Aditya-L1, L1, halo orbit - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

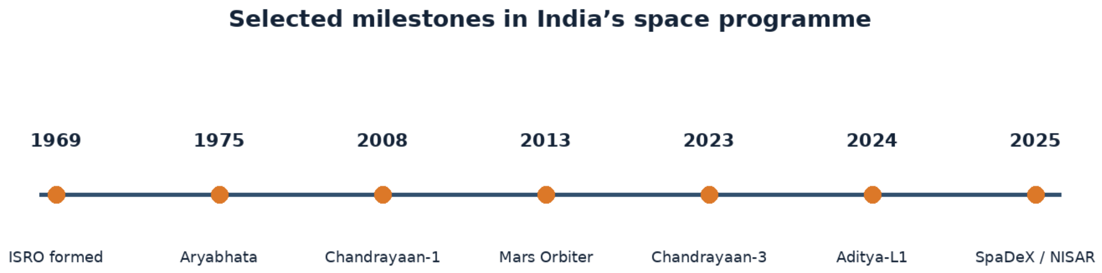


Figure 7.4 Selected milestones in India's space programme, arranged as a simple timeline.

Example

A solar storm can disturb radio communication or satellite operations, so monitoring the Sun has practical value.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Use a lamp, large ball and small bead to discuss a special region between Sun and Earth where forces and motion can support observation.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Why is continuous viewing of the Sun useful for studying sudden solar events?

What you have learnt

- The spacecraft launched on 2 September 2023 and entered its planned halo orbit on 6 January 2024.
- From near L1, Aditya-L1 can observe the Sun continuously with few interruptions by Earth.
- Aditya-L1 observes the Sun from a halo orbit near the Sun-Earth L1 region.

EXERCISES · ADITYA-L1

Exercises

A. Choose the correct answer

1. Which statement about aditya-l1 is correct?

A. Aditya-L1 is a rover driving on the Sun.
B. L1 is a launch pad on Earth.
C. Aditya-L1 studies the Sun from near the Sun-Earth L1 region.
D. Aditya-L1 studies only ocean tides.

2. What does the word "Aditya-L1" mean in this lesson?

A. a three-dimensional orbit around a Lagrange region
B. India's first dedicated space-based solar observatory
C. a useful gravitational region between Earth and the Sun
D. conditions in space influenced mainly by solar activity

3. Which observation or example best supports the lesson idea?

A. A solar storm can disturb radio communication or satellite operations, so monitoring the Sun has practical value.
B. A glass is filled with water.
C. A pencil is sharpened.

D. A notebook is closed on a table.

4. Which word means "India's first dedicated space-based solar observatory"?

A. L1

B. halo orbit

C. space weather

D. Aditya-L1

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.

B. Change several conditions at the same time and compare nothing.

C. Use a lamp, large ball and small bead to discuss a special region between Sun and Earth where forces and motion can support observation.

D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Aditya-l1 can be understood without observation, evidence or comparison.

B. Every object or event connected with aditya-l1 behaves in exactly the same way.

C. The main idea of aditya-l1 is unrelated to science or everyday experience.

D. Aditya-L1 observes the Sun from a halo orbit near the Sun-Earth L1 region.

B. Answer in one or two sentences

1. In one sentence, explain aditya-l1.
2. Write two important facts about aditya-l1.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is continuous viewing of the Sun useful for studying sudden solar events?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Aditya is a Sanskrit name associated with the Sun, and L1 refers to the first Lagrange region in the Sun-Earth system. In 2025, ISRO released Aditya-L1 datasets for scientific use and reported important solar observations.

UNIT 7 · CHAPTER 5

Gaganyaan

Learning outcomes

- Explain gaganyaan using simple scientific language.
- Use the words Gaganyaan and crew module correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Gaganyaan is India's human-spaceflight programme designed to demonstrate the ability to send Indian crew members to low Earth orbit and return them safely. As of June 2026, the programme remains in testing and preparation; no crewed Gaganyaan orbital mission has yet been completed.

Understanding the idea

The programme includes a human-rated launch vehicle, crew module, service module, life support, recovery systems and astronaut training.

The planned demonstration concept has described a crew of up to three people in an orbit of about 400 kilometres for roughly three days.

Tests examine escape systems, parachutes, propulsion, structures, electronics, recovery and many failure conditions.

In April 2026, ISRO conducted the second Integrated Air Drop Test using a simulated crew module to test parachute and recovery behaviour.

Read mission accounts as sequences: purpose, launch, journey, operations and results. A planned mission is not the same as a completed mission.

The important words in this chapter include Gaganyaan, crew module, re-entry. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

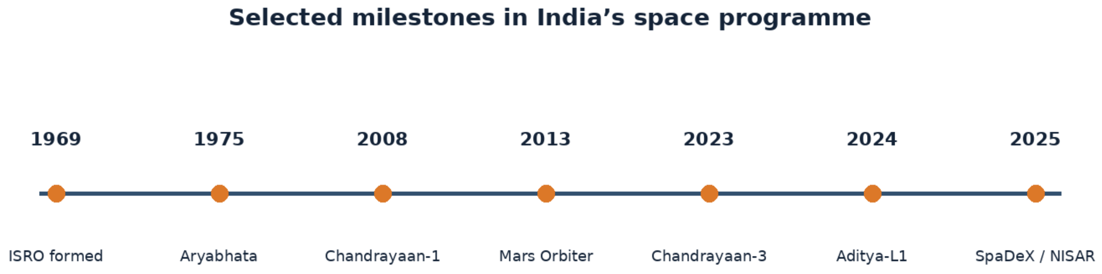


Figure 7.5 Selected milestones in India's space programme, arranged as a simple timeline.

Example

A safe human mission requires success at launch, in orbit, during re-entry, under parachutes and after splashdown recovery.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Create a safety chain with cards for launch, escape, life support, heat shield, parachutes and recovery. Remove one card and discuss the risk.

Think and discuss

Why must an uncrewed test be completed before placing astronauts aboard a new spacecraft?

What you have learnt

- The programme includes a human-rated launch vehicle, crew module, service module, life support, recovery systems and astronaut training.
- The planned demonstration concept has described a crew of up to three people in an orbit of about 400 kilometres for roughly three days.
- Gaganyaan is an active human-spaceflight programme still undergoing testing as of June 2026.

EXERCISES · GAGANYAAN

Exercises

A. Choose the correct answer

1. Which statement about gaganyaan is correct?

A. Gaganyaan has already landed people on the Moon.
B. Gaganyaan is an ocean-weather satellite.
C. Human spaceflight needs no escape or recovery system.
D. Gaganyaan is India's programme to develop safe crewed flight to low Earth orbit.

2. What does the word "Gaganyaan" mean in this lesson?

A. designed and verified to meet safety requirements for carrying people
B. India's human-spaceflight programme
C. the part of a spacecraft that carries astronauts
D. the return of a spacecraft into an atmosphere

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A glass is filled with water.
C. A safe human mission requires success at launch, in orbit, during re-entry, under parachutes and after splashdown recovery.
D. A notebook is closed on a table.

4. Which word means "India's human-spaceflight programme"?

A. Gaganyaan
B. crew module
C. re-entry
D. human-rated

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Create a safety chain with cards for launch, escape, life support, heat shield, parachutes and recovery. Remove one card and discuss the risk.

6. Which sentence gives the best summary of this lesson?

A. Gaganyaan can be understood without observation, evidence or comparison.
B. Gaganyaan is an active human-spaceflight programme still undergoing testing as of June 2026.
C. Every object or event connected with gaganyaan behaves in exactly the same way.
D. The main idea of gaganyaan is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain gaganyaan.
2. Write two important facts about gaganyaan.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why must an uncrewed test be completed before placing astronauts aboard a new spacecraft?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The launch vehicle used for Gaganyaan is a human-rated version of LVM3, often called HLVM3. Human-rating means designing and testing systems to meet much stricter safety and reliability requirements for carrying people.

UNIT 7 · CHAPTER 6

Great Indian Space Scientists

Learning outcomes

- Explain great indian space scientists using simple scientific language.
- Use the words scientist and engineer correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

India's space programme grew through the work of scientists, engineers, technicians, educators and leaders. Learning their stories shows that major missions are team achievements built over generations.

Understanding the idea

Vikram Sarabhai is widely regarded as the father of India's space programme and championed using space technology for national development.

Satish Dhawan strengthened ISRO's organisation, research culture and launch capabilities while supporting teams through success and failure.

U. R. Rao led major advances in Indian satellite technology and later chaired ISRO.

A. P. J. Abdul Kalam made important contributions to launch-vehicle and aerospace programmes and later became President of India.

Read mission accounts as sequences: purpose, launch, journey, operations and results. A planned mission is not the same as a completed mission.

Notice how the terms scientist, engineer, leadership appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

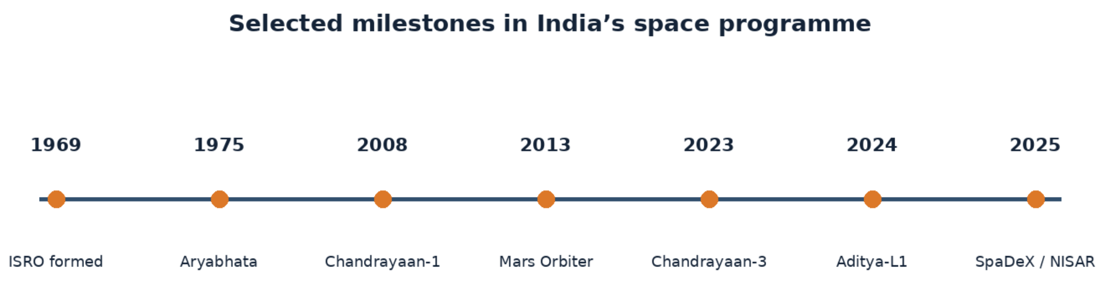


Figure 7.6 Selected milestones in India's space programme, arranged as a simple timeline.

Example

A mission needs many kinds of expertise: propulsion, electronics, software, structures, science, tracking, communication and management.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Choose one space scientist. Make a four-box biography: challenge, contribution, teamwork lesson and question you would ask.

Think and discuss

Why is it inaccurate to say that one person alone built a national space programme?

What you have learnt

- Vikram Sarabhai is widely regarded as the father of India's space programme and championed using space technology for national development.
- Satish Dhawan strengthened ISRO's organisation, research culture and launch capabilities while supporting teams through success and failure.
- Space missions succeed through long-term teamwork, leadership and many specialised skills.

EXERCISES · GREAT INDIAN SPACE SCIENTISTS

Exercises

A. Choose the correct answer

1. Which statement about great indian space scientists is correct?

A. Only astronauts contribute to a space mission.
B. Scientific leadership means avoiding all discussion of failure.
C. India’s space achievements were built by generations of scientists, engineers and teams.
D. One person designed and built every Indian spacecraft alone.

2. What does the word “scientist” mean in this lesson?

A. guiding people toward a shared goal
B. a person who designs and improves systems or solutions
C. a person who investigates questions using evidence
D. people combining different skills to achieve a task

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A notebook is closed on a table.
C. A mission needs many kinds of expertise: propulsion, electronics, software, structures, science, tracking, communication and management.
D. A pencil is sharpened.

4. Which word means “a person who investigates questions using evidence”?

A. engineer
B. scientist
C. leadership
D. teamwork

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Choose one space scientist. Make a four-box biography: challenge, contribution, teamwork lesson and question you would ask.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Great indian space scientists can be understood without observation, evidence or comparison.
B. Every object or event connected with great indian space scientists behaves in exactly the same way.
C. The main idea of great indian space scientists is unrelated to science or everyday experience.
D. Space missions succeed through long-term teamwork, leadership and many specialised skills.

B. Answer in one or two sentences

1. In one sentence, explain great indian space scientists.
2. Write two important facts about great indian space scientists.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is it inaccurate to say that one person alone built a national space programme?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Many Indian women scientists and engineers have held leading roles in missions including Mars, Moon and solar exploration. Good scientific leadership includes sharing credit, learning from failure and protecting teams while improving systems.

UNIT 7 · CHAPTER 7

India's Achievements in Space

Learning outcomes

- Explain india's achievements in space using simple scientific language.
- Use the words PSLV and LVM3 correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

India’s space achievements include reliable launch vehicles, large satellite programmes, planetary exploration, navigation, astronomy missions, docking demonstrations and growing human-spaceflight capability.

Understanding the idea

The PSLV became a widely used launcher for Earth-observation and scientific missions, while LVM3 carries heavier payloads.

Chandrayaan-3 achieved India’s first lunar soft landing, and Aditya-L1 became India’s first dedicated solar observatory.

SpaDeX demonstrated in-space docking in January 2025, followed by a second docking and power-transfer demonstration in April 2025.

The joint NASA-ISRO NISAR Earth-observation mission launched on 30 July 2025 and entered its science phase later that year.

Read mission accounts as sequences: purpose, launch, journey, operations and results. A planned mission is not the same as a completed mission.

Do not learn PSLV, LVM3, docking separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

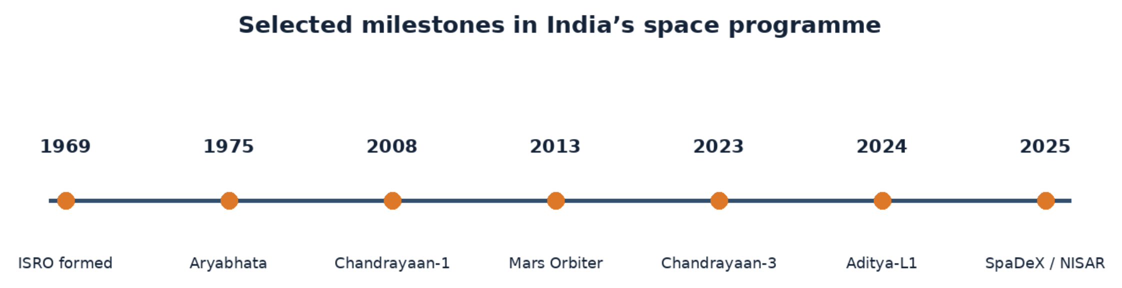


Figure 7.7 Selected milestones in India’s space programme, arranged as a simple timeline.

Example

Docking technology allows two spacecraft to meet and connect in orbit, which is useful for future stations and complex missions.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Sort achievement cards into launchers, satellites, science, human spaceflight and technology demonstrations.

Think and discuss

Which achievement is a technology demonstration rather than a planetary-science mission: SpaDeX or Chandrayaan-3? Explain.

What you have learnt

- The PSLV became a widely used launcher for Earth-observation and scientific missions, while LVM3 carries heavier payloads.
- Chandrayaan-3 achieved India’s first lunar soft landing, and Aditya-L1 became India’s first dedicated solar observatory.
- India’s space programme combines public services, exploration and new technologies.

EXERCISES · INDIA'S ACHIEVEMENTS IN SPACE

Exercises

A. Choose the correct answer

1. Which statement about india's achievements in space is correct?

A. India has achieved lunar landing, solar observation, docking and advanced Earth observation.
B. NISAR is a natural moon of Earth.
C. SpaDeX was a mission to land on Mars.
D. Every Indian space achievement is a crewed mission.

2. What does the word “PSLV” mean in this lesson?

A. India’s heavy-lift Launch Vehicle Mark-3
B. a joint NASA-ISRO radar Earth-observation mission
C. India’s Polar Satellite Launch Vehicle
D. bringing spacecraft together and connecting them in orbit

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A notebook is closed on a table.
C. A glass is filled with water.
D. Docking technology allows two spacecraft to meet and connect in orbit, which is useful for future stations and complex missions.

4. Which word means “India’s Polar Satellite Launch Vehicle”?

A. LVM3
B. docking
C. PSLV
D. NISAR

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Sort achievement cards into launchers, satellites, science, human spaceflight and technology demonstrations.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. India's achievements in space can be understood without observation, evidence or comparison.
B. India’s space programme combines public services, exploration and new technologies.
C. Every object or event connected with india's achievements in space behaves in exactly the same way.
D. The main idea of india's achievements in space is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain india's achievements in space.
2. Write two important facts about india's achievements in space.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Which achievement is a technology demonstration rather than a planetary-science mission: SpaDeX or Chandrayaan-3? Explain.
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: In 2025, ISRO astronaut Shubhanshu Shukla served as pilot on Axiom Mission 4 and completed an 18-day scientific mission that included time aboard the ISS. Space achievements include services on Earth as well as dramatic exploration missions.

UNIT 7 REVIEW

Review · ISRO & India’s Space Missions

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of introduction to isro using one example.
2. Why is a soft landing harder than simply reaching lunar orbit?
3. Write one correct statement and one common mistake about mangalyaan.
4. Explain the main idea of aditya-l1 using one example.
5. Why must an uncrewed test be completed before placing astronauts aboard a new spacecraft?
6. Write one correct statement and one common mistake about great indian space scientists.
7. Explain the main idea of india's achievements in space using one example.
8. Create one new question from this unit and write its answer.

B. Compare and connect

1. Write one connection between “Introduction to ISRO” and “India's Achievements in Space”.
2. Write one connection between “Chandrayaan Missions” and “Great Indian Space Scientists”.

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 7 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about introduction to isro is correct?
- A. ISRO studies only the Moon.
- B. ISRO is India’s national space agency, established in 1969.
- C. ISRO is a weather station with no spacecraft.
- D. ISRO was created to operate ordinary ships at sea.

2. Which statement about chandrayaan missions is correct?
- A. Chandrayaan-2 had no successful orbiter.
- B. A soft landing means crashing at high speed.
- C. Chandrayaan-3 was India’s first satellite of any kind.
- D. Chandrayaan-3 achieved a successful lunar soft landing in August 2023.

3. Which statement about mangalyaan is correct?
- A. Mangalyaan entered Mars orbit in September 2014 on India’s first attempt.
- B. Mangalyaan landed astronauts on Mars.
- C. Signals between Earth and Mars travel instantly.
- D. Mangalyaan orbited the Moon rather than Mars.

4. Which statement about aditya-l1 is correct?

A. Aditya-L1 is a rover driving on the Sun.
B. L1 is a launch pad on Earth.
C. Aditya-L1 studies the Sun from near the Sun-Earth L1 region.
D. Aditya-L1 studies only ocean tides.

5. Which statement about gaganyaan is correct?

A. Gaganyaan has already landed people on the Moon.
B. Gaganyaan is an ocean-weather satellite.
C. Human spaceflight needs no escape or recovery system.
D. Gaganyaan is India’s programme to develop safe crewed flight to low Earth orbit.

6. Which statement about great indian space scientists is correct?

A. Only astronauts contribute to a space mission.
B. Scientific leadership means avoiding all discussion of failure.
C. India’s space achievements were built by generations of scientists, engineers and teams.
D. One person designed and built every Indian spacecraft alone.

7. Which statement about india's achievements in space is correct?

A. India has achieved lunar landing, solar observation, docking and advanced Earth observation.
B. NISAR is a natural moon of Earth.
C. SpaDeX was a mission to land on Mars.
D. Every Indian space achievement is a crewed mission.

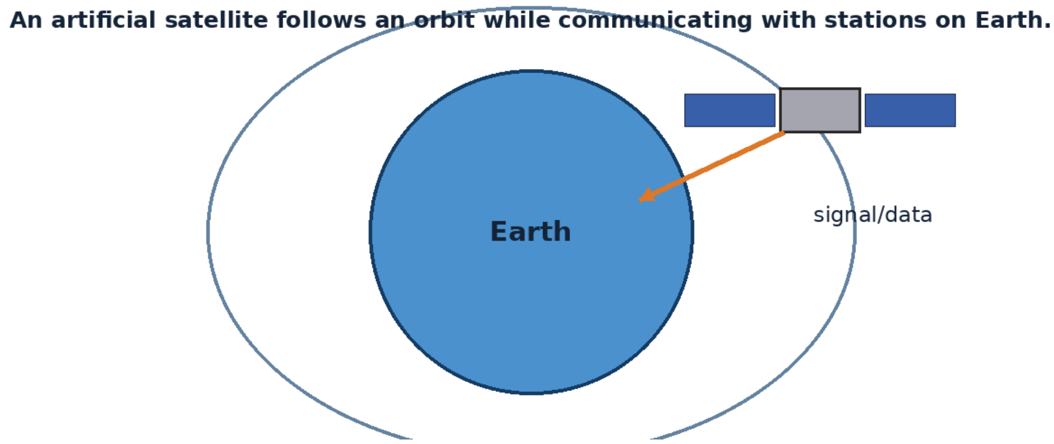
8. Which word means “the Indian Space Research Organisation”?

A. ISRO
B. space agency
C. ground station
D. sounding rocket

UNIT 8

Space Exploration

Space exploration combines courage, machines, planning and teamwork. Learners study astronauts, space stations and missions to the Moon and Mars through age-appropriate examples.



Chapters in this unit

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 8 · CHAPTER 1

Who are Astronauts?

Learning outcomes

- Explain who Astronauts are and what they do.
- Use the words astronaut and cosmonaut correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Astronauts are specially selected and trained people who travel to space or support spaceflight operations. Different countries use words such as astronaut, cosmonaut and taikonaut, but all describe human space travellers.

Understanding the idea

Astronaut candidates often bring experience in science, engineering, medicine, aviation or other demanding fields.

Training includes spacecraft systems, survival, robotics, experiments, teamwork, communication and physical conditioning.

Some astronauts pilot spacecraft, while others specialise in science, engineering or mission operations.

Astronauts represent large ground teams; no space traveller works alone.

Every mission depends on many roles. Astronauts are visible members of a much larger team of engineers, scientists, doctors and controllers.

Notice how the terms astronaut, cosmonaut, mission specialist appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

Example

An astronaut conducting a plant experiment must follow careful steps, record data and communicate with scientists on Earth.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Create a weekly “astronaut training” chart with safe goals for exercise, teamwork, reading, observation and clear communication.

Think and discuss

Which astronaut skill is most useful during an unexpected problem: memorisation, teamwork or clear reasoning? Explain.

What you have learnt

- Astronaut candidates often bring experience in science, engineering, medicine, aviation or other demanding fields.
- Training includes spacecraft systems, survival, robotics, experiments, teamwork, communication and physical conditioning.
- Astronauts are trained specialists who work as part of a much larger mission team.

EXERCISES · WHO ARE ASTRONAUTS?

Exercises

A. Choose the correct answer

1. Which statement about who are astronauts? is correct?

A. Astronauts combine specialised knowledge, physical training and teamwork.
B. Astronauts are chosen only because they can run fast.
C. Every astronaut has exactly the same job.
D. Astronauts operate without help from Earth.

2. What does the word “astronaut” mean in this lesson?

A. people on Earth who support a mission
B. a crew member trained for particular scientific or technical tasks
C. the Russian term commonly used for a space traveller
D. a person trained for space travel or work

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A notebook is closed on a table.
C. A pencil is sharpened.
D. An astronaut conducting a plant experiment must follow careful steps, record data and communicate with scientists on Earth.

4. Which word means “a person trained for space travel or work”?

A. cosmonaut
B. astronaut
C. mission specialist
D. ground team

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Create a weekly “astronaut training” chart with safe goals for exercise, teamwork, reading, observation and clear communication.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Astronauts can be understood without observation, evidence or comparison.
B. Every object or event connected with astronauts behaves in exactly the same way.
C. Astronauts are trained specialists who work as part of a much larger mission team.
D. The main idea of astronauts is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain who are astronauts?
2. Write two important facts about astronauts
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Which astronaut skill is most useful during an unexpected problem: memorisation, teamwork or clear reasoning? Explain.
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Rakesh Sharma became the first Indian citizen in space in 1984 aboard the Soviet Salyut 7 space station. Shubhanshu Shukla became the first Indian to visit the International Space Station during Axiom Mission 4 in 2025.

UNIT 8 · CHAPTER 2

Space Stations

Learning outcomes

- Explain space stations using simple scientific language.
- Use the words space station and ISS correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A space station is a large spacecraft designed for people to live and work in orbit for extended periods. It acts as a laboratory, home, workshop and test site.

Understanding the idea

The International Space Station, or ISS, has been continuously occupied since November 2000.

The ISS orbits Earth roughly every 90 minutes and usually carries an international crew.

Experiments use microgravity to study biology, materials, fluids, medicine, technology and Earth observation.

Space stations need power, life support, communication, docking ports, exercise equipment and regular resupply.

Every mission depends on many roles. Astronauts are visible members of a much larger team of engineers, scientists, doctors and controllers.

Do not learn space station, ISS, module separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

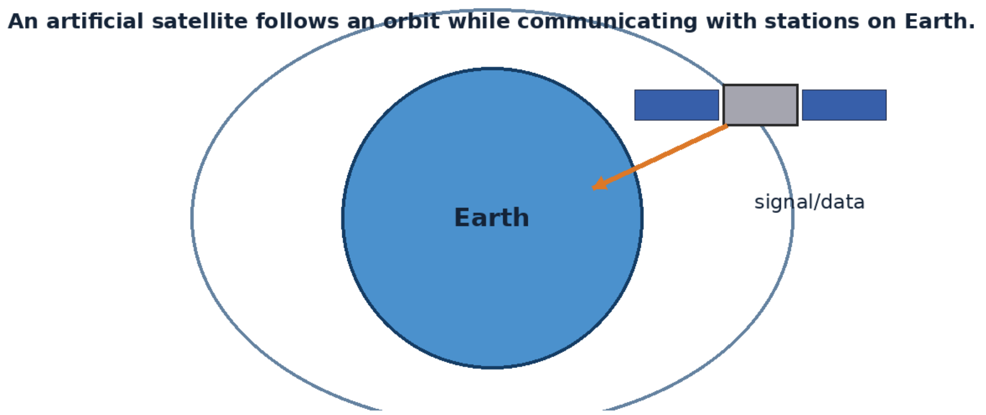


Figure 8.2 An artificial satellite follows an orbit and sends or receives information.

Example

Astronauts exercise about two hours per day on many long missions to help protect muscles and bones.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Design a space-station module on graph paper. Include sleep, work, exercise, food, hygiene and emergency areas.

Think and discuss

Why is recycling water especially important on a space station?

What you have learnt

- The International Space Station, or ISS, has been continuously occupied since November 2000.
- The ISS orbits Earth roughly every 90 minutes and usually carries an international crew.
- A space station supports long-duration living, science and technology testing in orbit.

EXERCISES · SPACE STATIONS

Exercises

A. Choose the correct answer

1. Which statement about space stations is correct?

A. Space stations need no air, power or water systems.
B. Space stations are orbiting laboratories and homes for crews.
C. The ISS completes one orbit each year.
D. A space station rests on a tower above Earth.

2. What does the word “space station” mean in this lesson?

A. one section of a larger spacecraft or station
B. the International Space Station
C. a spacecraft designed for people to live and work in orbit
D. delivery of food, equipment, fuel or other materials

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A pencil is sharpened.
C. Astronauts exercise about two hours per day on many long missions to help protect muscles and bones.
D. A glass is filled with water.

4. Which word means “a spacecraft designed for people to live and work in orbit”?

A. ISS
B. module
C. space station
D. resupply

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Design a space-station module on graph paper. Include sleep, work, exercise, food, hygiene and emergency areas.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. A space station supports long-duration living, science and technology testing in orbit.
B. Space stations can be understood without observation, evidence or comparison.
C. Every object or event connected with space stations behaves in exactly the same way.
D. The main idea of space stations is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain space stations.
2. Write two important facts about space stations.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is recycling water especially important on a space station?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The ISS is larger than a six-bedroom house in living and working space and travels around Earth about sixteen times per day. China operates the Tiangong space station, another permanently crew-capable laboratory in low Earth orbit.

UNIT 8 · CHAPTER 3

Moon Exploration Missions

Learning outcomes

- Explain moon exploration missions using simple scientific language.
- Use the words orbiter and lander correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Moon exploration uses orbiters, impactors, landers, rovers, sample-return spacecraft and human missions. Each type answers different questions and tests different technologies.

Understanding the idea

The Apollo programme landed astronauts on the Moon six times between 1969 and 1972.

Robotic missions from several countries have mapped the Moon, studied its soil and searched for water-related materials.

Orbiters observe large areas, while landers and rovers examine specific sites in detail.

Future missions study polar ice, geology and technologies needed for longer stays.

Every mission depends on many roles. Astronauts are visible members of a much larger team of engineers, scientists, doctors and controllers.

The vocabulary of this chapter - orbiter, lander, rover - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

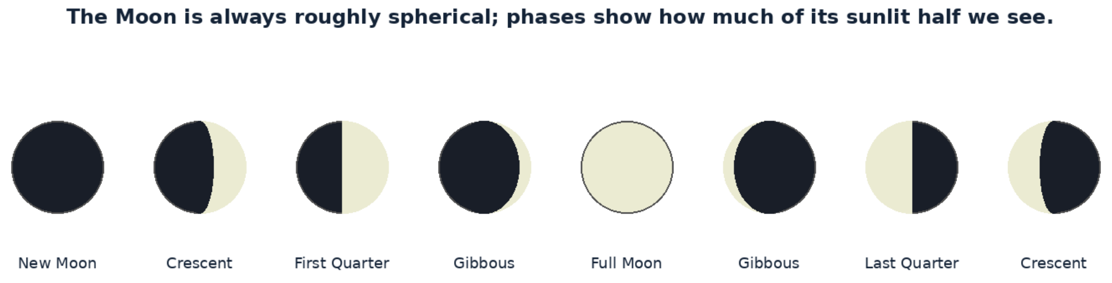


Figure 8.3 Moon phases show the changing portion of the Moon's sunlit half visible from Earth.

Example

An orbiter can map a whole region repeatedly, but a rover can touch, drill or analyse rocks at one site.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Assign teams different mission types and ask each to choose one instrument and one question it can answer.

Think and discuss

Why might scientists send both an orbiter and a rover to study the same world?

What you have learnt

- The Apollo programme landed astronauts on the Moon six times between 1969 and 1972.
- Robotic missions from several countries have mapped the Moon, studied its soil and searched for water-related materials.
- Different mission types provide different scales and kinds of lunar evidence.

EXERCISES · MOON EXPLORATION MISSIONS

Exercises

A. Choose the correct answer

1. Which statement about moon exploration missions is correct?
A. A rover studies the whole Moon from far away.
B. Orbiters, landers, rovers and sample-return missions study the Moon in different ways.
C. Moon samples cannot be studied after returning to Earth.
D. An orbiter must always land after one circle.

2. What does the word “orbiter” mean in this lesson?
A. a mobile robot that travels on a surface
B. a spacecraft designed to reach a surface
C. a mission that brings material back to Earth
D. a spacecraft that travels around a world

3. Which observation or example best supports the lesson idea?
A. A notebook is closed on a table.
B. A glass is filled with water.
C. A pencil is sharpened.
D. An orbiter can map a whole region repeatedly, but a rover can touch, drill or analyse rocks at one site.

4. Which word means “a spacecraft that travels around a world”?
A. lander
B. rover
C. sample return
D. orbiter

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Assign teams different mission types and ask each to choose one instrument and one question it can answer.

6. Which sentence gives the best summary of this lesson?
A. Moon exploration missions can be understood without observation, evidence or comparison.
B. Every object or event connected with moon exploration missions behaves in exactly the same way.
C. Different mission types provide different scales and kinds of lunar evidence.
D. The main idea of moon exploration missions is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain moon exploration missions.
2. Write two important facts about moon exploration missions.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why might scientists send both an orbiter and a rover to study the same world?

2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Samples brought back by Apollo and robotic missions are still studied with new instruments decades later. Permanently shadowed lunar craters can remain extremely cold and may preserve water ice.

UNIT 8 · CHAPTER 4

Mars Exploration Missions

Learning outcomes

- Explain mars exploration missions using simple scientific language.
- Use the words Mars and rover correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Mars missions investigate a cold, rocky planet that once had rivers, lakes and a wetter climate. Spacecraft study its atmosphere, surface, interior and potential for ancient microbial life.

Understanding the idea

Orbiters map Mars, relay communications and observe weather and surface changes.

Landers and rovers analyse rocks, soil, atmosphere and underground clues at selected sites.

Mars has a thin carbon-dioxide atmosphere, polar ice caps, giant volcanoes and deep canyons.

Because signals take minutes to travel, Mars robots need some ability to navigate and protect themselves.

Every mission depends on many roles. Astronauts are visible members of a much larger team of engineers, scientists, doctors and controllers.

The important words in this chapter include Mars, rover, autonomous. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

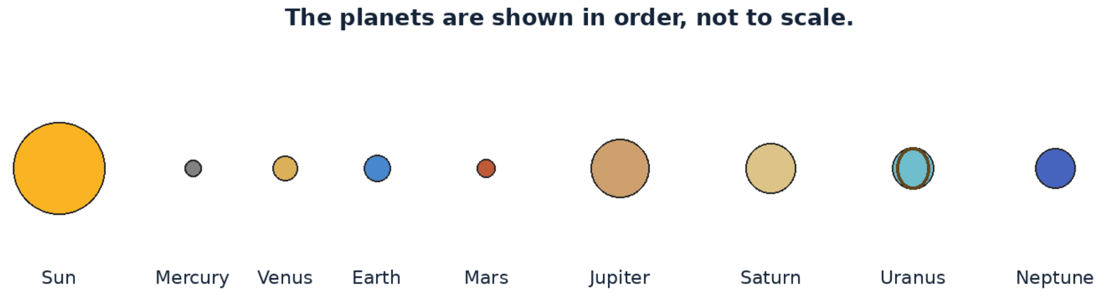


Figure 8.4 The Sun and eight planets shown in order. Sizes and distances are not to scale.

Example

A rover can receive a daily plan from Earth but may stop automatically if its cameras detect a dangerous rock.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Build a paper Mars map with obstacles. Write step-by-step rover commands, then test them without changing instructions mid-run.

Think and discuss

Why is sending a human-controlled joystick signal continuously to Mars impractical?

What you have learnt

- Orbiters map Mars, relay communications and observe weather and surface changes.
- Landers and rovers analyse rocks, soil, atmosphere and underground clues at selected sites.
- Mars exploration combines orbital mapping with careful surface investigation and autonomous robotics.

Exercises

A. Choose the correct answer

1. Which statement about mars exploration missions is correct?

A. Every Mars mission carries astronauts.
B. Mars missions use orbiters and robots to study geology, atmosphere and past water.
C. Mars robots can be controlled with no signal delay.
D. Mars has a thick oxygen atmosphere like Earth.

2. What does the word “Mars” mean in this lesson?

A. to receive and pass along a signal
B. a mobile robotic surface explorer
C. the fourth planet from the Sun
D. able to make some decisions using onboard systems

3. Which observation or example best supports the lesson idea?

A. A rover can receive a daily plan from Earth but may stop automatically if its cameras detect a dangerous rock.
B. A pencil is sharpened.
C. A notebook is closed on a table.
D. A glass is filled with water.

4. Which word means “the fourth planet from the Sun”?

A. Mars
B. rover
C. autonomous
D. relay

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Build a paper Mars map with obstacles. Write step-by-step rover commands, then test them without changing instructions mid-run.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Mars exploration combines orbital mapping with careful surface investigation and autonomous robotics.
B. Mars exploration missions can be understood without observation, evidence or comparison.
C. Every object or event connected with mars exploration missions behaves in exactly the same way.
D. The main idea of mars exploration missions is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain mars exploration missions.
2. Write two important facts about mars exploration missions.

3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is sending a human-controlled joystick signal continuously to Mars impractical?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: NASA's Ingenuity helicopter made the first powered, controlled flights on another planet in 2021. Mars dust can cover solar panels and also create large regional or global storms.

UNIT 8 · CHAPTER 5

Famous Space Missions

Learning outcomes

- Explain famous space missions using simple scientific language.
- Use the words space age and space telescope correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Famous space missions are remembered because they opened new paths, answered important questions or demonstrated difficult technologies. Their value comes from evidence and learning, not fame alone.

Understanding the idea

Sputnik 1 became the first artificial satellite in 1957, beginning the space age.

Apollo 11 achieved the first human landing on the Moon in 1969.

Voyager 1 and 2 explored the outer planets and continue sending information from beyond the heliosphere.

Space telescopes such as Hubble and the James Webb Space Telescope study distant stars, galaxies and planetary systems above much of Earth's atmosphere.

Every mission depends on many roles. Astronauts are visible members of a much larger team of engineers, scientists, doctors and controllers.

Notice how the terms space age, space telescope, heliosphere appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

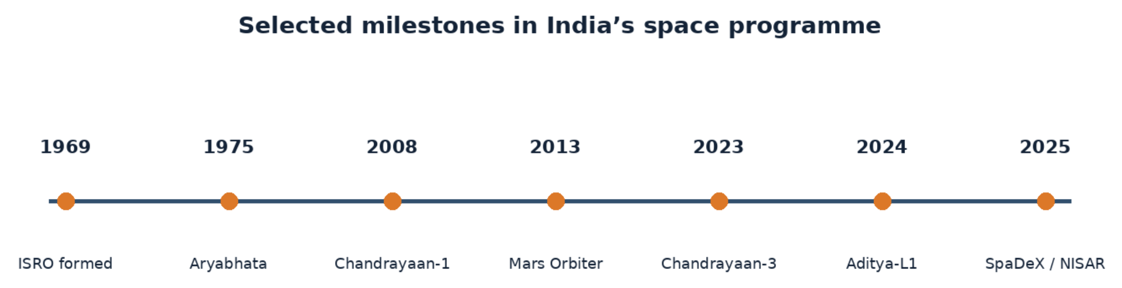


Figure 8.5 Selected milestones in India's space programme, arranged as a simple timeline.

Example

A mission can become famous for a "first," a major discovery, a long operating life or a new technology.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Choose one mission and make a mission badge showing its destination, tool, achievement and date.

Think and discuss

Which is more important when judging a mission: popularity or the evidence and capability it produced?

What you have learnt

- Sputnik 1 became the first artificial satellite in 1957, beginning the space age.
- Apollo 11 achieved the first human landing on the Moon in 1969.
- Space missions become milestones by achieving new capabilities and knowledge.

EXERCISES · FAMOUS SPACE MISSIONS

Exercises

A. Choose the correct answer

1. Which statement about famous space missions is correct?
A. A mission is scientifically important only if it carries people.
B. Voyager explored only Earth's oceans.
C. Famous missions are milestones because of discoveries, firsts or new capabilities.
D. Sputnik 1 was the first person on the Moon.

2. What does the word "space age" mean in this lesson?
A. the period beginning with the first artificial satellites
B. an important stage or achievement
C. a telescope operating above Earth's atmosphere
D. the broad region influenced by the solar wind

3. Which observation or example best supports the lesson idea?
A. A glass is filled with water.
B. A mission can become famous for a "first," a major discovery, a long operating life or a new technology.
C. A pencil is sharpened.
D. A notebook is closed on a table.

4. Which word means "the period beginning with the first artificial satellites"?
A. space telescope
B. space age
C. heliosphere
D. milestone

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Guess once and do not record anything.
B. Choose one mission and make a mission badge showing its destination, tool, achievement and date.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?
A. Famous space missions can be understood without observation, evidence or comparison.
B. Every object or event connected with famous space missions behaves in exactly the same way.
C. Space missions become milestones by achieving new capabilities and knowledge.
D. The main idea of famous space missions is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain famous space missions.

2. Write two important facts about famous space missions.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Which is more important when judging a mission: popularity or the evidence and capability it produced?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Voyager 1 is the most distant human-made object from Earth as of 2026. The golden records aboard both Voyager spacecraft carry sounds and images representing life and culture on Earth.

UNIT 8 REVIEW

Review · Space Exploration

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of who are astronauts? using one example.
2. Why is recycling water especially important on a space station?
3. Write one correct statement and one common mistake about moon exploration missions.
4. Explain the main idea of mars exploration missions using one example.
5. Which is more important when judging a mission: popularity or the evidence and capability it produced?
6. Create one new question from this unit and write its answer.

B. Compare and connect

1. Write one connection between “Who are Astronauts?” and “Famous Space Missions”.
2. Write one connection between “Space Stations” and “Mars Exploration Missions”.

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 8 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about who are astronauts? is correct?
- A. Astronauts combine specialised knowledge, physical training and teamwork.
- B. Astronauts are chosen only because they can run fast.
- C. Every astronaut has exactly the same job.
- D. Astronauts operate without help from Earth.

2. Which statement about space stations is correct?
- A. Space stations need no air, power or water systems.
- B. Space stations are orbiting laboratories and homes for crews.
- C. The ISS completes one orbit each year.
- D. A space station rests on a tower above Earth.

3. Which statement about moon exploration missions is correct?
- A. A rover studies the whole Moon from far away.
- B. Orbiters, landers, rovers and sample-return missions study the Moon in different ways.
- C. Moon samples cannot be studied after returning to Earth.
- D. An orbiter must always land after one circle.

4. Which statement about mars exploration missions is correct?
- A. Every Mars mission carries astronauts.
- B. Mars missions use orbiters and robots to study geology, atmosphere and past water.

C. Mars robots can be controlled with no signal delay.
D. Mars has a thick oxygen atmosphere like Earth.

5. Which statement about famous space missions is correct?

A. A mission is scientifically important only if it carries people.
B. Voyager explored only Earth's oceans.
C. Famous missions are milestones because of discoveries, firsts or new capabilities.
D. Sputnik 1 was the first person on the Moon.

6. Which word means “a person trained for space travel or work”?

A. cosmonaut
B. astronaut
C. mission specialist
D. ground team

7. Which word means “a spacecraft designed for people to live and work in orbit”?

A. ISS
B. module
C. space station
D. resupply

8. Which word means “a spacecraft that travels around a world”?

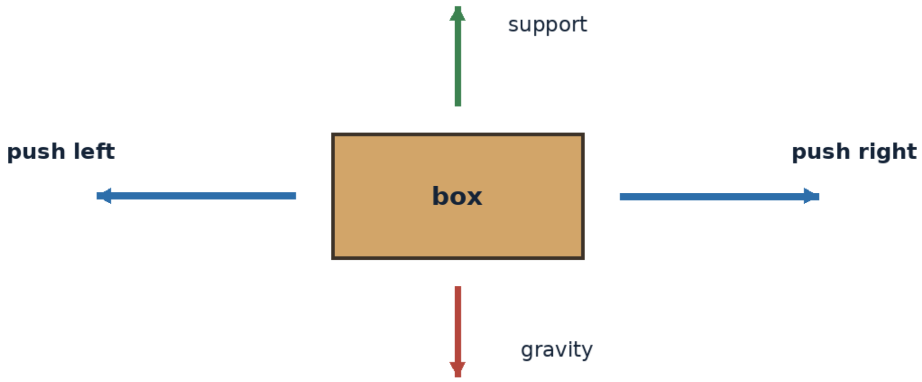
A. lander
B. rover
C. sample return
D. orbiter

UNIT 9

Basic Physics for Space Science

Space science rests on basic ideas from physics. Gravity, force, motion, light, heat, machines and the atmosphere are introduced through observations that children can make safely.

Forces are pushes or pulls. Their size and direction affect motion.



Chapters in this unit

1. Gravity
2. Force
3. Motion
4. Light
5. Shadows
6. Heat
7. Simple Machines
8. Air and Atmosphere

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 9 · CHAPTER 1

Gravity

Learning outcomes

- Explain gravity using simple scientific language.
- Use the words gravity and mass correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Gravity is the attraction between objects that have mass. It gives objects weight near Earth, holds the atmosphere and oceans, shapes planets and guides moons, planets and spacecraft in orbit.

Understanding the idea

More massive objects produce stronger gravitational effects, while greater distance weakens the attraction.

Earth pulls objects toward its centre, which we experience as “down.”

An orbit is a continuous fall around a curved world, not a place where gravity disappears.

The Moon’s gravity helps create tides, and the Sun’s gravity holds the solar system together.

Ask what force or energy causes a change. Use simple measurements and fair comparisons whenever possible.

Do not learn gravity, mass, weight separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

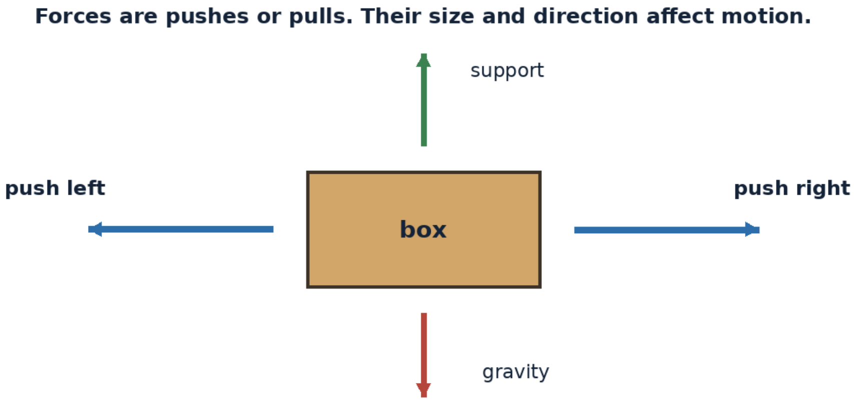


Figure 9.1 Forces are pushes or pulls. Their size and direction can change motion.

Example

When a ball is dropped, Earth pulls the ball downward and the ball also pulls Earth upward by a tiny amount.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Drop a flat sheet of paper and a crumpled sheet. Compare air resistance, then discuss that gravity acts on both.

Think and discuss

Why do astronauts float if gravity is still acting on their spacecraft?

What you have learnt

- More massive objects produce stronger gravitational effects, while greater distance weakens the attraction.
- Earth pulls objects toward its centre, which we experience as “down.”
- Gravity attracts masses and can bend motion into orbits.

EXERCISES · GRAVITY

Exercises

A. Choose the correct answer

1. Which statement about gravity is correct?

A. Objects in orbit are beyond all gravity.
B. Gravity holds moons, planets and spacecraft in orbit.
C. Gravity exists only on Earth’s surface.
D. Gravity pushes masses away from each other.

2. What does the word “gravity” mean in this lesson?

A. the amount of matter in an object
B. motion controlled mainly by gravity
C. the force of gravity on an object
D. the attraction between objects with mass

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A notebook is closed on a table.
C. When a ball is dropped, Earth pulls the ball downward and the ball also pulls Earth upward by a tiny amount.
D. A glass is filled with water.

4. Which word means “the attraction between objects with mass”?

A. mass
B. weight
C. gravity
D. free fall

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Drop a flat sheet of paper and a crumpled sheet. Compare air resistance, then discuss that gravity acts on both.

6. Which sentence gives the best summary of this lesson?

A. Gravity can be understood without observation, evidence or comparison.
B. Every object or event connected with gravity behaves in exactly the same way.
C. The main idea of gravity is unrelated to science or everyday experience.
D. Gravity attracts masses and can bend motion into orbits.

B. Answer in one or two sentences

1. In one sentence, explain gravity.
2. Write two important facts about gravity.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why do astronauts float if gravity is still acting on their spacecraft?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Astronauts in orbit feel weightless because they are in free fall, even though Earth's gravity there remains strong. Gravity helps gather gas and dust into stars and planets.

UNIT 9 · CHAPTER 2

Force

Learning outcomes

- Explain force using simple scientific language.
- Use the words force and balanced forces correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A force is a push or pull that can change an object's motion or shape. Forces have both strength and direction.

Understanding the idea

Balanced forces do not change an object's overall motion, while unbalanced forces cause acceleration.

Contact forces include friction and pushes; non-contact forces include gravity and magnetic force.

A rocket's thrust is a force, and air drag is a force acting against motion through the atmosphere.

Force is measured in newtons, named after scientist Isaac Newton.

Ask what force or energy causes a change. Use simple measurements and fair comparisons whenever possible.

The vocabulary of this chapter - force, balanced forces, unbalanced force - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

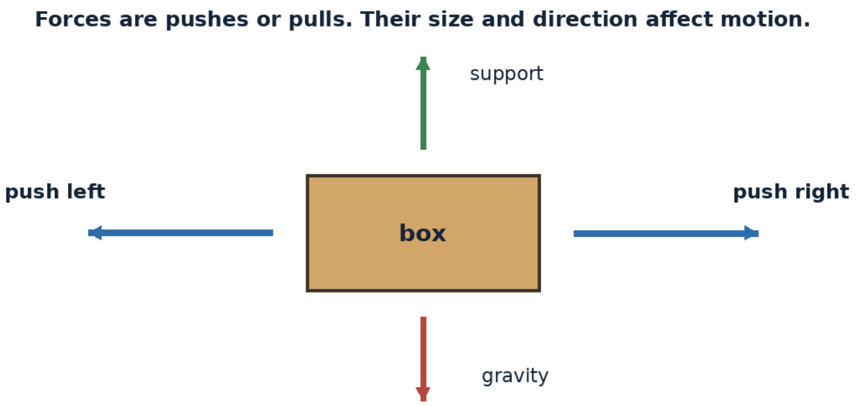


Figure 9.2 Forces are pushes or pulls. Their size and direction can change motion.

Example

A book resting on a table has downward weight balanced by an upward support force from the table.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Push an empty box gently and then more strongly. Compare how the change in motion depends on force.

Think and discuss

Can an object be moving even when the forces on it are balanced? Give an example.

What you have learnt

- Balanced forces do not change an object's overall motion, while unbalanced forces cause acceleration.
- Contact forces include friction and pushes; non-contact forces include gravity and magnetic force.
- Unbalanced force changes velocity; balanced force does not require an object to be at rest.

EXERCISES · FORCE

Exercises

A. Choose the correct answer

1. Which statement about force is correct?

A. Gravity is a type of contact force.
B. Balanced forces always make an object stop.
C. An unbalanced force changes an object's motion.
D. A force has no direction.

2. What does the word "force" mean in this lesson?

A. a push or pull
B. a force resisting sliding or motion between surfaces
C. a total force that causes acceleration
D. forces that cancel so there is no change in velocity

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A book resting on a table has downward weight balanced by an upward support force from the table.
C. A notebook is closed on a table.
D. A pencil is sharpened.

4. Which word means "a push or pull"?

A. balanced forces
B. unbalanced force
C. friction
D. force

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Push an empty box gently and then more strongly. Compare how the change in motion depends on force.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Force can be understood without observation, evidence or comparison.
B. Unbalanced force changes velocity; balanced force does not require an object to be at rest.
C. Every object or event connected with force behaves in exactly the same way.
D. The main idea of force is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain force.
2. Write two important facts about force.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Can an object be moving even when the forces on it are balanced? Give an example.
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: A force can change direction without greatly changing speed, as gravity does for a near-circular orbit. Friction can be helpful, such as for walking, or unwanted, such as heating moving machinery.

UNIT 9 · CHAPTER 3

Motion

Learning outcomes

- Explain motion using simple scientific language.
- Use the words motion and speed correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Motion is a change in position over time. To describe motion clearly, we compare an object with a reference point and may measure distance, time, speed and direction.

Understanding the idea

Speed tells how much distance is covered in a certain time.

Velocity includes both speed and direction.

Acceleration is any change in velocity, including speeding up, slowing down or turning.

An object can move at constant speed while accelerating if its direction is changing, as in circular motion.

Ask what force or energy causes a change. Use simple measurements and fair comparisons whenever possible.

The important words in this chapter include motion, speed, velocity. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

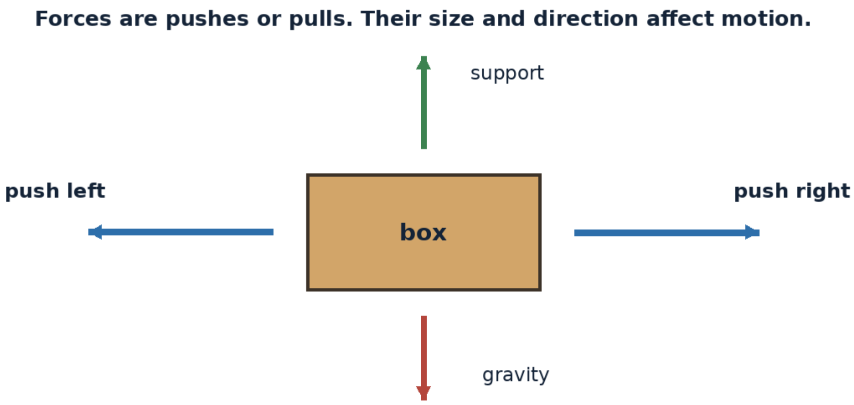


Figure 9.3 Forces are pushes or pulls. Their size and direction can change motion.

Example

A satellite in circular orbit can keep nearly the same speed while gravity continuously changes its direction.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Measure how far a toy car travels in five seconds on two surfaces. Compare average speeds.

Think and discuss

A car turns a corner without speeding up. Has its velocity changed? Explain.

What you have learnt

- Speed tells how much distance is covered in a certain time.
- Velocity includes both speed and direction.
- Motion must be described relative to a reference point and can change in speed or direction.

Exercises

A. Choose the correct answer

1. Which statement about motion is correct?

A. Motion can be described without any reference point.
B. Velocity means speed only and never direction.
C. Acceleration includes changes in speed or direction.
D. Turning at constant speed is not acceleration.

2. What does the word "motion" mean in this lesson?

A. a change in velocity
B. distance travelled per unit time
C. a change in position over time
D. speed in a stated direction

3. Which observation or example best supports the lesson idea?

A. A satellite in circular orbit can keep nearly the same speed while gravity continuously changes its direction.
B. A pencil is sharpened.
C. A glass is filled with water.
D. A notebook is closed on a table.

4. Which word means "a change in position over time"?

A. motion
B. speed
C. velocity
D. acceleration

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Measure how far a toy car travels in five seconds on two surfaces. Compare average speeds.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Motion can be understood without observation, evidence or comparison.
B. Every object or event connected with motion behaves in exactly the same way.
C. The main idea of motion is unrelated to science or everyday experience.
D. Motion must be described relative to a reference point and can change in speed or direction.

B. Answer in one or two sentences

1. In one sentence, explain motion.
2. Write two important facts about motion.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. A car turns a corner without speeding up. Has its velocity changed? Explain.
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: There is no single "still" position for everything; motion depends on the chosen reference point. Earth rotates and revolves while the Sun moves through the Milky Way.

UNIT 9 · CHAPTER 4

Light

Learning outcomes

- Explain light using simple scientific language.
- Use the words light and visible light correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Light is a form of energy that travels as electromagnetic waves. Visible light is the small range our eyes detect, while space science also uses radio, infrared, ultraviolet, X-ray and gamma-ray observations.

Understanding the idea

Light travels extremely fast in a vacuum, about 300,000 kilometres per second.

Objects are visible when light from them enters our eyes, either emitted by the object or reflected from it.

Different materials can transmit, absorb, reflect, scatter or bend light.

A spectrum separates light by wavelength and can reveal temperature and chemical composition.

Ask what force or energy causes a change. Use simple measurements and fair comparisons whenever possible.

Notice how the terms light, visible light, spectrum appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

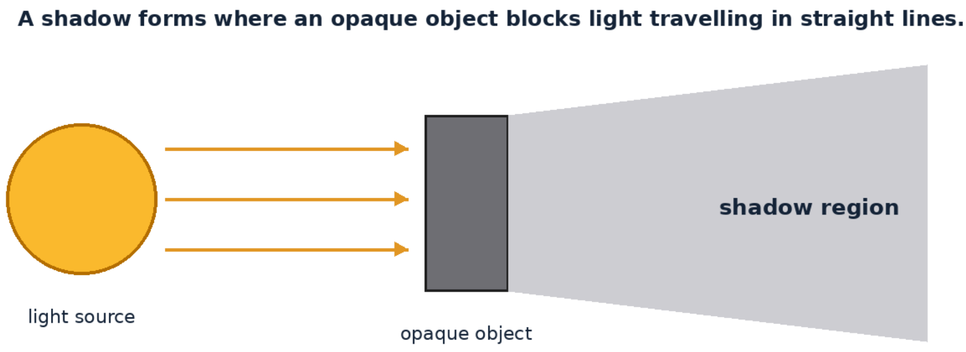


Figure 9.4 A shadow forms where an opaque object blocks light travelling in straight lines.

Example

The Moon is visible because it reflects sunlight, while the Sun is visible because it emits light.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Shine a torch onto foil, paper and clear plastic. Compare reflection, absorption and transmission.

Think and discuss

Why can a radio telescope study space even though radio waves are invisible to our eyes?

What you have learnt

- Light travels extremely fast in a vacuum, about 300,000 kilometres per second.
- Objects are visible when light from them enters our eyes, either emitted by the object or reflected from it.
- Light carries energy and information across space in many wavelengths.

EXERCISES · LIGHT

Exercises

A. Choose the correct answer

1. Which statement about light is correct?

A. Astronomers study many kinds of electromagnetic light, not only visible light.
B. Light needs ordinary air to cross space.
C. The Moon produces all of its visible light.
D. Radio waves cannot carry information because eyes cannot see them.

2. What does the word “light” mean in this lesson?

A. the wavelengths detected by human eyes
B. light arranged by wavelength or energy
C. to bounce light from a surface
D. electromagnetic energy that travels through space

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A glass is filled with water.
C. The Moon is visible because it reflects sunlight, while the Sun is visible because it emits light.
D. A notebook is closed on a table.

4. Which word means “electromagnetic energy that travels through space”?

A. visible light
B. light
C. spectrum
D. reflect

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Shine a torch onto foil, paper and clear plastic. Compare reflection, absorption and transmission.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Light can be understood without observation, evidence or comparison.
B. Light carries energy and information across space in many wavelengths.
C. Every object or event connected with light behaves in exactly the same way.
D. The main idea of light is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain light.
2. Write two important facts about light.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can a radio telescope study space even though radio waves are invisible to our eyes?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Light from the Sun takes about eight minutes to reach Earth, so we always see the Sun as it was minutes ago. A rainbow forms when water droplets bend, reflect and separate sunlight into colours.

UNIT 9 · CHAPTER 5

Shadows

Learning outcomes

- Explain shadows using simple scientific language.
- Use the words opaque and transparent correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

A shadow forms when an opaque object blocks light from reaching a surface. Shadow size, direction and sharpness depend on the positions and sizes of the light source, object and screen.

Understanding the idea

An opaque object blocks most visible light, a transparent material lets much light through and a translucent material spreads some light.

A small or distant light source often creates a sharper shadow than a large nearby source.

The darkest central shadow is called the umbra, while a partly shaded region is the penumbra.

Eclipses are large-scale shadow events involving the Sun, Earth and Moon.

Ask what force or energy causes a change. Use simple measurements and fair comparisons whenever possible.

Do not learn opaque, transparent, umbra separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

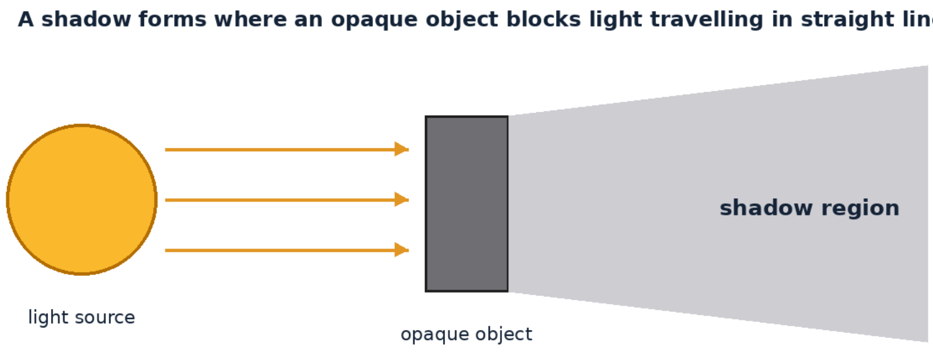


Figure 9.5 A shadow forms where an opaque object blocks light travelling in straight lines.

Example

A person's outdoor shadow is long when the Sun is low and shorter when the Sun is high.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Move a toy closer to and farther from a torch while keeping the screen fixed. Record shadow size.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Why does moving an object closer to a lamp usually make its shadow larger?

What you have learnt

- An opaque object blocks most visible light, a transparent material lets much light through and a translucent material spreads some light.
- A small or distant light source often creates a sharper shadow than a large nearby source.
- Shadows reveal the geometry of a light source, object and receiving surface.

EXERCISES · SHADOWS

Exercises

A. Choose the correct answer

1. Which statement about shadows is correct?

A. Shadow direction never changes.
B. A shadow forms when an opaque object blocks light.
C. A shadow is a black substance released by objects.
D. Transparent glass always makes a completely dark shadow.

2. What does the word "opaque" mean in this lesson?

A. the darkest central part of a shadow
B. blocking most visible light
C. a region of partial shadow
D. allowing much light to pass through clearly

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A pencil is sharpened.
C. A notebook is closed on a table.
D. A person's outdoor shadow is long when the Sun is low and shorter when the Sun is high.

4. Which word means "blocking most visible light"?

A. transparent
B. umbra
C. opaque
D. penumbra

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Move a toy closer to and farther from a torch while keeping the screen fixed. Record shadow size.

6. Which sentence gives the best summary of this lesson?

A. Shadows can be understood without observation, evidence or comparison.
B. Every object or event connected with shadows behaves in exactly the same way.
C. The main idea of shadows is unrelated to science or everyday experience.
D. Shadows reveal the geometry of a light source, object and receiving surface.

B. Answer in one or two sentences

1. In one sentence, explain shadows.
2. Write two important facts about shadows.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why does moving an object closer to a lamp usually make its shadow larger?

2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The Sun is a large light source, so eclipse shadows can contain both umbra and penumbra. Astronomers used shadows in ancient times to estimate time, direction and even Earth's size.

UNIT 9 · CHAPTER 6

Heat

Learning outcomes

- Explain heat using simple scientific language.
- Use the words heat and temperature correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Heat is energy transferred because of a temperature difference. In space science, heating and cooling affect spacecraft, planets, atmospheres and living systems.

Understanding the idea

Conduction transfers energy through direct contact, convection through moving fluids and radiation through electromagnetic waves.

Space is mostly a vacuum, so spacecraft mainly gain and lose heat by radiation rather than air convection.

A spacecraft surface in sunlight can become hot while a shaded surface becomes very cold.

Thermal-control systems use coatings, insulation, radiators and heaters to keep equipment within safe temperatures.

Ask what force or energy causes a change. Use simple measurements and fair comparisons whenever possible.

The vocabulary of this chapter - heat, temperature, conduction - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

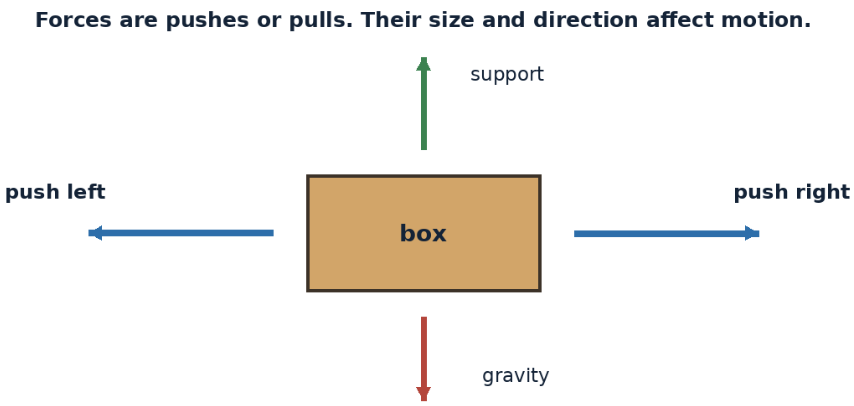


Figure 9.6 Forces are pushes or pulls. Their size and direction can change motion.

Example

A shiny emergency blanket reflects thermal radiation and can help reduce heat loss.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Place equal ice cubes on metal and wood surfaces. Observe which melts faster and discuss conduction.

Think and discuss

Why can a spacecraft overheat in sunlight even though outer space is often described as cold?

What you have learnt

- Conduction transfers energy through direct contact, convection through moving fluids and radiation through electromagnetic waves.
- Space is mostly a vacuum, so spacecraft mainly gain and lose heat by radiation rather than air convection.
- In vacuum, radiation is the main way spacecraft exchange heat with their surroundings.

EXERCISES · HEAT

Exercises

A. Choose the correct answer

1. Which statement about heat is correct?

A. Spacecraft mainly exchange heat by radiation in the vacuum of space.
B. Heat and temperature mean exactly the same thing.
C. Convection works through completely empty space.
D. No object can become hot in space.

2. What does the word “heat” mean in this lesson?

A. heat transfer through contact
B. energy transferred because of a temperature difference
C. a measure related to how energetic particles are
D. energy transfer by electromagnetic waves

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A notebook is closed on a table.
C. A glass is filled with water.
D. A shiny emergency blanket reflects thermal radiation and can help reduce heat loss.

4. Which word means “energy transferred because of a temperature difference”?

A. temperature
B. conduction
C. radiation
D. heat

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Place equal ice cubes on metal and wood surfaces. Observe which melts faster and discuss conduction.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Heat can be understood without observation, evidence or comparison.
B. In vacuum, radiation is the main way spacecraft exchange heat with their surroundings.
C. Every object or event connected with heat behaves in exactly the same way.
D. The main idea of heat is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain heat.
2. Write two important facts about heat.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can a spacecraft overheat in sunlight even though outer space is often described as cold?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Multi-layer insulation gives many spacecraft a gold or silver foil-like appearance. Temperature is a measure related to particle energy; heat is energy being transferred.

UNIT 9 · CHAPTER 7

Simple Machines

Learning outcomes

- Explain simple machines using simple scientific language.
- Use the words simple machine and lever correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Simple machines change the size or direction of a force. The traditional six are lever, wheel and axle, pulley, inclined plane, wedge and screw.

Understanding the idea

Simple machines do not remove work; they trade force for distance or change direction.

Levers and pulleys help lift or position loads, while wheels reduce resistance during transport.

Screws convert turning motion into forward motion and are used widely in spacecraft assembly.

Robotic arms combine joints, motors and machine principles to move equipment in space.

Ask what force or energy causes a change. Use simple measurements and fair comparisons whenever possible.

The important words in this chapter include simple machine, lever, pulley. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

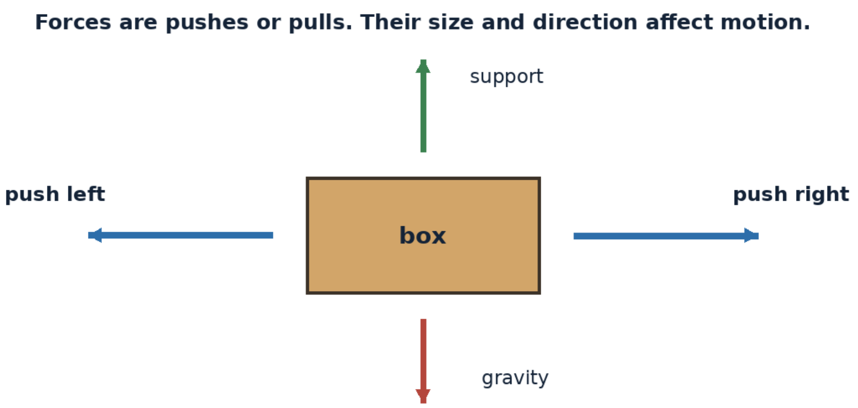


Figure 9.7 Forces are pushes or pulls. Their size and direction can change motion.

Example

A ramp makes it possible to raise a heavy box using less force over a longer distance.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Build a lever with a ruler and eraser fulcrum. Move the fulcrum and compare the force needed to lift a small object.

Think and discuss

Why does a longer ramp require less force but more travel distance?

What you have learnt

- Simple machines do not remove work; they trade force for distance or change direction.
- Levers and pulleys help lift or position loads, while wheels reduce resistance during transport.
- Simple machines trade force, distance and direction while conserving the overall work idea.

EXERCISES · SIMPLE MACHINES

Exercises

A. Choose the correct answer

1. Which statement about simple machines is correct?

A. A ramp always needs more force than lifting straight up.
B. A lever has no turning point.
C. Simple machines can reduce required force by increasing distance or changing direction.
D. Simple machines create energy from nothing.

2. What does the word “simple machine” mean in this lesson?

A. a grooved wheel used with a rope or cable
B. a basic device that changes force or motion
C. a rigid bar turning around a fulcrum
D. a sloping surface used to raise or lower objects

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A notebook is closed on a table.
C. A ramp makes it possible to raise a heavy box using less force over a longer distance.
D. A glass is filled with water.

4. Which word means “a basic device that changes force or motion”?

A. simple machine
B. lever
C. pulley
D. inclined plane

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Build a lever with a ruler and eraser fulcrum. Move the fulcrum and compare the force needed to lift a small object.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Simple machines can be understood without observation, evidence or comparison.
B. Every object or event connected with simple machines behaves in exactly the same way.
C. The main idea of simple machines is unrelated to science or everyday experience.
D. Simple machines trade force, distance and direction while conserving the overall work idea.

B. Answer in one or two sentences

1. In one sentence, explain simple machines.
2. Write two important facts about simple machines.

3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why does a longer ramp require less force but more travel distance?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The Canadarm2 robotic arm on the ISS can move equipment and assist astronauts during some operations. A bolt and nut are related to the screw simple machine because the spiral thread converts rotation into linear movement.

UNIT 9 · CHAPTER 8

Air and Atmosphere

Learning outcomes

- Explain air and atmosphere using simple scientific language.
- Use the words air and atmosphere correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Air is the mixture of gases around us, and Earth's atmosphere is the entire gaseous envelope surrounding the planet. It supports life, weather, flight and protection from harmful space conditions.

Understanding the idea

Dry air near sea level is mostly nitrogen and oxygen, with small amounts of other gases.

Air has mass and pressure even though it is usually invisible.

Atmospheric density decreases with height, making breathing and aircraft flight harder at high altitude.

The atmosphere burns up many small meteoroids and absorbs or scatters some harmful radiation.

Ask what force or energy causes a change. Use simple measurements and fair comparisons whenever possible.

Notice how the terms air, atmosphere, air pressure appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

The atmosphere becomes thinner with height; boundaries between layers are gradual.

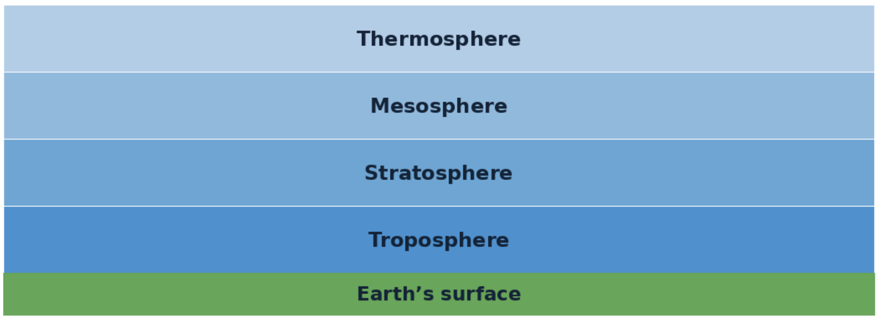


Figure 9.8 Earth's atmosphere becomes thinner with height and is described in broad layers.

Example

A sealed packet may puff up on a high mountain because outside air pressure is lower.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Push an upside-down cup containing tissue straight into water. The dry tissue shows that air occupies space.

Think and discuss

Why do astronauts need pressurised spacecraft and suits even when sunlight is available?

What you have learnt

- Dry air near sea level is mostly nitrogen and oxygen, with small amounts of other gases.
- Air has mass and pressure even though it is usually invisible.
- Earth's atmosphere is a thin, protective mixture of gases with mass and pressure.

EXERCISES · AIR AND ATMOSPHERE

Exercises

A. Choose the correct answer

1. Which statement about air and atmosphere is correct?
A. The atmosphere becomes denser with height.
B. Earth's atmosphere supports life and protects the surface.
C. Air has no mass or pressure.
D. Astronauts can breathe ordinary space because sunlight is present.

2. What does the word "air" mean in this lesson?
A. force produced by air pushing on surfaces
B. how much mass is packed into a volume
C. the mixture of gases around us
D. the gases surrounding a planet

3. Which observation or example best supports the lesson idea?
A. A pencil is sharpened.
B. A sealed packet may puff up on a high mountain because outside air pressure is lower.
C. A glass is filled with water.
D. A notebook is closed on a table.

4. Which word means "the mixture of gases around us"?
A. atmosphere
B. air
C. air pressure
D. density

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Push an upside-down cup containing tissue straight into water. The dry tissue shows that air occupies space.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?
A. Air and atmosphere can be understood without observation, evidence or comparison.
B. Earth's atmosphere is a thin, protective mixture of gases with mass and pressure.
C. Every object or event connected with air and atmosphere behaves in exactly the same way.
D. The main idea of air and atmosphere is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain air and atmosphere.
2. Write two important facts about air and atmosphere.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why do astronauts need pressurised spacecraft and suits even when sunlight is available?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: If Earth were the size of a classroom globe, most of the atmosphere would be represented by a very thin coating. Mars has an atmosphere too, but it is much thinner and mostly carbon dioxide.

UNIT 9 REVIEW

Review · Basic Physics for Space Science

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of gravity using one example.
2. Can an object be moving even when the forces on it are balanced? Give an example.
3. Write one correct statement and one common mistake about motion.
4. Explain the main idea of light using one example.
5. Why does moving an object closer to a lamp usually make its shadow larger?
6. Write one correct statement and one common mistake about heat.
7. Explain the main idea of simple machines using one example.
8. Why do astronauts need pressurised spacecraft and suits even when sunlight is available?

B. Compare and connect

1. Write one connection between “Gravity” and “Air and Atmosphere”.
2. Write one connection between “Force” and “Simple Machines”.

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 9 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about gravity is correct?
- A. Objects in orbit are beyond all gravity.
- B. Gravity holds moons, planets and spacecraft in orbit.
- C. Gravity exists only on Earth’s surface.
- D. Gravity pushes masses away from each other.

2. Which statement about force is correct?
- A. Gravity is a type of contact force.
- B. Balanced forces always make an object stop.
- C. An unbalanced force changes an object’s motion.
- D. A force has no direction.

3. Which statement about motion is correct?
- A. Motion can be described without any reference point.
- B. Velocity means speed only and never direction.
- C. Acceleration includes changes in speed or direction.
- D. Turning at constant speed is not acceleration.

4. Which statement about light is correct?
- A. Astronomers study many kinds of electromagnetic light, not only visible light.
- B. Light needs ordinary air to cross space.

C. The Moon produces all of its visible light.
D. Radio waves cannot carry information because eyes cannot see them.

5. Which statement about shadows is correct?
A. Shadow direction never changes.
B. A shadow forms when an opaque object blocks light.
C. A shadow is a black substance released by objects.
D. Transparent glass always makes a completely dark shadow.

6. Which statement about heat is correct?
A. Spacecraft mainly exchange heat by radiation in the vacuum of space.
B. Heat and temperature mean exactly the same thing.
C. Convection works through completely empty space.
D. No object can become hot in space.

7. Which statement about simple machines is correct?
A. A ramp always needs more force than lifting straight up.
B. A lever has no turning point.
C. Simple machines can reduce required force by increasing distance or changing direction.
D. Simple machines create energy from nothing.

8. Which statement about air and atmosphere is correct?
A. The atmosphere becomes denser with height.
B. Earth’s atmosphere supports life and protects the surface.
C. Air has no mass or pressure.
D. Astronauts can breathe ordinary space because sunlight is present.

UNIT 10

Scientific Reasoning & Logical Aptitude

Olympiad success depends on more than remembering facts. This unit develops observation, classification, pattern recognition, sequencing, mathematical reasoning and logical thinking.

Find the rule, test every position, then choose the missing item.

○ △ ○ △ ○ **?**

Answer: △

Chapters in this unit

1. Observation Skills
2. Picture Reasoning
3. Pattern Recognition
4. Classification
5. Sequencing
6. Simple Mathematical Reasoning
7. Logical Thinking

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 10 · CHAPTER 1

Observation Skills

Learning outcomes

- Explain observation skills using simple scientific language.
- Use the words observation and inference correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Observation is the careful collection of information using senses and measuring tools. Strong observers separate what they actually notice from what they guess or conclude.

Understanding the idea

An observation might describe colour, number, position, time, shape, sound or measurement.

An inference is an explanation based on observations and prior knowledge.

Repeated observations help reveal changes and reduce mistakes.

Tables, labelled sketches and checklists make observations easier to compare.

Do not guess from appearance alone. Identify the rule, test it against every item and reject an answer that breaks the rule.

The vocabulary of this chapter - observation, inference, measurement - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

Figure 10.1 A pattern question is solved by finding a rule and checking that it fits every position.

Example

“The Moon is low in the east at 7 p.m.” is an observation; “it will rain because the Moon is low” is an unsupported inference.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Study a tray of ten objects for thirty seconds, cover it and list details. Repeat using a checklist and compare accuracy.

Think and discuss

How can two people observe the same event but record different details?

What you have learnt

- An observation might describe colour, number, position, time, shape, sound or measurement.
- An inference is an explanation based on observations and prior knowledge.
- Record what is seen or measured before deciding what it means.

EXERCISES · OBSERVATION SKILLS

Exercises

A. Choose the correct answer

1. Which statement about observation skills is correct?

A. An observation is always a guess about the cause.
B. A scientific observation records what is actually seen or measured.
C. One quick look is always more reliable than repeated checks.
D. Precise records should avoid numbers and directions.

2. What does the word “observation” mean in this lesson?

A. an explanation drawn from observations and knowledge
B. closeness to the correct value or description
C. a quantity found using a standard unit or tool
D. information gathered by watching or measuring

3. Which observation or example best supports the lesson idea?

A. “The Moon is low in the east at 7 p.m.” is an observation; “it will rain because the Moon is low” is an unsupported inference.
B. A glass is filled with water.
C. A notebook is closed on a table.
D. A pencil is sharpened.

4. Which word means “information gathered by watching or measuring”?

A. inference
B. measurement
C. accuracy
D. observation

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Study a tray of ten objects for thirty seconds, cover it and list details. Repeat using a checklist and compare accuracy.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Record what is seen or measured before deciding what it means.
B. Observation skills can be understood without observation, evidence or comparison.
C. Every object or event connected with observation skills behaves in exactly the same way.
D. The main idea of observation skills is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain observation skills.
2. Write two important facts about observation skills.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. How can two people observe the same event but record different details?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Scientists often photograph or measure the same target many times so they can detect small changes. A precise observation avoids vague words such as “somewhere” or “a lot” when a number or direction is available.

UNIT 10 · CHAPTER 2

Picture Reasoning

Learning outcomes

- Explain picture reasoning using simple scientific language.
- Use the words rotation and reflection correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Picture reasoning means using visual clues to solve a problem. The solver compares shapes, positions, directions, sizes, counts and changes rather than relying only on words.

Understanding the idea

First identify what stays the same and what changes from picture to picture.

Check rotation, reflection, movement, addition, removal and colour or shading patterns.

Ignore decorative details that do not affect the rule.

Test the chosen rule against every picture, not only the first two.

Do not guess from appearance alone. Identify the rule, test it against every item and reject an answer that breaks the rule.

The important words in this chapter include rotation, reflection, visual clue. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

Figure 10.2 A pattern question is solved by finding a rule and checking that it fits every position.

Example

If an arrow turns a quarter-turn clockwise in each box, the missing arrow should continue the same turn.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Draw a three-picture rule using triangles and dots. Ask a partner to complete the fourth picture and explain the rule.

Think and discuss

How can you tell whether a shape has rotated or been reflected?

What you have learnt

- First identify what stays the same and what changes from picture to picture.
- Check rotation, reflection, movement, addition, removal and colour or shading patterns.
- A picture answer is strong only when one consistent rule explains every step.

EXERCISES · PICTURE REASONING

Exercises

A. Choose the correct answer

1. Which statement about picture reasoning is correct?

A. The best rule may change randomly in every box.
B. Decorative colour is always the only clue.
C. Picture reasoning uses consistent visual rules such as rotation, reflection or counting.
D. A reflection and a rotation are exactly the same change.

2. What does the word "rotation" mean in this lesson?

A. a feature in a picture that helps solve a problem
B. following the same rule throughout
C. a flipped mirror image
D. turning around a point

3. Which observation or example best supports the lesson idea?

A. A notebook is closed on a table.
B. A glass is filled with water.
C. If an arrow turns a quarter-turn clockwise in each box, the missing arrow should continue the same turn.
D. A pencil is sharpened.

4. Which word means "turning around a point"?

A. rotation
B. reflection
C. visual clue
D. consistent

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Draw a three-picture rule using triangles and dots. Ask a partner to complete the fourth picture and explain the rule.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Picture reasoning can be understood without observation, evidence or comparison.
B. Every object or event connected with picture reasoning behaves in exactly the same way.
C. A picture answer is strong only when one consistent rule explains every step.
D. The main idea of picture reasoning is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain picture reasoning.
2. Write two important facts about picture reasoning.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. How can you tell whether a shape has rotated or been reflected?

2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Many visual puzzles include two rules at once, such as rotation plus an increasing number of dots. Looking at answer options can reveal which feature the question expects you to track.

UNIT 10 · CHAPTER 3

Pattern Recognition

Learning outcomes

- Explain pattern recognition using simple scientific language.
- Use the words pattern and sequence correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Pattern recognition is finding a rule in a repeated or changing sequence. Patterns may use numbers, shapes, colours, sounds, movements or positions.

Understanding the idea

A repeating pattern cycles through the same block, such as red-blue-blue.

A growing pattern adds or changes something in a regular way.

Number patterns may use addition, subtraction, doubling, halving or alternating rules.

A correct rule must predict earlier and later terms, not only one missing item.

Do not guess from appearance alone. Identify the rule, test it against every item and reject an answer that breaks the rule.

Notice how the terms pattern, sequence, repeating appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

Figure 10.3 A pattern question is solved by finding a rule and checking that it fits every position.

Example

The sequence 2, 4, 8, 16 follows a “multiply by two” rule.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Build a repeating pattern with objects, then hide one item. Ask a partner to name the missing item and rule.

Think and discuss

Why is “the next number is 10” weaker than “add 2 each time, so the next number is 10”?

What you have learnt

- A repeating pattern cycles through the same block, such as red-blue-blue.
- A growing pattern adds or changes something in a regular way.
- State the rule as well as the missing item.

Exercises

A. Choose the correct answer

1. Which statement about pattern recognition is correct?
A. A rule needs to fit just one pair of terms.
B. A pattern has no rule.
C. Pattern recognition identifies a rule that consistently predicts the sequence.
D. Only colours can form patterns.

2. What does the word “pattern” mean in this lesson?
A. returning in the same cycle
B. a pattern that increases or changes regularly
C. items placed in a particular order
D. an arrangement or change following a rule

3. Which observation or example best supports the lesson idea?
A. The sequence 2, 4, 8, 16 follows a “multiply by two” rule.
B. A glass is filled with water.
C. A pencil is sharpened.
D. A notebook is closed on a table.

4. Which word means “an arrangement or change following a rule”?
A. sequence
B. pattern
C. repeating
D. growing pattern

5. Which action is the best way to investigate the lesson idea safely and carefully?
A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Build a repeating pattern with objects, then hide one item. Ask a partner to name the missing item and rule.

6. Which sentence gives the best summary of this lesson?
A. State the rule as well as the missing item.
B. Pattern recognition can be understood without observation, evidence or comparison.
C. Every object or event connected with pattern recognition behaves in exactly the same way.
D. The main idea of pattern recognition is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain pattern recognition.
2. Write two important facts about pattern recognition.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is “the next number is 10” weaker than “add 2 each time, so the next number is 10”?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Astronomers recognise patterns in repeated observations, such as Moon phases and variable-star brightness. Some sequences have more than one possible rule unless enough terms or conditions are given.

UNIT 10 · CHAPTER 4

Classification

Learning outcomes

- Explain classification using simple scientific language.
- Use the words classification and characteristic correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Classification means grouping objects by shared characteristics. A good classification rule is clear, useful and applied consistently.

Understanding the idea

The same objects can be classified in different valid ways depending on the question.

Scientists classify planets by composition, position, size or other properties.

An odd-one-out question requires identifying the rule shared by the other items.

A classification can use more than one level, such as space objects, then stars and non-stars.

Do not guess from appearance alone. Identify the rule, test it against every item and reject an answer that breaks the rule.

Do not learn classification, characteristic, category separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

Figure 10.4 A pattern question is solved by finding a rule and checking that it fits every position.

Example

Mercury, Venus, Earth and Mars can be grouped as rocky planets, while Jupiter is the odd one out in that set.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Sort cards labelled Sun, Earth, Moon, Mars, comet and satellite using two different rules. Explain both.

Safety: Do not look directly at the Sun. Carry out outdoor work with responsible adult supervision.

Think and discuss

Can one object belong to several useful groups at the same time? Give an example.

What you have learnt

- The same objects can be classified in different valid ways depending on the question.
- Scientists classify planets by composition, position, size or other properties.
- Classification depends on a stated characteristic and must be consistent.

EXERCISES · CLASSIFICATION

Exercises

A. Choose the correct answer

1. Which statement about classification is correct?

A. There is only one possible way to group any objects.
B. An odd-one-out question needs no shared rule.
C. Classification groups objects using clear shared characteristics.
D. A classification rule may change for each item.

2. What does the word “classification” mean in this lesson?

A. grouping by shared characteristics
B. a feature used for comparison
C. the item that does not share the chosen rule
D. a group defined by a rule

3. Which observation or example best supports the lesson idea?

A. Mercury, Venus, Earth and Mars can be grouped as rocky planets, while Jupiter is the odd one out in that set.
B. A notebook is closed on a table.
C. A glass is filled with water.
D. A pencil is sharpened.

4. Which word means “grouping by shared characteristics”?

A. characteristic
B. category
C. classification
D. odd one out

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Sort cards labelled Sun, Earth, Moon, Mars, comet and satellite using two different rules. Explain both.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Classification can be understood without observation, evidence or comparison.
B. Every object or event connected with classification behaves in exactly the same way.
C. Classification depends on a stated characteristic and must be consistent.
D. The main idea of classification is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain classification.
2. Write two important facts about classification.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Can one object belong to several useful groups at the same time? Give an example.
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Biologists, geologists and astronomers all classify objects, but their categories change when new evidence appears. Pluto’s classification changed because astronomers agreed on a more precise planet definition, not because Pluto itself changed.

UNIT 10 · CHAPTER 5

Sequencing

Learning outcomes

- Explain sequencing using simple scientific language.
- Use the words sequence and chronological correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Sequencing means placing events, stages or objects in a logical order. The order may follow time, size, position, cause and effect or a repeating cycle.

Understanding the idea

Time sequences use clues such as first, next, before, after and finally.

Process sequences arrange steps so each one prepares for the next.

Cycles return to the beginning, so any stage can be chosen as the starting point if the order remains correct.

Astronomy sequences include planet order, Moon phases and mission stages.

Do not guess from appearance alone. Identify the rule, test it against every item and reject an answer that breaks the rule.

The vocabulary of this chapter - sequence, chronological, cycle - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

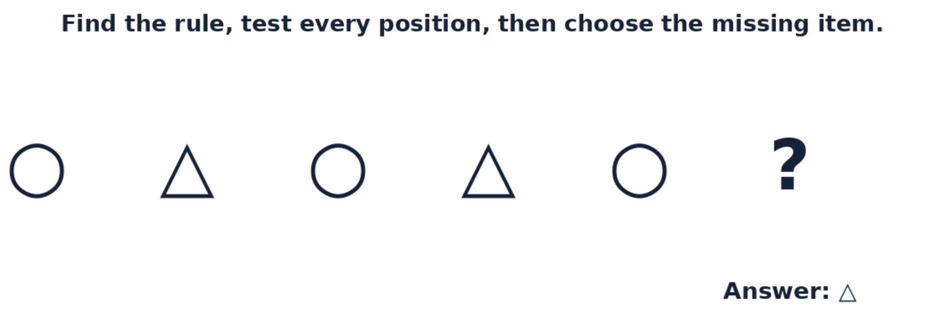


Figure 10.5 A pattern question is solved by finding a rule and checking that it fits every position.

Example

Launch, stage separation, orbit insertion and satellite deployment form a mission sequence.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Write five steps for preparing a safe sky observation. Cut them apart, mix them and ask a partner to reorder them.

Think and discuss

How is a cycle different from a one-way sequence?

What you have learnt

- Time sequences use clues such as first, next, before, after and finally.
- Process sequences arrange steps so each one prepares for the next.
- Use clue words and cause-effect links to place stages in a defensible order.

EXERCISES · SEQUENCING

Exercises

A. Choose the correct answer

1. Which statement about sequencing is correct?

A. The final step must always happen first.
B. Cycles never repeat.
C. Sequencing arranges events or objects by a clear order such as time or process.
D. A sequence is a random collection with no order.

2. What does the word "sequence" mean in this lesson?

A. a relationship in which one event helps produce another
B. a sequence that repeats
C. arranged by time
D. an ordered set of items or events

3. Which observation or example best supports the lesson idea?

A. Launch, stage separation, orbit insertion and satellite deployment form a mission sequence.
B. A pencil is sharpened.
C. A notebook is closed on a table.
D. A glass is filled with water.

4. Which word means "an ordered set of items or events"?

A. chronological
B. cycle
C. cause and effect
D. sequence

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Write five steps for preparing a safe sky observation. Cut them apart, mix them and ask a partner to reorder them.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Use clue words and cause-effect links to place stages in a defensible order.
B. Sequencing can be understood without observation, evidence or comparison.
C. Every object or event connected with sequencing behaves in exactly the same way.
D. The main idea of sequencing is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain sequencing.
2. Write two important facts about sequencing.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. How is a cycle different from a one-way sequence?

2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: A cause can come before an effect even when the effect is noticed more strongly. Mission checklists use strict sequences because doing a correct action at the wrong time can be unsafe.

UNIT 10 · CHAPTER 6

Simple Mathematical Reasoning

Learning outcomes

- Explain simple mathematical reasoning using simple scientific language.
- Use the words estimate and unit correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Mathematical reasoning uses numbers, shapes, measurements and relationships to solve problems. The goal is to choose operations because they match the situation, not because a familiar number appears.

Understanding the idea

Read the question, identify known information and decide what must be found.

Estimate before calculating so an unreasonable answer is easier to notice.

Units matter: kilometres, seconds, degrees and kilograms describe different quantities.

Tables and diagrams can reveal patterns that are hard to see in a long sentence.

Do not guess from appearance alone. Identify the rule, test it against every item and reject an answer that breaks the rule.

The important words in this chapter include estimate, unit, operation. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

Figure 10.6 A pattern question is solved by finding a rule and checking that it fits every position.

Example

If a satellite completes 16 orbits in one day, it completes about 32 orbits in two days, assuming the rate stays the same.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Estimate how many minutes you spend outdoors in one week, then calculate and compare.

Think and discuss

Why should an answer of “20 kilograms” be rejected if the question asks for time?

What you have learnt

- Read the question, identify known information and decide what must be found.
- Estimate before calculating so an unreasonable answer is easier to notice.
- Match the operation and unit to the meaning of the problem, then check reasonableness.

EXERCISES · SIMPLE MATHEMATICAL REASONING

Exercises

A. Choose the correct answer

1. Which statement about simple mathematical reasoning is correct?

A. Checking an answer is unnecessary.
B. The largest number in the question must always be added.
C. Units do not matter in an answer.
D. Mathematical reasoning includes choosing operations, using units and checking reasonableness.

2. What does the word “estimate” mean in this lesson?

A. a sensible approximate value
B. making sense for the situation
C. a standard used to express a measurement
D. a mathematical action such as addition or division

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A glass is filled with water.
C. A notebook is closed on a table.
D. If a satellite completes 16 orbits in one day, it completes about 32 orbits in two days, assuming the rate stays the same.

4. Which word means “a sensible approximate value”?

A. estimate
B. unit
C. operation
D. reasonable

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Estimate how many minutes you spend outdoors in one week, then calculate and compare.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Simple mathematical reasoning can be understood without observation, evidence or comparison.
B. Every object or event connected with simple mathematical reasoning behaves in exactly the same way.
C. Match the operation and unit to the meaning of the problem, then check reasonableness.
D. The main idea of simple mathematical reasoning is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain simple mathematical reasoning.
2. Write two important facts about simple mathematical reasoning.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why should an answer of “20 kilograms” be rejected if the question asks for time?

2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Astronomers often use powers of ten because space distances and sizes vary enormously. An estimate can be more useful than an exact answer when the measurements themselves are approximate.

UNIT 10 · CHAPTER 7

Logical Thinking

Learning outcomes

- Explain logical thinking using simple scientific language.
- Use the words logic and conclusion correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Logical thinking means reaching conclusions through clear rules and evidence. It includes finding assumptions, testing possibilities and rejecting contradictions.

Understanding the idea

A conclusion follows only when the given facts support it.

Words such as all, some, none, must and may have different meanings.

A counterexample is one example that proves a statement such as “all objects have this property” is false.

Elimination can solve a problem by removing options that contradict clues.

Do not guess from appearance alone. Identify the rule, test it against every item and reject an answer that breaks the rule.

Notice how the terms logic, conclusion, assumption appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

Figure 10.7 A pattern question is solved by finding a rule and checking that it fits every position.

Example

If all planets orbit the Sun and Earth is a planet, then Earth orbits the Sun. The conclusion follows from the two statements.

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Write four objects and three clues that identify exactly one object. Test the puzzle on a partner.

Think and discuss

What is the difference between "must be true" and "could be true"?

What you have learnt

- A conclusion follows only when the given facts support it.
- Words such as all, some, none, must and may have different meanings.
- A logical conclusion must follow from the stated evidence without hidden assumptions.

EXERCISES · LOGICAL THINKING

Exercises

A. Choose the correct answer

1. Which statement about logical thinking is correct?

A. Logical conclusions must follow from the given facts and rules.
B. "May" and "must" have identical meanings.
C. A conclusion is correct whenever it sounds interesting.
D. One counterexample can never challenge an "all" statement.

2. What does the word "logic" mean in this lesson?

A. an idea accepted without being stated or proven
B. reasoning according to clear rules
C. an example that shows a general claim is false
D. a result reached from evidence or premises

3. Which observation or example best supports the lesson idea?

A. If all planets orbit the Sun and Earth is a planet, then Earth orbits the Sun. The conclusion follows from the two statements.
B. A glass is filled with water.
C. A notebook is closed on a table.
D. A pencil is sharpened.

4. Which word means "reasoning according to clear rules"?

A. conclusion
B. logic
C. assumption
D. counterexample

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Write four objects and three clues that identify exactly one object. Test the puzzle on a partner.

6. Which sentence gives the best summary of this lesson?

A. A logical conclusion must follow from the stated evidence without hidden assumptions.
B. Logical thinking can be understood without observation, evidence or comparison.
C. Every object or event connected with logical thinking behaves in exactly the same way.
D. The main idea of logical thinking is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain logical thinking.
2. Write two important facts about logical thinking.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. What is the difference between "must be true" and "could be true"?

2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: A statement can be true but still not follow from the facts given in a particular puzzle. Scientific explanations remain open to revision when new evidence conflicts with an old model.

UNIT 10 REVIEW

Review · Scientific Reasoning & Logical Aptitude

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of observation skills using one example.
2. How can you tell whether a shape has rotated or been reflected?
3. Write one correct statement and one common mistake about pattern recognition.
4. Explain the main idea of classification using one example.
5. How is a cycle different from a one-way sequence?
6. Write one correct statement and one common mistake about simple mathematical reasoning.
7. Explain the main idea of logical thinking using one example.
8. Create one new question from this unit and write its answer.

B. Compare and connect

1. Write one connection between “Observation Skills” and “Logical Thinking”.
2. Write one connection between “Picture Reasoning” and “Simple Mathematical Reasoning”.

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 10 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about observation skills is correct?
- A. An observation is always a guess about the cause.
- B. A scientific observation records what is actually seen or measured.
- C. One quick look is always more reliable than repeated checks.
- D. Precise records should avoid numbers and directions.

2. Which statement about picture reasoning is correct?
- A. The best rule may change randomly in every box.
- B. Decorative colour is always the only clue.
- C. Picture reasoning uses consistent visual rules such as rotation, reflection or counting.
- D. A reflection and a rotation are exactly the same change.

3. Which statement about pattern recognition is correct?
- A. A rule needs to fit just one pair of terms.
- B. A pattern has no rule.
- C. Pattern recognition identifies a rule that consistently predicts the sequence.
- D. Only colours can form patterns.

4. Which statement about classification is correct?
- A. There is only one possible way to group any objects.
- B. An odd-one-out question needs no shared rule.
- C. Classification groups objects using clear shared characteristics.
- D. A classification rule may change for each item.

5. Which statement about sequencing is correct?

A. The final step must always happen first.
B. Cycles never repeat.
C. Sequencing arranges events or objects by a clear order such as time or process.
D. A sequence is a random collection with no order.

6. Which statement about simple mathematical reasoning is correct?

A. Checking an answer is unnecessary.
B. The largest number in the question must always be added.
C. Units do not matter in an answer.
D. Mathematical reasoning includes choosing operations, using units and checking reasonableness.

7. Which statement about logical thinking is correct?

A. Logical conclusions must follow from the given facts and rules.
B. “May” and “must” have identical meanings.
C. A conclusion is correct whenever it sounds interesting.
D. One counterexample can never challenge an “all” statement.

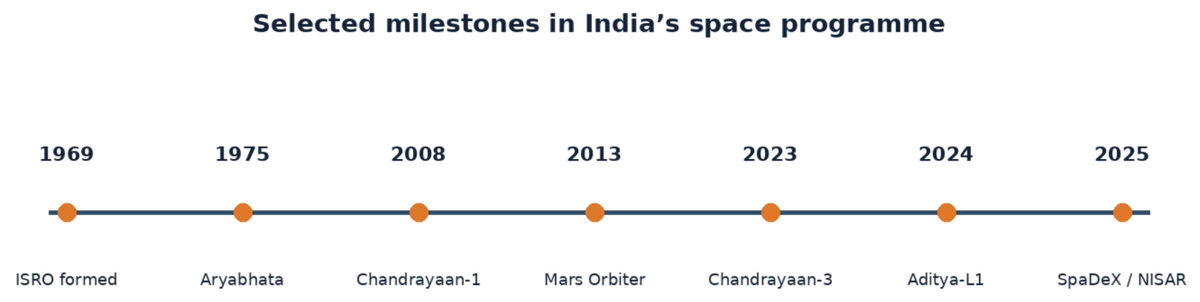
8. Which word means “information gathered by watching or measuring”?

A. inference
B. measurement
C. accuracy
D. observation

UNIT 11

General Space Awareness

Space news changes with time. This unit teaches learners to recognise reliable current information, understand major discoveries and place surprising facts in the right scientific context.



Chapters in this unit

1. Current Space Events
2. Famous Astronauts
3. Planets in the News
4. Important Space Discoveries
5. Amazing Space Facts

At the end of the unit, complete the mixed review before checking the answer key.

UNIT 11 · CHAPTER 1

Current Space Events

Learning outcomes

- Explain current space events using simple scientific language.
- Use the words current event and status correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Space events change quickly, so a good handbook states its update date and separates completed achievements from plans. This lesson summarises selected developments confirmed up to 29 June 2026.

Understanding the idea

ISRO completed the first SpaDeX docking on 16 January 2025 and later demonstrated a second docking and short power transfer in April 2025.

The joint NASA-ISRO NISAR satellite launched on 30 July 2025 and entered its science phase later that year to study changes across Earth using radar.

ISRO astronaut Shubhanshu Shukla served as pilot on Axiom Mission 4 in 2025 and completed an 18-day scientific mission involving the International Space Station.

Gaganyaan testing continued in 2026, including the second Integrated Air Drop Test of a simulated crew module on 10 April 2026.

Check the date, the source and the exact status word. “Planned”, “launched”, “testing” and “completed” describe different stages.

The important words in this chapter include current event, status, docking. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

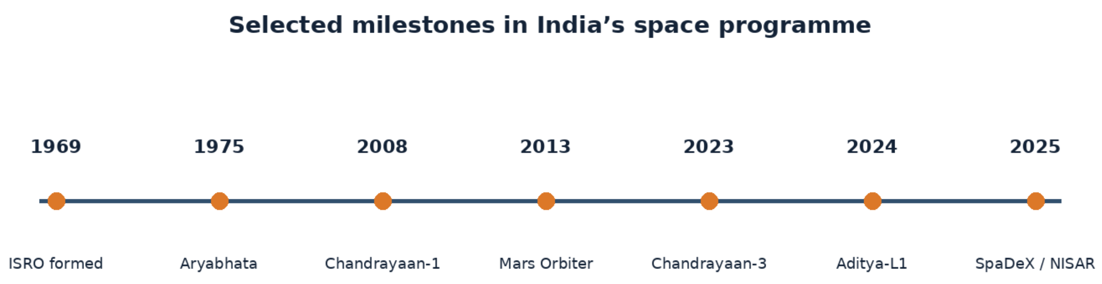


Figure 11.1 Selected milestones in India’s space programme, arranged as a simple timeline.

Example

A news headline saying “mission announced” does not mean “mission completed.” Check the event date, source and exact verb.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Choose one current event. Write four lines: what happened, when, who confirmed it and why it matters.

Think and discuss

Which words show different mission status: launched, tested, planned, entered science phase and completed?

What you have learnt

- ISRO completed the first SpaDeX docking on 16 January 2025 and later demonstrated a second docking and short power transfer in April 2025.
- The joint NASA-ISRO NISAR satellite launched on 30 July 2025 and entered its science phase later that year to study changes across Earth using radar.
- Current space awareness requires dates, reliable sources and careful status words.

Exercises

A. Choose the correct answer

1. Which statement about current space events is correct?

A. A current space report must distinguish completed, ongoing and planned events.
B. A planned mission should be described as already completed.
C. Social-media rumours are stronger evidence than official mission updates.
D. Dates are unnecessary in current-affairs reporting.

2. What does the word “current event” mean in this lesson?

A. the mission period focused on planned scientific observations
B. the present stage or condition of a mission
C. bringing spacecraft together and connecting them
D. a recent occurrence important enough to report

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.
B. A news headline saying “mission announced” does not mean “mission completed.” Check the event date, source and exact verb.
C. A pencil is sharpened.
D. A notebook is closed on a table.

4. Which word means “a recent occurrence important enough to report”?

A. current event
B. status
C. docking
D. science phase

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Ignore the date, time, direction and evidence.
D. Choose one current event. Write four lines: what happened, when, who confirmed it and why it matters.

6. Which sentence gives the best summary of this lesson?

A. Current space events can be understood without observation, evidence or comparison.
B. Current space awareness requires dates, reliable sources and careful status words.
C. Every object or event connected with current space events behaves in exactly the same way.
D. The main idea of current space events is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain current space events.
2. Write two important facts about current space events.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Which words show different mission status: launched, tested, planned, entered science phase and completed?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Aditya-L1 continued returning solar observations and public scientific datasets after entering halo orbit in January 2024. Current-awareness pages should be reviewed before reprinting because mission status can change after publication.

UNIT 11 · CHAPTER 2

Famous Astronauts

Learning outcomes

- Explain famous astronauts using simple scientific language.
- Use the words pioneer and crew correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Famous astronauts are remembered for exploration, science, courage or leadership, but their achievements always depended on large teams. A balanced list includes people from different countries, roles and periods.

Understanding the idea

Yuri Gagarin became the first human in space in 1961.

Valentina Tereshkova became the first woman in space in 1963.

Neil Armstrong and Buzz Aldrin became the first humans to walk on the Moon in 1969, while Michael Collins orbited above.

Rakesh Sharma became the first Indian citizen in space in 1984; Kalpana Chawla, born in India, later became a NASA astronaut and flew on two Space Shuttle missions.

Check the date, the source and the exact status word. “Planned”, “launched”, “testing” and “completed” describe different stages.

Notice how the terms pioneer, crew, commander appear in the explanation. A scientific word is useful only when it helps us describe an observation, comparison or cause clearly.

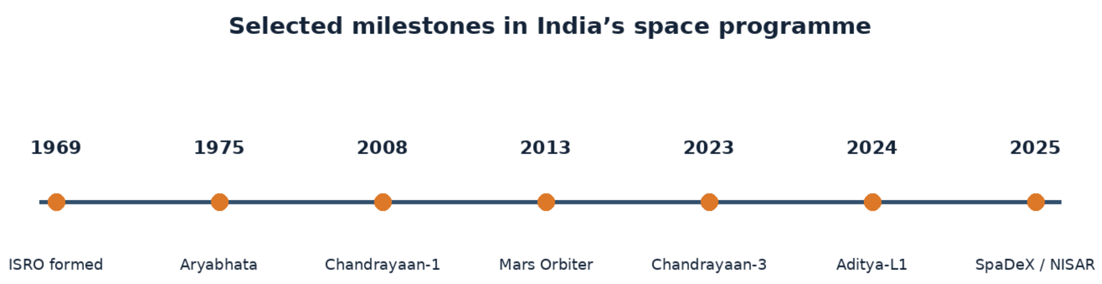


Figure 11.2 Selected milestones in India’s space programme, arranged as a simple timeline.

Example

A crew member who remains in orbit can be as essential as those who land, because mission roles are interdependent.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Choose one astronaut and write a “mission contribution” paragraph that names the team and the person’s specific role.

Think and discuss

Why should Michael Collins be included when discussing the first Moon landing even though he did not walk on the surface?

What you have learnt

- Yuri Gagarin became the first human in space in 1961.
- Valentina Tereshkova became the first woman in space in 1963.
- Astronaut achievements are individual milestones inside large cooperative missions.

EXERCISES · FAMOUS ASTRONAUTS

Exercises

A. Choose the correct answer

1. Which statement about famous astronauts is correct?

A. Only people who walk on another world count as astronauts.
B. A spacecraft crew does not need support from Earth.
C. Rakesh Sharma was the first human in space.
D. Astronaut milestones are achieved through specialised roles and teamwork.

2. What does the word “pioneer” mean in this lesson?

A. a person who helps open a new field or path
B. the people working together on a spacecraft
C. the specific responsibility held by a team member
D. the person responsible for leading a mission crew

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.
B. A crew member who remains in orbit can be as essential as those who land, because mission roles are interdependent.
C. A glass is filled with water.
D. A notebook is closed on a table.

4. Which word means “a person who helps open a new field or path”?

A. crew
B. pioneer
C. commander
D. mission role

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Choose one astronaut and write a “mission contribution” paragraph that names the team and the person’s specific role.
B. Guess once and do not record anything.
C. Change several conditions at the same time and compare nothing.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Famous astronauts can be understood without observation, evidence or comparison.
B. Every object or event connected with famous astronauts behaves in exactly the same way.
C. The main idea of famous astronauts is unrelated to science or everyday experience.

D. Astronaut achievements are individual milestones inside large cooperative missions.

B. Answer in one or two sentences

1. In one sentence, explain famous astronauts.
2. Write two important facts about famous astronauts.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why should Michael Collins be included when discussing the first Moon landing even though he did not walk on the surface?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Peggy Whitson has spent more cumulative days in space than any other American and commanded multiple missions. Astronaut biographies include training, teamwork and scientific work, not only launch-day fame.

UNIT 11 · CHAPTER 3

Planets in the News

Learning outcomes

- Explain planets in the news using simple scientific language.
- Use the words exoplanet and transit correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Planets enter the news when spacecraft arrive, telescopes discover new clues or unusual viewing events occur. Reliable reports explain what was measured and avoid turning early hints into certain claims.

Understanding the idea

Mars remains a major target for orbiters, rovers and studies of past water and climate.

Jupiter and its icy moons are studied because moons such as Europa may contain subsurface oceans.

Venus missions investigate its thick atmosphere, extreme greenhouse heating and hidden surface.

Exoplanet news concerns planets around other stars, where scientists measure size, orbit, mass or atmospheric clues.

Check the date, the source and the exact status word. “Planned”, “launched”, “testing” and “completed” describe different stages.

Do not learn exoplanet, transit, evidence separately from the concept. Say each word in a complete sentence and connect it with evidence from the example.

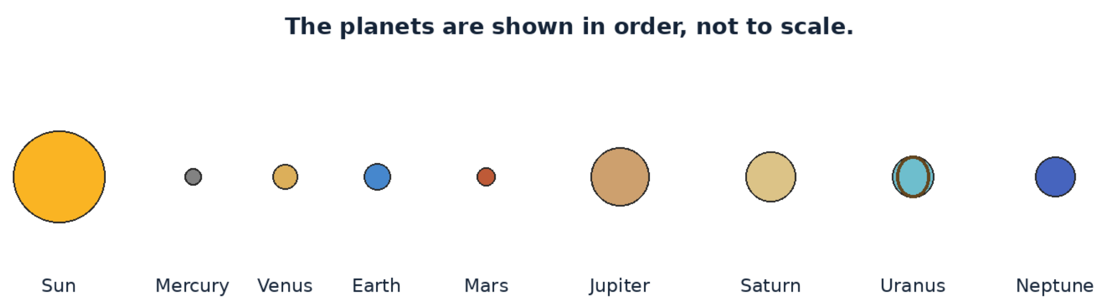


Figure 11.3 The Sun and eight planets shown in order. Sizes and distances are not to scale.

Example

"A planet may have conditions suitable for liquid water" is not the same as "life has been found."

Why this example helps: This situation shows which detail matters most. State that clue first and then use the lesson concept to explain it.

Activity

Take a space headline and underline words that show uncertainty, such as may, possible, candidate or evidence.

Think and discuss

Why should a science report distinguish a measured fact from an interpretation?

What you have learnt

- Mars remains a major target for orbiters, rovers and studies of past water and climate.
- Jupiter and its icy moons are studied because moons such as Europa may contain subsurface oceans.
- Planet news is strongest when it identifies the observation, evidence and uncertainty.

EXERCISES · PLANETS IN THE NEWS

Exercises

A. Choose the correct answer

1. Which statement about planets in the news is correct?

A. Exoplanets all orbit the Sun.

B. Responsible planet news separates observations from interpretations and uncertainty.

C. Science headlines need no supporting measurements.

D. Every possible clue is a confirmed discovery of life.

2. What does the word "exoplanet" mean in this lesson?

A. the amount that remains unknown or imprecise

B. information supporting an explanation

C. a passage in front of a star that slightly reduces its observed light

D. a planet orbiting a star beyond the Sun

3. Which observation or example best supports the lesson idea?

A. A glass is filled with water.

B. A notebook is closed on a table.

C. "A planet may have conditions suitable for liquid water" is not the same as "life has been found."

D. A pencil is sharpened.

4. Which word means "a planet orbiting a star beyond the Sun"?

A. transit

B. evidence

C. exoplanet

D. uncertainty

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.

B. Take a space headline and underline words that show uncertainty, such as may, possible, candidate or evidence.

C. Change several conditions at the same time and compare nothing.

D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Planets in the news can be understood without observation, evidence or comparison.

B. Planet news is strongest when it identifies the observation, evidence and uncertainty.

C. Every object or event connected with planets in the news behaves in exactly the same way.
D. The main idea of planets in the news is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain planets in the news.
2. Write two important facts about planets in the news.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why should a science report distinguish a measured fact from an interpretation?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: Most known exoplanets are discovered indirectly by measuring their effects on stars rather than seeing the planets as clear photographs. A transit occurs when a planet passes in front of its star from our viewpoint and blocks a tiny fraction of starlight.

UNIT 11 · CHAPTER 4

Important Space Discoveries

Learning outcomes

- Explain important space discoveries using simple scientific language.
- Use the words discovery and gravitational wave correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Space discoveries change our understanding when new evidence reveals an object, process or relationship. Discoveries may come from a new telescope, a careful reanalysis or a mission reaching an unexplored place.

Understanding the idea

Chandrayaan-1 observations contributed strong evidence for water-related molecules on the Moon.

Thousands of exoplanets have shown that planetary systems are common and diverse.

Gravitational waves were directly detected in 2015, opening a new way to observe violent cosmic events.

Images of black-hole surroundings and detailed observations by modern space telescopes test ideas about gravity, galaxies and star formation.

Check the date, the source and the exact status word. “Planned”, “launched”, “testing” and “completed” describe different stages.

The vocabulary of this chapter - discovery, gravitational wave, confirmation - gives us a precise way to talk about what is happening. This precision is especially useful in Olympiad questions with similar-looking options.

A galaxy is a vast system of stars, gas, dust and dark matter held together by gravity.

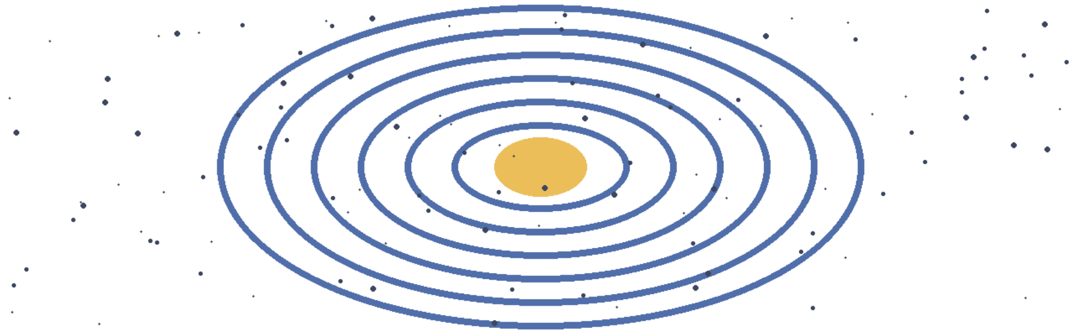


Figure 11.4 A galaxy contains enormous numbers of stars, together with gas, dust and dark matter.

Example

A discovery is often a chain: instrument measurement, data checking, independent confirmation and scientific explanation.

Why this example helps: It places the idea in a familiar situation. Identify the observation, name the concept and explain the link between them.

Activity

Invent a “discovery claim” and list the evidence needed before it should be accepted.

Think and discuss

Why is independent checking important before announcing a surprising discovery?

What you have learnt

- Chandrayaan-1 observations contributed strong evidence for water-related molecules on the Moon.
- Thousands of exoplanets have shown that planetary systems are common and diverse.
- Scientific discoveries are supported by tested evidence and often create new questions.

EXERCISES · IMPORTANT SPACE DISCOVERIES

Exercises

A. Choose the correct answer

1. Which statement about important space discoveries is correct?

A. An important discovery requires reliable evidence, checking and explanation.
B. A discovery is true as soon as one person guesses it.
C. Space discoveries can come only from astronauts.
D. Independent checking weakens science.

2. What does the word “discovery” mean in this lesson?

A. additional evidence showing a result is reliable
B. new knowledge supported by evidence
C. a tool used to observe or measure
D. a ripple in spacetime produced by accelerating massive objects

3. Which observation or example best supports the lesson idea?

A. A discovery is often a chain: instrument measurement, data checking, independent confirmation and scientific explanation.
B. A glass is filled with water.
C. A notebook is closed on a table.
D. A pencil is sharpened.

4. Which word means “new knowledge supported by evidence”?

A. gravitational wave
B. confirmation
C. instrument
D. discovery

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.
B. Change several conditions at the same time and compare nothing.
C. Invent a “discovery claim” and list the evidence needed before it should be accepted.
D. Ignore the date, time, direction and evidence.

6. Which sentence gives the best summary of this lesson?

A. Important space discoveries can be understood without observation, evidence or comparison.
B. Every object or event connected with important space discoveries behaves in exactly the same way.
C. The main idea of important space discoveries is unrelated to science or everyday experience.
D. Scientific discoveries are supported by tested evidence and often create new questions.

B. Answer in one or two sentences

1. In one sentence, explain important space discoveries.
2. Write two important facts about important space discoveries.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why is independent checking important before announcing a surprising discovery?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The cosmic microwave background is ancient light that provides evidence about the early universe. Many discoveries are made by teams sharing data across countries and observatories.

UNIT 11 · CHAPTER 5

Amazing Space Facts

Learning outcomes

- Explain amazing space facts using simple scientific language.
- Use the words scale and density correctly in a sentence.
- Apply the idea to an observation, activity or Olympiad question.

Let us begin

Amazing facts are most useful when they are accurate, explained and placed in context. A surprising number should tell us something about scale, motion, matter or the limits of human experience.

Understanding the idea

Sunlight reaches Earth in about eight minutes, while light from the nearest star beyond the Sun takes more than four years.

A day on Venus, measured by one rotation, is longer than a Venus year.

Saturn’s average density is lower than water, although no ocean could realistically hold the planet.

A teaspoon of neutron-star material would have an enormous mass on Earth, but such material cannot be removed and kept in that state under ordinary conditions.

Check the date, the source and the exact status word. “Planned”, “launched”, “testing” and “completed” describe different stages.

The important words in this chapter include scale, density, neutron star. These words are tools for explaining the idea, not an isolated list to memorise. Use them while describing the example and the activity.

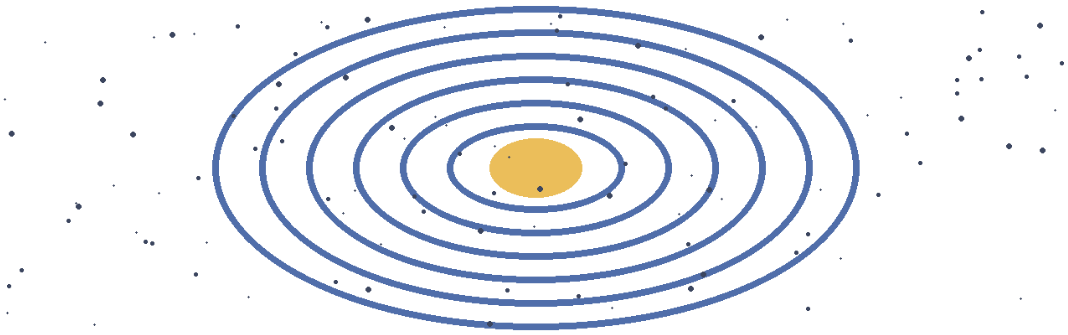


Figure 11.5 A galaxy contains enormous numbers of stars, together with gas, dust and dark matter.

Example

A comparison is clearer than a huge number alone: one light-year is about 63,000 times the average Earth-Sun distance.

Why this example helps: The example turns an abstract idea into a sequence that can be followed. In an Olympiad answer, use the evidence in the situation rather than guessing from a keyword.

Activity

Choose one amazing fact and verify it using two reliable science sources. Rewrite it with a clear comparison.

Think and discuss

Why can a true fact still be misleading when its conditions or units are omitted?

What you have learnt

- Sunlight reaches Earth in about eight minutes, while light from the nearest star beyond the Sun takes more than four years.
- A day on Venus, measured by one rotation, is longer than a Venus year.
- A good space fact includes correct units, conditions and context.

EXERCISES · AMAZING SPACE FACTS

Exercises

A. Choose the correct answer

1. Which statement about amazing space facts is correct?

A. Space is perfectly empty everywhere.
B. Every planet has a day shorter than its year.
C. Amazing space facts should include accurate units, conditions and context.
D. A surprising statement needs no checking.

2. What does the word “scale” mean in this lesson?

A. an extremely dense remnant of a massive star
B. the relative size, distance or time used for comparison
C. mass packed into a given volume
D. information needed to understand a fact correctly

3. Which observation or example best supports the lesson idea?

A. A pencil is sharpened.

B. A comparison is clearer than a huge number alone: one light-year is about 63,000 times the average Earth-Sun distance.

C. A glass is filled with water.

D. A notebook is closed on a table.

4. Which word means “the relative size, distance or time used for comparison”?

A. scale

B. density

C. neutron star

D. context

5. Which action is the best way to investigate the lesson idea safely and carefully?

A. Guess once and do not record anything.

B. Change several conditions at the same time and compare nothing.

C. Ignore the date, time, direction and evidence.

D. Choose one amazing fact and verify it using two reliable science sources. Rewrite it with a clear comparison.

6. Which sentence gives the best summary of this lesson?

A. Amazing space facts can be understood without observation, evidence or comparison.

B. A good space fact includes correct units, conditions and context.

C. Every object or event connected with amazing space facts behaves in exactly the same way.

D. The main idea of amazing space facts is unrelated to science or everyday experience.

B. Answer in one or two sentences

1. In one sentence, explain amazing space facts.
2. Write two important facts about amazing space facts.
3. How does the everyday example in this lesson show the main idea?

C. Think and apply

1. Why can a true fact still be misleading when its conditions or units are omitted?
2. Draw or label a simple diagram that explains the main relationship in this lesson.

At the end of the lesson

Further reading: The footprints left by Apollo astronauts can remain for a very long time because the Moon has no wind or rain to erase them quickly. Space is not completely empty; it contains particles, radiation, magnetic fields and large clouds of gas and dust.

UNIT 11 REVIEW

Review · General Space Awareness

Use the ideas from more than one chapter. Give a reason whenever the question asks why or how.

A. Recall and explain

1. Explain the main idea of current space events using one example.
2. Why should Michael Collins be included when discussing the first Moon landing even though he did not walk on the surface?
3. Write one correct statement and one common mistake about planets in the news.
4. Explain the main idea of important space discoveries using one example.
5. Why can a true fact still be misleading when its conditions or units are omitted?
6. Create one new question from this unit and write its answer.

B. Compare and connect

1. Write one connection between “Current Space Events” and “Amazing Space Facts”.
2. Write one connection between “Famous Astronauts” and “Important Space Discoveries”.

C. Observation or drawing task

Create one labelled diagram containing at least four ideas from this unit. Under the diagram, write two sentences explaining how the ideas are connected.

UNIT 11 OLYMPIAD PRACTICE

Mixed multiple-choice questions

1. Which statement about current space events is correct?
A. A current space report must distinguish completed, ongoing and planned events.
B. A planned mission should be described as already completed.
C. Social-media rumours are stronger evidence than official mission updates.
D. Dates are unnecessary in current-affairs reporting.

2. Which statement about famous astronauts is correct?
A. Only people who walk on another world count as astronauts.
B. A spacecraft crew does not need support from Earth.
C. Rakesh Sharma was the first human in space.
D. Astronaut milestones are achieved through specialised roles and teamwork.

3. Which statement about planets in the news is correct?
A. Exoplanets all orbit the Sun.
B. Responsible planet news separates observations from interpretations and uncertainty.
C. Science headlines need no supporting measurements.
D. Every possible clue is a confirmed discovery of life.

4. Which statement about important space discoveries is correct?
A. An important discovery requires reliable evidence, checking and explanation.
B. A discovery is true as soon as one person guesses it.
C. Space discoveries can come only from astronauts.
D. Independent checking weakens science.

5. Which statement about amazing space facts is correct?
A. Space is perfectly empty everywhere.
B. Every planet has a day shorter than its year.
C. Amazing space facts should include accurate units, conditions and context.
D. A surprising statement needs no checking.

6. Which word means “a recent occurrence important enough to report”?
A. current event
B. status
C. docking
D. science phase

7. Which word means “a person who helps open a new field or path”?
A. crew
B. pioneer
C. commander
D. mission role

8. Which word means “a planet orbiting a star beyond the Sun”?
A. transit
B. evidence
C. exoplanet
D. uncertainty

Model Olympiad Paper 1

Foundation Level · 40 questions · Choose one correct option for each question. This practice paper is not a claim about the official examination pattern.

1. Which statement about earth’s rotation is correct?

A. Rotation causes the Moon’s phases.
B. Earth rotates from west to east once in about 24 hours.
C. Earth rotates once every 365 years.
D. Earth rotates from east to west.

2. Which statement about asteroids is correct?

A. Asteroids are stars that have stopped shining.
B. All asteroids lie on Earth’s surface.
C. Most asteroids are rocky or metallic bodies orbiting the Sun.
D. The asteroid belt is a solid ring with no gaps.

3. Which statement about day and night is correct?

A. Earth becomes dark because the Sun switches off.
B. Day and night are caused by the Moon blocking the Sun every evening.
C. Day and night happen because clouds cover half of Earth.
D. Day and night are caused by Earth rotating on its axis.

4. Which word means “grouping by shared characteristics”?

A. characteristic
B. category
C. classification
D. odd one out

5. Which statement about shadows is correct?

A. Shadow direction never changes.
B. A shadow forms when an opaque object blocks light.
C. A shadow is a black substance released by objects.
D. Transparent glass always makes a completely dark shadow.

6. Which word means “a vehicle or engine that produces thrust by expelling material”?

A. thrust
B. exhaust
C. oxidiser
D. rocket

7. Which word means “designed to be used more than once”?

A. booster
B. recovery
C. refurbish
D. reusable

8. Which statement about dwarf planets is correct?

A. A dwarf planet is always a moon.
B. A dwarf planet orbits the Sun but has not cleared its orbital neighbourhood.
C. A dwarf planet does not orbit the Sun.
D. A dwarf planet must be smaller than every asteroid.

Model Olympiad Paper 1 · Continued

9. Which statement about heat is correct?

A. Spacecraft mainly exchange heat by radiation in the vacuum of space.
B. Heat and temperature mean exactly the same thing.
C. Convection works through completely empty space.
D. No object can become hot in space.

10. Which word means "electromagnetic energy that travels through space"?

A. visible light
B. light
C. spectrum
D. reflect

11. Which word means "a spacecraft that travels around a world"?

A. lander
B. rover
C. sample return
D. orbiter

12. Which statement about logical thinking is correct?

A. Logical conclusions must follow from the given facts and rules.
B. "May" and "must" have identical meanings.
C. A conclusion is correct whenever it sounds interesting.
D. One counterexample can never challenge an "all" statement.

13. Which statement about famous astronauts is correct?

A. Only people who walk on another world count as astronauts.
B. A spacecraft crew does not need support from Earth.
C. Rakesh Sharma was the first human in space.
D. Astronaut milestones are achieved through specialised roles and teamwork.

14. Which statement about the milky way galaxy is correct?

A. The Milky Way is another name for the solar system.
B. The solar system is located inside the Milky Way galaxy.
C. The Milky Way contains only eight stars.
D. Earth lies at the exact centre of the Milky Way.

15. Which statement about introduction to isro is correct?

A. ISRO studies only the Moon.
B. ISRO is India's national space agency, established in 1969.
C. ISRO is a weather station with no spacecraft.
D. ISRO was created to operate ordinary ships at sea.

16. Which word means "a small rocky or metallic body orbiting the Sun"?

A. asteroid
B. asteroid belt
C. near-Earth object
D. metallic

Model Olympiad Paper 1 · Continued

17. Which statement about air and atmosphere is correct?

A. The atmosphere becomes denser with height.
B. Earth’s atmosphere supports life and protects the surface.
C. Air has no mass or pressure.
D. Astronauts can breathe ordinary space because sunlight is present.

18. Which word means “a push or pull”?

A. balanced forces
B. unbalanced force
C. friction
D. force

19. Which statement about important space discoveries is correct?

A. An important discovery requires reliable evidence, checking and explanation.
B. A discovery is true as soon as one person guesses it.
C. Space discoveries can come only from astronauts.
D. Independent checking weakens science.

20. Which statement about solar and lunar eclipses is correct?

A. Ordinary sunglasses make direct Sun viewing safe.
B. A solar eclipse happens when the Moon passes between the Sun and Earth.
C. A lunar eclipse happens when Venus blocks the Moon.
D. A solar eclipse occurs at every full Moon.

21. Which statement about the eight planets is correct?

A. Our solar system has eight planets orbiting the Sun.
B. Our solar system has nine official planets including Pluto.
C. Planets produce their own visible light like stars.
D. Every planet has a solid rocky surface.

22. Which word means “the period beginning with the first artificial satellites”?

A. space telescope
B. space age
C. heliosphere
D. milestone

23. Which word means “Mercury, Venus, Earth and Mars”?

A. outer planets
B. orbital period
C. inner planets
D. asteroid belt

24. Which word means “India’s first dedicated space-based solar observatory”?

A. L1
B. halo orbit
C. space weather
D. Aditya-L1

Model Olympiad Paper 1 · Continued

25. Which statement about natural satellites is correct?

A. A natural satellite is always built by engineers.
B. A natural satellite is a moon that orbits a larger body.
C. Moons orbit only the Sun and never planets.
D. Every planet has exactly one moon.

26. Which statement about order of the planets is correct?

A. Neptune is the closest planet to the Sun.
B. Earth is the third planet from the Sun, between Venus and Mars.
C. Mars is farther from the Sun than Jupiter.
D. Earth is between Jupiter and Saturn.

27. Which statement about simple machines is correct?

A. A ramp always needs more force than lifting straight up.
B. A lever has no turning point.
C. Simple machines can reduce required force by increasing distance or changing direction.
D. Simple machines create energy from nothing.

28. Which word means “a type of electromagnetic wave used for communication”?

A. uplink
B. downlink
C. geostationary orbit
D. radio wave

29. Which word means “a recent occurrence important enough to report”?

A. current event
B. status
C. docking
D. science phase

30. Which word means “the relative size, distance or time used for comparison”?

A. scale
B. density
C. neutron star
D. context

31. Which statement about sunrise and sunset is correct?

A. The Sun appears to rise in the east and set in the west because Earth rotates.
B. Sunrise happens in the west everywhere.
C. Sunset occurs because the Sun becomes smaller.
D. The Sun circles Earth once every day.

32. Which word means “spinning around an axis”?

A. axis
B. time zone
C. longitude
D. rotation

Model Olympiad Paper 1 · Continued

33. Which statement about simple mathematical reasoning is correct?

A. Checking an answer is unnecessary.
B. The largest number in the question must always be added.
C. Units do not matter in an answer.
D. Mathematical reasoning includes choosing operations, using units and checking reasonableness.

34. Which word means “north, south, east and west”?

A. compass
B. magnetic field
C. cardinal directions
D. north-east

35. Which statement about famous space missions is correct?

A. A mission is scientifically important only if it carries people.
B. Voyager explored only Earth’s oceans.
C. Famous missions are milestones because of discoveries, firsts or new capabilities.
D. Sputnik 1 was the first person on the Moon.

36. Which statement about current space events is correct?

A. A current space report must distinguish completed, ongoing and planned events.
B. A planned mission should be described as already completed.
C. Social-media rumours are stronger evidence than official mission updates.
D. Dates are unnecessary in current-affairs reporting.

37. Which statement about observing the sky is correct?

A. A useful observation needs only a colourful drawing.
B. Direction does not matter when describing the sky.
C. It is safe to view the Sun through ordinary sunglasses.
D. A scientific sky record includes date, time, direction and observing conditions.

38. Which statement about light is correct?

A. Astronomers study many kinds of electromagnetic light, not only visible light.
B. Light needs ordinary air to cross space.
C. The Moon produces all of its visible light.
D. Radio waves cannot carry information because eyes cannot see them.

39. Which statement about great indian space scientists is correct?

A. Only astronauts contribute to a space mission.
B. Scientific leadership means avoiding all discussion of failure.
C. India’s space achievements were built by generations of scientists, engineers and teams.
D. One person designed and built every Indian spacecraft alone.

40. Which statement about gravity is correct?

A. Objects in orbit are beyond all gravity.
B. Gravity holds moons, planets and spacecraft in orbit.
C. Gravity exists only on Earth’s surface.
D. Gravity pushes masses away from each other.

Model Olympiad Paper 2

Foundation Level · 40 questions · Choose one correct option for each question. This practice paper is not a claim about the official examination pattern.

1. Which word means “the prepared site from which a rocket launches”?

A. flame trench
B. launch pad
C. mission control
D. safe zone

2. Which statement about natural and artificial satellites is correct?

A. Artificial satellites are human-made spacecraft placed in orbit for specific tasks.
B. Artificial satellites can work without power or communication.
C. A satellite stays up because gravity disappears in space.
D. Every satellite is a natural moon.

3. Which statement about sequencing is correct?

A. The final step must always happen first.
B. Cycles never repeat.
C. Sequencing arranges events or objects by a clear order such as time or process.
D. A sequence is a random collection with no order.

4. Which word means “the name of India’s lunar mission series”?

A. soft landing
B. Chandrayaan
C. lander
D. rover

5. Which statement about planets in the news is correct?

A. Exoplanets all orbit the Sun.
B. Responsible planet news separates observations from interpretations and uncertainty.
C. Science headlines need no supporting measurements.
D. Every possible clue is a confirmed discovery of life.

6. Which statement about our sun is correct?

A. The Sun is the star at the centre of our solar system.
B. The Sun is made mainly of rock and ice.
C. The Sun shines by reflecting light from the Moon.
D. The Sun is a planet orbiting Earth.

7. Which word means “the distance light travels in one year”?

A. Proxima Centauri
B. star system
C. light-year
D. red dwarf

8. Which statement about what is a rocket? is correct?

A. A rocket is pulled upward by the Moon.
B. A rocket works only by pushing against air.
C. Rockets need wings to flap in space.
D. A rocket produces thrust by pushing exhaust backward.

Model Olympiad Paper 2 · Continued

9. Which statement about weather satellites is correct?

A. Weather satellites help monitor clouds and storms from space.
B. Forecasts use only one photograph and no other data.
C. Weather satellites create all weather on Earth.
D. A polar satellite remains fixed over one city.

10. Which statement about earth: our home planet is correct?

A. Earth is the largest planet in the solar system.
B. Earth is the third planet from the Sun and the only known world with life.
C. Earth has no atmosphere or liquid water.
D. Earth is the closest planet to the Sun.

11. Which statement about who are astronauts? is correct?

A. Astronauts combine specialised knowledge, physical training and teamwork.
B. Astronauts are chosen only because they can run fast.
C. Every astronaut has exactly the same job.
D. Astronauts operate without help from Earth.

12. Which word means “the regular rise and fall of sea level”?

A. high tide
B. tide
C. spring tide
D. neap tide

13. Which statement about moon exploration missions is correct?

A. A rover studies the whole Moon from far away.
B. Orbiters, landers, rovers and sample-return missions study the Moon in different ways.
C. Moon samples cannot be studied after returning to Earth.
D. An orbiter must always land after one circle.

14. Which statement about how rockets reach space is correct?

A. A rocket must point straight upward for the whole flight.
B. A spacecraft reaches orbit by gaining very high sideways speed.
C. Gravity stops completely at the edge of space.
D. Orbit is achieved only by rising above clouds.

15. Which statement about seasons is correct?

A. Seasons are caused by the Moon changing shape.
B. Every place on Earth has the same season at the same time.
C. Seasons happen because Earth is much nearer the Sun in summer.
D. Earth’s tilt and revolution cause the seasons.

16. Which statement about brightness of stars is correct?

A. The brightest-looking star must always be the largest.
B. Distance never affects how bright a star looks.
C. A star’s apparent brightness depends on luminosity, distance and intervening material.
D. All stars give out exactly the same amount of energy.

Model Olympiad Paper 2 · Continued

17. Which statement about other galaxies is correct?

A. A galaxy contains only one star and one planet.
B. Galaxies exist in many shapes and contain enormous numbers of stars.
C. Galaxies are found only inside the solar system.
D. Every galaxy is identical to the Milky Way.

18. Which statement about classification is correct?

A. There is only one possible way to group any objects.
B. An odd-one-out question needs no shared rule.
C. Classification groups objects using clear shared characteristics.
D. A classification rule may change for each item.

19. Which statement about force is correct?

A. Gravity is a type of contact force.
B. Balanced forces always make an object stop.
C. An unbalanced force changes an object's motion.
D. A force has no direction.

20. Which statement about comets is correct?

A. A comet is a burning star falling through Earth's atmosphere.
B. Comets are made only of liquid water.
C. A comet's tail always points behind its motion.
D. A comet is an icy body that can form a coma and tails near the Sun.

21. Which statement about directions is correct?

A. North, south, east and west are the four cardinal directions.
B. A compass always points east-west.
C. Directions change whenever a person turns around.
D. Up, down, near and far are the four cardinal directions.

22. Which word means "a small natural rock or metal object moving in space"?

A. meteor
B. meteorite
C. meteoroid
D. meteor shower

23. Which statement about what are stars? is correct?

A. All stars have the same temperature and size.
B. Stars are small points attached to the sky.
C. Stars are hot glowing spheres that produce energy through fusion.
D. Stars shine only by reflecting sunlight.

24. Which statement about seasons revisited is correct?

A. A solstice happens every day at noon.
B. Solstices and equinoxes mark important points in Earth's seasonal cycle.
C. An equinox means the Sun stops shining.
D. Seasons are identical in every climate.

Model Olympiad Paper 2 · Continued

25. Which statement about phases of the moon is correct?

A. The Moon makes more light during a full Moon.
B. Moon phases happen because clouds cover the Moon.
C. Earth's shadow causes every Moon phase.
D. Moon phases are caused by changing viewing geometry as the Moon orbits Earth.

26. Which word means "the spinning of an object around its axis"?

A. rotation
B. axis
C. daylight
D. cycle

27. Which word means "the third planet from the Sun and our home world"?

A. atmosphere
B. Earth
C. magnetic field
D. habitable

28. Which word means "a large nearly round body orbiting a star that has cleared its orbital neighbourhood"?

A. planet
B. rocky planet
C. gas giant
D. ice giant

29. Which statement about satellites in everyday life is correct?

A. Satellite data cannot help people on Earth.
B. Satellites support communication, navigation, weather, mapping and disaster response.
C. Satellites are useful only to astronauts.
D. Every satellite has exactly the same mission.

30. Which statement about motion is correct?

A. Motion can be described without any reference point.
B. Velocity means speed only and never direction.
C. Acceleration includes changes in speed or direction.
D. Turning at constant speed is not acceleration.

31. Which statement about understanding the universe is correct?

A. The universe is a small empty shell around Earth.
B. The universe contains only the Milky Way.
C. The universe includes all space, time, matter and energy.
D. Scientists understand every part of the universe completely.

32. Which statement about chandrayaan missions is correct?

A. Chandrayaan-2 had no successful orbiter.
B. A soft landing means crashing at high speed.
C. Chandrayaan-3 was India's first satellite of any kind.
D. Chandrayaan-3 achieved a successful lunar soft landing in August 2023.

Model Olympiad Paper 2 · Continued

33. Which word means “a sensible approximate value”?

A. estimate
B. unit
C. operation
D. reasonable

34. Which statement about introduction to constellations is correct?

A. Every constellation is visible all night from every place.
B. Constellations are groups of stars tied together in space.
C. Constellations organise the sky into named regions and recognised patterns.
D. There are only twelve constellations in the whole sky.

35. Which statement about earth’s revolution is correct?

A. Earth completes one revolution every 24 hours.
B. Earth’s orbit is centred on the Moon.
C. Revolution means spinning on an axis.
D. Earth completes one revolution around the Sun in about 365.25 days.

36. Which word means “the eyes becoming more sensitive in low light”?

A. dark adaptation
B. cloud cover
C. record
D. solar filter

37. Which word means “a planet orbiting a star beyond the Sun”?

A. transit
B. evidence
C. exoplanet
D. uncertainty

38. Which word means “the apparent shape of the Moon’s illuminated part”?

A. waxing
B. waning
C. crescent
D. phase

39. Which word means “the boundary between the lit and dark halves of a world”?

A. local time
B. terminator
C. midday
D. midnight

40. Which statement about day and night revisited is correct?

A. The moving day-night boundary crosses Earth as the planet rotates.
B. Every city experiences noon at the same moment.
C. The same half of Earth is permanently dark.
D. Sunrise is caused by the Sun moving around Earth.

Answer Key

Use the key after attempting the questions. For short-answer and drawing questions, compare the explanation with the lesson text and check whether the scientific reason is clear.

Unit 1 Review: U1R1-B U1R2-C U1R3-C U1R4-A U1R5-A U1R6-D U1R7-D U1R8-C
Unit 1, Chapter 1: What is Astronomy?: 1-B 2-C 3-B 4-C 5-D 6-D
Unit 1, Chapter 2: The Sky Around Us: 1-C 2-C 3-D 4-D 5-A 6-B
Unit 1, Chapter 3: Day and Night: 1-C 2-C 3-D 4-A 5-B 6-D 3-C 26-A
Unit 1, Chapter 4: Sunrise and Sunset: 1-A 2-A 3-C 4-B 5-C 6-B 31-A
Unit 1, Chapter 5: Directions: 1-A 2-B 3-B 4-C 5-D 6-D 34-C 21-A
Unit 1, Chapter 6: Seasons: 1-D 2-D 3-C 4-D 5-A 6-B 15-D
Unit 1, Chapter 7: Observing the Sky: 1-D 2-B 3-C 4-A 5-B 6-D 37-D 36-A
Unit 1, Chapter 8: Stars Visible at Night: 1-C 2-B 3-C 4-B 5-C 6-B
Unit 1, Chapter 9: Introduction to Constellations: 1-C 2-C 3-C 4-C 5-D 6-D 34-C
Unit 1, Chapter 10: Astronomy in Everyday Life: 1-C 2-C 3-D 4-D 5-A 6-B
Unit 1, Chapter 11: Simple Astronomical Observations: 1-B 2-C 3-C 4-A 5-B 6-D
Unit 2 Review: U2R1-A U2R2-A U2R3-B U2R4-B U2R5-B U2R6-C U2R7-D U2R8-A
Unit 2, Chapter 1: Our Sun: 1-A 2-C 3-C 4-D 5-B 6-A 6-A
Unit 2, Chapter 2: The Eight Planets: 1-A 2-C 3-D 4-A 5-C 6-C 21-A 28-A
Unit 2, Chapter 3: Dwarf Planets: 1-B 2-C 3-C 4-B 5-D 6-A 8-B
Unit 2, Chapter 4: Order of the Planets: 1-B 2-C 3-D 4-C 5-A 6-C 23-C 26-B
Unit 2, Chapter 5: Natural Satellites: 1-B 2-B 3-A 4-D 5-B 6-A 25-B
Unit 2, Chapter 6: Asteroids: 1-C 2-B 3-A 4-A 5-C 6-C 2-C 16-A
Unit 2, Chapter 7: Comets: 1-D 2-B 3-D 4-B 5-D 6-A 20-D
Unit 2, Chapter 8: Meteors and Meteorites: 1-A 2-C 3-A 4-C 5-A 6-C 22-C
Unit 2, Chapter 9: Characteristics of the Planets: 1-D 2-C 3-D 4-D 5-B 6-A
Unit 3 Review: U3R1-C U3R2-C U3R3-B U3R4-B U3R5-B U3R6-C U3R7-A U3R8-B
Unit 3, Chapter 1: What are Stars?: 1-C 2-A 3-B 4-A 5-D 6-B 23-C
Unit 3, Chapter 2: Brightness of Stars: 1-C 2-A 3-D 4-B 5-A 6-D 16-C
Unit 3, Chapter 3: The Nearest Stars: 1-B 2-A 3-C 4-C 5-B 6-B 7-C
Unit 3, Chapter 4: The Milky Way Galaxy: 1-B 2-D 3-A 4-D 5-C 6-D 14-B
Unit 3, Chapter 5: Other Galaxies: 1-B 2-C 3-D 4-A 5-D 6-B 17-B
Unit 3, Chapter 6: Understanding the Universe: 1-C 2-C 3-B 4-B 5-A 6-D 31-C
Unit 4 Review: U4R1-B U4R2-B U4R3-B U4R4-D U4R5-A U4R6-B U4R7-C U4R8-B
Unit 4, Chapter 1: Earth: Our Home Planet: 1-B 2-B 3-C 4-B 5-B 6-C 10-B 27-B
Unit 4, Chapter 2: Continents and Oceans: 1-B 2-B 3-A 4-C 5-C 6-A
Unit 4, Chapter 3: Earth's Rotation: 1-B 2-C 3-A 4-D 5-D 6-C 1-B 32-D
Unit 4, Chapter 4: Earth's Revolution: 1-D 2-B 3-D 4-A 5-A 6-A 35-D
Unit 4, Chapter 5: Day and Night Revisited: 1-A 2-C 3-B 4-B 5-B 6-C 39-B 40-A
Unit 4, Chapter 6: Seasons Revisited: 1-B 2-D 3-C 4-C 5-C 6-A 24-B
Unit 4, Chapter 7: Phases of the Moon: 1-C 2-D 3-B 4-D 5-D 6-C 25-C 38-D
Unit 4, Chapter 8: Solar and Lunar Eclipses: 1-B 2-D 3-A 4-A 5-A 6-A 20-B
Unit 4, Chapter 9: Tides: 1-A 2-A 3-A 4-B 5-B 6-C 12-B
Unit 5 Review: U5R1-A U5R2-C U5R3-A U5R4-A U5R5-B U5R6-C U5R7-D U5R8-A
Unit 5, Chapter 1: Natural and Artificial Satellites: 1-A 2-A 3-A 4-C 5-D 6-D 2-A
Unit 5, Chapter 2: Communication Satellites: 1-C 2-A 3-D 4-D 5-A 6-B 28-D
Unit 5, Chapter 3: Weather Satellites: 1-A 2-B 3-C 4-A 5-B 6-D 9-A
Unit 5, Chapter 4: GPS and Navigation: 1-A 2-A 3-A 4-B 5-C 6-B
Unit 5, Chapter 5: Satellites in Everyday Life: 1-B 2-D 3-B 4-C 5-D 6-D 29-B
Unit 6 Review: U6R1-D U6R2-B U6R3-D U6R4-C U6R5-C U6R6-D U6R7-A U6R8-B
Unit 6, Chapter 1: What is a Rocket?: 1-D 2-C 3-D 4-D 5-B 6-A 6-D 8-D
Unit 6, Chapter 2: How Rockets Reach Space: 1-B 2-A 3-B 4-A 5-C 6-C 14-B

Unit 6, Chapter 3: Launch Pads: 1-D 2-C 3-B 4-B 5-D 6-A 1-B
Unit 6, Chapter 4: Astronauts: 1-C 2-A 3-D 4-C 5-A 6-C
Unit 6, Chapter 5: Reusable Rockets: 1-C 2-A 3-A 4-D 5-B 6-A 7-D
Unit 7 Review: U7R1-B U7R2-D U7R3-A U7R4-C U7R5-D U7R6-C U7R7-A U7R8-A
Unit 7, Chapter 1: Introduction to ISRO: 1-B 2-C 3-D 4-A 5-D 6-B 15-B
Unit 7, Chapter 2: Chandrayaan Missions: 1-D 2-B 3-B 4-B 5-A 6-D 4-B 32-D
Unit 7, Chapter 3: Mangalyaan: 1-A 2-C 3-C 4-C 5-B 6-B
Unit 7, Chapter 4: Aditya-L1: 1-C 2-B 3-A 4-D 5-C 6-D 24-D
Unit 7, Chapter 5: Gaganyaan: 1-D 2-B 3-C 4-A 5-D 6-B
Unit 7, Chapter 6: Great Indian Space Scientists: 1-C 2-C 3-C 4-B 5-A 6-D 39-C
Unit 7, Chapter 7: India's Achievements in Space: 1-A 2-C 3-D 4-C 5-B 6-B
Unit 8 Review: U8R1-A U8R2-A U8R3-B U8R4-B U8R5-C U8R6-B U8R7-C U8R8-D
Unit 8, Chapter 1: Who are Astronauts?: 1-A 2-D 3-D 4-B 5-B 6-C 11-A
Unit 8, Chapter 2: Space Stations: 1-A 2-C 3-C 4-C 5-C 6-A
Unit 8, Chapter 3: Moon Exploration Missions: 1-B 2-D 3-D 4-D 5-D 6-C 11-D 13-B
Unit 8, Chapter 4: Mars Exploration Missions: 1-B 2-C 3-A 4-A 5-A 6-A
Unit 8, Chapter 5: Famous Space Missions: 1-C 2-A 3-B 4-B 5-B 6-C 22-B 35-C
Unit 9 Review: U9R1-B U9R2-C U9R3-A U9R4-D U9R5-B U9R6-A U9R7-C U9R8-B
Unit 9, Chapter 1: Gravity: 1-B 2-D 3-C 4-C 5-D 6-D 40-B
Unit 9, Chapter 2: Force: 1-C 2-A 3-B 4-D 5-A 6-B 18-D 19-C
Unit 9, Chapter 3: Motion: 1-A 2-C 3-A 4-A 5-B 6-D 30-A
Unit 9, Chapter 4: Light: 1-D 2-D 3-C 4-B 5-C 6-B 10-B 38-D
Unit 9, Chapter 5: Shadows: 1-B 2-B 3-D 4-C 5-D 6-D 5-B
Unit 9, Chapter 6: Heat: 1-A 2-B 3-D 4-D 5-A 6-B 9-A
Unit 9, Chapter 7: Simple Machines: 1-C 2-B 3-C 4-A 5-B 6-D 27-C
Unit 9, Chapter 8: Air and Atmosphere: 1-B 2-C 3-B 4-B 5-C 6-B 17-B
Unit 10 Review: U10R1-B U10R2-C U10R3-C U10R4-B U10R5-C U10R6-D U10R7-A U10R8-D
Unit 10, Chapter 1: Observation Skills: 1-B 2-D 3-A 4-D 5-B 6-A
Unit 10, Chapter 2: Picture Reasoning: 1-C 2-D 3-C 4-A 5-C 6-C
Unit 10, Chapter 3: Pattern Recognition: 1-C 2-D 3-A 4-B 5-D 6-A
Unit 10, Chapter 4: Classification: 1-B 2-A 3-A 4-C 5-A 6-C 4-C 18-B
Unit 10, Chapter 5: Sequencing: 1-C 2-D 3-A 4-D 5-B 6-A 3-C
Unit 10, Chapter 6: Simple Mathematical Reasoning: 1-D 2-A 3-D 4-A 5-C 6-C 33-D 33-A
Unit 10, Chapter 7: Logical Thinking: 1-A 2-B 3-A 4-B 5-D 6-A 12-A
Unit 11 Review: U11R1-A U11R2-C U11R3-B U11R4-A U11R5-C U11R6-A U11R7-B U11R8-C
Unit 11, Chapter 1: Current Space Events: 1-A 2-D 3-B 4-A 5-D 6-B 29-A 36-A
Unit 11, Chapter 2: Famous Astronauts: 1-C 2-A 3-B 4-B 5-A 6-D 13-C
Unit 11, Chapter 3: Planets in the News: 1-B 2-D 3-C 4-C 5-B 6-B 5-B 37-C
Unit 11, Chapter 4: Important Space Discoveries: 1-A 2-B 3-A 4-D 5-C 6-D 19-A
Unit 11, Chapter 5: Amazing Space Facts: 1-C 2-B 3-B 4-A 5-D 6-B 30-A

Glossary

acceleration: a change in velocity
accuracy: closeness to the correct value or description
Aditya-L1: India's first dedicated space-based solar observatory
air: the mixture of gases around us
air pressure: force produced by air pushing on surfaces
alignment: a nearly straight arrangement of objects
apparent brightness: how bright an object looks from a particular place
apparent motion: motion that seems to happen because the observer or Earth is moving
artificial satellite: a human-made object placed in orbit
assumption: an idea accepted without being stated or proven
asterism: a familiar star pattern that is not necessarily a whole constellation
asteroid: a small rocky or metallic body orbiting the Sun
asteroid belt: a region containing many rocky bodies between Mars and Jupiter
astronaut: a person trained to travel or work in space
astronomer: a scientist who studies the universe
astronomical unit: Earth's average distance from the Sun
astronomy: the scientific study of space and objects beyond Earth
atmosphere: the layer of gases surrounding Earth
autonomous: able to make some decisions using onboard systems
autonomy: the ability to perform some tasks without immediate human control
axis: an imaginary line about which an object rotates
balanced forces: forces that cancel so there is no change in velocity
booster: a rocket stage that adds thrust, especially early in flight
calendar: a system for organising days, months and years
cardinal directions: north, south, east and west
category: a group defined by a rule
cause and effect: a relationship in which one event helps produce another
Ceres: a dwarf planet in the asteroid belt

Glossary · Continued

Chandrayaan: the name of India's lunar mission series
characteristic: a feature used to describe or identify something
chronological: arranged by time
classification: grouping by shared characteristics
climate: the usual long-term weather pattern of a place
cloud cover: the amount of sky covered by clouds
cluster: a group of galaxies held together by gravity
coma: the cloud around an active comet nucleus
comet: an icy, dusty body orbiting the Sun
commander: the person responsible for leading a mission crew
compass: an instrument used to find direction
conclusion: a result reached from evidence or premises
conduction: heat transfer through contact
confirmation: additional evidence showing a result is reliable
consistent: following the same rule throughout
constellation: an officially named region of the sky
context: information needed to understand a fact correctly
continent: one of Earth's very large land regions
core: the central region of an object
cosmonaut: the Russian term commonly used for a space traveller
counterexample: an example that shows a general claim is false
crater: a bowl-shaped hollow often made by an impact
crescent: a thin curved illuminated shape
crew: the people working together on a spacecraft
crew module: the part of a spacecraft that carries astronauts
current event: a recent occurrence important enough to report
cycle: a pattern that repeats
dark adaptation: the eyes becoming more sensitive in low light

Glossary · Continued

data: recorded facts or measurements
daylight: the part of the day when a place receives sunlight
density: how much mass is packed into a volume
direct sunlight: sunlight arriving more nearly straight onto a surface
disaster monitoring: observing dangerous events to support warnings and response
discovery: new knowledge supported by evidence
distance: the space between two points
docking: bringing spacecraft together and connecting them in orbit
downlink: a signal sent from a satellite to Earth
drag: resistance caused by moving through air or another fluid
dwarf planet: a nearly round body orbiting the Sun that has not cleared its orbital neighbourhood
Earth: the third planet from the Sun and our home world
Earth observation: measuring and imaging Earth from aircraft or spacecraft
eclipse: an event in which one object blocks light or enters a shadow
elliptical galaxy: a rounded or oval galaxy with little visible spiral structure
engineer: a person who designs and improves systems or solutions
equinox: a time when neither hemisphere is strongly tilted toward the Sun
estimate: a sensible approximate value
evidence: facts or measurements used to support an explanation
exhaust: hot gases expelled by an engine
exoplanet: a planet orbiting a star beyond the Sun
expansion: the increase of distances between widely separated galaxies as space changes
flame trench: a channel that directs hot exhaust away
force: a push or pull
forecast: a scientific prediction of future weather
free fall: motion controlled mainly by gravity
friction: a force resisting sliding or motion between surfaces
fusion: the joining of small atomic nuclei that releases energy

Glossary · Continued

Gaganyaan: India’s human-spaceflight programme
galactic centre: the central region of a galaxy
galaxy: a huge gravitational system of stars, gas, dust and dark matter
gas giant: a very large planet made mostly of gases
geostationary: appearing fixed over one region of Earth
geostationary orbit: an orbit where a satellite appears fixed above one point on the equator
gnomon: the part of a sundial that casts a shadow
GPS: the United States global satellite navigation system
gravitational wave: a ripple in spacetime produced by accelerating massive objects
gravity: the attraction between objects with mass
ground station: a facility that communicates with spacecraft
ground team: people on Earth who support a mission
growing pattern: a pattern that increases or changes regularly
habitable: suitable for living things
halo orbit: a three-dimensional orbit around a Lagrange region
heat: energy transferred because of a temperature difference
heliosphere: the broad region influenced by the solar wind
hemisphere: one half of Earth
high tide: a time when sea level is relatively high
horizon: the line where the land or sea seems to meet the sky
human-rated: designed and verified to meet safety requirements for carrying people
ice giant: a giant planet rich in water, ammonia and methane materials
inclined plane: a sloping surface used to raise or lower objects
inference: an explanation drawn from observations and knowledge
inner planets: Mercury, Venus, Earth and Mars
instrument: a tool used to observe or measure
interplanetary: travelling or operating between planets
irregular galaxy: a galaxy without a regular spiral or elliptical shape

Glossary · Continued

ISRO: the Indian Space Research Organisation
ISS: the International Space Station
L1: a useful gravitational region between Earth and the Sun
lander: a spacecraft designed to reach and operate on a surface
launch pad: the prepared site from which a rocket launches
leadership: guiding people toward a shared goal
lever: a rigid bar turning around a fulcrum
life cycle: the stages through which an object changes over time
life support: systems that provide conditions needed for survival
light: electromagnetic energy that travels through space
light pollution: extra artificial light that brightens the night sky
light-year: the distance light travels in one year
local time: clock time for a particular location
logic: reasoning according to clear rules
longitude: imaginary north-south lines used to locate places east or west
luminosity: the total energy a star gives out per second
luminous: giving out light
lunar eclipse: the Moon passes through Earth's shadow
LVM3: India's heavy-lift Launch Vehicle Mark-3
magnetic field: an invisible region where magnetic forces act
magnitude: a number system astronomers use to describe brightness
Mangalyaan: India's Mars Orbiter Mission
map: a representation of a place or surface
Mars: the fourth planet from the Sun
mass: the amount of matter in an object
measurement: a quantity found using a standard unit or tool
metallic: containing or made from metal
meteor: the streak of light produced during atmospheric entry

Glossary · Continued

meteor shower: many meteors seen when Earth crosses a debris stream
meteorite: material from a meteoroid that reaches the ground
meteoroid: a small natural rock or metal object moving in space
meteorologist: a scientist who studies weather and the atmosphere
microgravity: a condition of continuous free fall that makes objects seem nearly weightless
midday: the middle part of the day when the Sun is highest
midnight: the middle of the night
milestone: an important stage or achievement
Milky Way: the galaxy containing our solar system
mission control: the team and facility that monitor and manage a mission
mission role: the specific responsibility held by a team member
mission specialist: a crew member trained for particular scientific or technical tasks
model: a simplified representation used to explain or test an idea
module: one section of a larger spacecraft or station
monsoon: a seasonal wind system often linked with major rainfall changes
moon: a common name for a natural satellite
motion: a change in position over time
natural satellite: a naturally formed object that orbits a larger body
NavIC: India's regional satellite navigation system
navigation: finding position and direction of travel
neap tide: a tide with a smaller-than-usual range near quarter Moon
near-Earth object: an asteroid or comet whose orbit brings it near Earth's orbit
neighbourhood: the region near an object's orbit
neutron star: an extremely dense remnant of a massive star
NISAR: a joint NASA-ISRO radar Earth-observation mission
north-east: the direction halfway between north and east
nucleus: the solid central part of a comet
observable universe: the region whose signals have had time to reach us

Glossary · Continued

observation: information gathered by carefully watching or measuring
ocean: a vast body of salt water
odd one out: the item that does not share the chosen rule
opaque: blocking most visible light
operation: a mathematical action such as addition or division
orbit: the curved path of one object around another
orbital period: the time an object takes to complete one orbit
orbiter: a spacecraft that travels around a world
Orion: a well-known constellation visible from many parts of Earth
outer planets: Jupiter, Saturn, Uranus and Neptune
oxidiser: a chemical that allows fuel to burn without using outside air
pattern: an arrangement that can be recognised
payload: the instruments or equipment that perform a mission
penumbra: a region of partial shadow
phase: the apparent shape of the Moon’s illuminated part
pioneer: a person who helps open a new field or path
planet: a large nearly round body orbiting a star that has cleared its orbital neighbourhood
plasma: a very hot state of matter containing charged particles
polar orbit: an orbit that passes near both poles
Proxima Centauri: the nearest known star beyond the Sun
PSLV: India’s Polar Satellite Launch Vehicle
pulley: a grooved wheel used with a rope or cable
radiation: energy transfer by electromagnetic waves
radio wave: a type of electromagnetic wave used for communication
re-entry: the return of a spacecraft into an atmosphere
reasonable: making sense for the situation
receiver: a device that detects and uses incoming signals
record: to write, draw or save information

Glossary · Continued

recovery: bringing hardware back safely
red dwarf: a small, relatively cool and faint type of star
reflect: to bounce light from a surface
reflection: a flipped mirror image
refurbish: to inspect, repair and prepare for reuse
relay: to receive and pass along a signal
remote sensing: collecting information about an object or area from a distance
repeat: to do again in the same way
repeating: returning in the same cycle
resupply: delivery of food, equipment, fuel or other materials
reusable: designed to be used more than once
revolution: the motion of one object travelling around another
ring system: many particles orbiting a planet in flat bands
rocket: a vehicle or engine that produces thrust by expelling material
rocky planet: a planet made mainly of rock and metal with a solid surface
rotation: the spinning of an object around its axis
rotation period: the time a world takes to spin once
rover: a mobile robotic vehicle that travels on another world
safe zone: an area kept clear to reduce risk
sample return: a mission that brings material back to Earth
satellite: an object that moves around a larger body
satellite bus: the supporting structure and systems of a spacecraft
satellite image: a picture or measurement of Earth made from orbit
scale: the relative size, distance or time used for comparison
scatter: to send light in many directions
science phase: the mission period focused on planned scientific observations
scientist: a person who investigates questions using evidence
season: a part of the year with a typical pattern of weather and daylight

Glossary · Continued

sequence: items placed in a particular order
simple machine: a basic device that changes force or motion
soft landing: a controlled landing slow enough to avoid destructive impact
solar eclipse: the Moon blocks some or all of the Sun from view
solar filter: a special certified filter made for safe Sun observation
solar system: the Sun and all objects held in orbit around it
solar wind: a stream of charged particles flowing from the Sun
solstice: a time when a hemisphere is most tilted toward or away from the Sun
sounding rocket: a research rocket that makes a suborbital flight
space age: the period beginning with the first artificial satellites
space agency: an organisation that plans and carries out space activities
space station: a spacecraft designed for people to live and work in orbit
space telescope: a telescope operating above Earth's atmosphere
space weather: conditions in space influenced mainly by solar activity
spacewalk: work performed outside a spacecraft in space
spectrum: light arranged by wavelength or energy
speed: distance travelled per unit time
spiral arm: a curved region of stars and gas in a spiral galaxy
spiral galaxy: a galaxy with a disc and curved arms
spring tide: a tide with a larger-than-usual range near new or full Moon
star: a hot glowing ball of gas that produces energy and light
star system: two or more stars related by gravity
status: the present stage or condition of a mission
suborbital: reaching space without completing an orbit
sunrise: the appearance of the Sun above the eastern horizon
sunset: the disappearance of the Sun below the western horizon
teamwork: people combining different skills to achieve a task
technology: tools and methods created using scientific knowledge

Glossary · Continued

tectonic plate: a slowly moving piece of Earth’s rigid outer layer
temperature: a measure related to how energetic particles are
terminator: the boundary between the lit and dark halves of a world
thrust: the pushing force that accelerates a rocket
tide: the regular rise and fall of sea level
tilt: a slant from an upright position
time delay: the travel time of a signal between distant places
time zone: a region using the same standard clock time
timing: precise measurement and distribution of time
transit: a passage in front of a star that slightly reduces its observed light
transparent: allowing much light to pass through clearly
twilight: the partly lit time before sunrise or after sunset
twinkle: rapid changes in the apparent brightness or position of a star
umbra: the darkest central part of a shadow
unbalanced force: a total force that causes acceleration
uncertainty: the amount that remains unknown or imprecise
unit: a standard used to express a measurement
universe: all space, time, matter and energy
uplink: a signal sent from Earth to a satellite
velocity: speed in a particular direction
visibility: how easily something can be seen
visible light: the wavelengths detected by human eyes
visual clue: a feature in a picture that helps solve a problem
waning: the illuminated part appears to shrink
waxing: the illuminated part appears to grow
weight: the force of gravity on an object
year: the time Earth takes to orbit the Sun

Scientific Source Notes

The student chapters use simple language and therefore do not carry academic references on every page. The following primary organisations were used to verify mission status and scientific facts. Current information was reviewed through 29 June 2026.

- **Indian Space Research Organisation (ISRO):** Mission pages and official updates for Chandrayaan, Mars Orbiter Mission, Aditya-L1, Gaganyaan, SpaDeX, NISAR and Axiom Mission 4.
- **National Aeronautics and Space Administration (NASA):** Solar System Exploration, Earth science, NISAR, Moon, Mars, stars and galaxies.
- **European Space Agency (ESA):** Age-appropriate background on space science, launch vehicles and exploration.
- **International Astronomical Union (IAU):** Definitions and naming conventions for planets, dwarf planets and astronomical objects.
- **National Oceanic and Atmospheric Administration (NOAA):** Weather satellites, atmosphere, tides and Earth observation.

Time-sensitive 2025-2026 updates used in this edition

SpaDeX: first docking on 16 January 2025; undocking in March 2025; second docking and power transfer in April 2025.

NISAR: launched on 30 July 2025 and entered its science phase in November 2025.

Axiom Mission 4: Shubhanshu Shukla launched in June 2025 and completed an 18-day scientific mission involving the International Space Station.

Gaganyaan: development and qualification testing continued; the second Integrated Air Drop Test was completed on 10 April 2026. The crewed mission is not described as completed.